Optimal Carbon Tax with Dynamic Skill Heterogeneity

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July 23, 2026†
Work in Progress: Latest version

Abstract

How should a government price carbon when it cares about redistribution and emissions? I extend the dynamic Mirrlees model to two goods, one polluting, with privately observed skills that evolve over the life cycle. Under weakly separable preferences, the optimal carbon wedge equals the social cost of carbon while the labor and savings wedges keep their standard forms. The below-Pigou prescription of the second-best tradition disappears once the income tax is set optimally. Because actual carbon taxes are linear, I also consider the problem when the instrument must be uniform and find that the restriction constrains redistribution but not externality correction. In a US calibration, the optimal energy wedge rises with skill, so the regressivity of real carbon taxes comes from their linearity. However, a flat tax at the social cost of carbon captures 98% of the welfare gain of the full optimum, so the income tax should focus on redistribution and the carbon price on externality correction.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*
JEL Classification: *D62; D82; H21; H23*

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†I thank my advisor Guillaume Sublet for his invaluable guidance and support, as well as the members of his workshop and the participants of the CIREQ Weekly PhD Student Research Workshop, the McGill University microbreakfast, the University of Montréal macro brownbag, the CIREQ-CIRANO-RRECQ Environmental and Natural Resource Economics Workshop, the 19th CIREQ PhD Students' Conference, the 59th CEA Annual Conference, the 35th CREEA Annual Conference, the 65th SCSE Conference, the 60th CEA Annual Conference, the 25th LAGV conference, and the 7th WCERE conference for their comments and feedback. I am grateful for the support and funding from the Fonds de Recherche du Québec Société et Culture (FRQSC) through the Doctoral Training Scholarships program. DOI: https://doi.org/10.69777/329014.

1 Introduction

Carbon pricing is the economist's favorite climate policy, and the voter's least.France abandoned its carbon tax plans after the *gilets jaunes* protests and Canada repealed its consumer carbon tax in 2025 after it became an electoral wedge issue on the grounds that the tax unfairly burdens low-income households. This disconnect might in part be driven by the fact that most carbon policy is informed by integrated assessment models (Environment and Climate Change Canada (2016), Interagency Working Group on Social Cost of Greenhouse Gases (2021)), which are representative agent models that are not designed to address this concern.

Whether carbon taxes are regressive is a topic of intense study, with one aspect of the debate relating to the choice of measure. When households are ranked by annual income, taxes on gasoline and other energy appear to fall more heavily on the poor. However, when households are ranked by lifetime income, the burden appears much flatter because households smooth consumption and today's low earners include the young and temporarily unfortunate (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is thus a lifetime object, not an annual one. It's therefore important to consider the timing of an optimal carbon tax.

However, most research focuses on the incidence of existing carbon taxes rather than asking how they should be designed in the first place. Barrage (2019) and Douenne et al. (2026) are the exceptions, but the first ties the tax system to an *ad hoc* revenue target rather than optimal redistribution and the second restricts attention to linear instruments even though income taxes are in practice non-linear. How then should a government price carbon when it cares about redistribution, and does the presence of pollution change how income should be taxed?

The answer turns on how the income tax is modeled. The Ramsey tradition treats it as a schedule of linear instruments and usually finds that carbon should be priced below the Pigouvian level because the carbon tax compounds the distortions the rest of the tax system already imposes. The Mirrleesian tradition allows the government to levy nonlinear income taxes and usually finds that the carbon tax should be set at the Pigouvian level, barring exceptions in the structure of preferences.

I take the Mirrleesian approach into a dynamic setting with two goods, one clean and one dirty, in which agents draw privately observed skill shocks over their lifecycle and their consumption of the dirty good feeds a stock externality. To my knowledge, this is the first dynamic-Mirrleesian derivation of the optimal carbon tax. Compared to the static irrelevance theorems of Kaplow (2012) and Jacobs and Mooij (2015), dynamics introduce savings and history-dependent wedges.

I find that the static irrelevance result holds in a dynamic environment. Under weak separability, the carbon wedge equals the social cost of carbon, the labor wedge keeps the Golosov et al. (2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Redistributing through the optimal nonlinear income tax restores the split between corrective and redistributive taxation.

That is, the carbon price should not be adjusted to perform redistribution.

I then quantify the separation by solving the optimal mechanism. The optimal wedge on energy is the social cost of carbon scaled up by a history-dependent Corlett-Hague premium: it starts at the Pigouvian level, rises with age as the labor wedge does, and ends the working life about 9% above the Pigouvian price. That premium is however small in welfare terms. A flat carbon tax at the social cost of carbon captures 98% of the welfare gain of the full optimum and the wedge's entire Mirrleesian fine structure is worth only a few thousandths of a percent of consumption. This near-sufficiency is the separation in welfare terms: pricing carbon at the social cost of carbon and redistributing through the income tax is close to optimal even when preferences are nonseparable and energy is a leisure complement.

The deviations from the Pigouvian benchmark that do arise are small, signed by how each good complements labor, and progressive in skill. Late in the working life the optimal energy wedge reaches about 70% for the top skill quintile against 57% for the bottom, both above the Pigouvian level near 50%. The optimal carbon wedge rises with skill and the regressivity of real carbon taxes is therefore a property of flat instruments. This is the normative counterpart the positive incidence debate does not answer to, reaching from the Mirrleesian side the near-Pigouvian result that Douenne et al. (2026) and Douenne et al. (2025) reach from the Ramsey side.

These results assume every margin is set optimally. Real systems set them one at a time, and Section 7 prices the carbon tax against a labor, savings, or commodity wedge frozen away from its optimum. The deviations are signed. A frozen labor wedge held above its optimum pulls the carbon price below the social cost of carbon, and one held below pushes it above, by the redistribution the schedule fails to carry. A frozen savings wedge cannot move the carbon price in one direction; it only tilts the price over the life cycle, above the social cost of carbon at some ages and below it at others. And when the carbon price itself is legislated below the social cost of carbon, the case governments actually face, the optimal labor wedge rises by the marginal budget share of energy times the uncollected damage: the income tax collects the carbon the carbon price misses. The optimal linear instruments of the Ramsey tradition sit inside this family, at the levels where the frozen wedge's shadow price averages to zero.

Section 2 places the paper among the Ramsey, static-Mirrleesian, and dynamic public-finance literatures. Sections 3 to 5 build the two-good dynamic Mirrlees economy and derive the separation and the modified social cost of carbon. Section 6 restricts the carbon tax to a uniform rate, the form real carbon taxes take, and prices the welfare cost of the restriction. Section 7 freezes the labor, savings, and commodity wedges in turn and re-derives the rest against them. Section 8 solves the mechanism numerically on US data. Section 9 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price and draws on a third that has the tools to adjudicate between them, yet has never been aimed at the environment. Should a government facing a distortionary tax system price carbon below marginal damages because the environmental tax compounds distortions elsewhere, above marginal damages because the environmental tax can be used to ease distortions elsewhere, or exactly at marginal damages?

The Ramsey tradition typically finds that the optimal carbon tax should lie below marginal damages, but that the structure of the optimal income tax is unchanged (Sandmo 1975; Bovenberg and Mooij 1994; Bovenberg and Smulders 1996). Barrage (2019) extends their static results with a dynamic climate-economy Ramsey model and finds that the optimal carbon tax should be 8-24% below the first-best Pigouvian level in the presence of distortionary income taxation. However, this analysis assumes a representative agent, focusing on efficiency and ignoring any distributional concerns.

Adding heterogeneous agents to this model overturns this result. Douenne et al. (2026) extend Barrage's dynamic Ramsey model to heterogeneous agents, taking inspiration from the model developed in Werning (2007), and find that the optimal carbon tax is nearly Pigouvian. Douenne et al. (2025) show that this is robust to the inclusion of uninsurable income risk. Whether such a flat carbon tax is regressive and whether revenue recycling is progressive is still an open debate (Klenert et al. 2018; Goulder et al. 2019) and a parallel policy and ethics literature presses for differentiated carbon taxation (Chancel 2022; Dietsch 2024).

Without personalized carbon taxes, the exact restrictions on a uniform carbon tax reshape both its level and incidence. Using a heterogeneous agent microsimulation model calibrated to German data and focusing on revenue-neutral carbon tax reforms, Ploeg et al. (2025) find that the carbon tax should exceed marginal damages when agents have high inequality aversion. In a model calibrated to German data where households differ in income and energy efficiency, Hänsel et al. (2022) show that the first-best keeps the carbon tax uniform at the Pigouvian level and carries all redistribution through lump-sum transfers conditioned on the household's income–energy type while leaving the income tax undistorted.

Separation can also fail for a reason orthogonal to instrument constraints. When climate damages fall unequally across households and compensation is incomplete, the social cost of carbon itself becomes distribution-dependent (Dennig et al. 2015; Kornek et al. 2021). I hold damages homogeneous to isolate the redistributive-instrument channel, so the separation I study is the Atkinson–Stiglitz one.

A static Mirrleesian literature establishes that separation and overturns with it the Ramsey results: the below-damages result is not intrinsic to distortionary taxation but rather an artifact of the linear-instrument restriction. Once the income tax is the optimal nonlinear schedule of Mirrlees (1971), the logic of Atkinson and Stiglitz (1976) applies and differential commodity taxes are redundant under weak separability. Cremer et al. (1998), Cremer et al. (2001), and Cremer et al. (2010) develop the environmental version of the problem where a nonlinear income tax alongside a strictly Pigouvian levy is efficient, so the public-funds correction of the Ramsey tradition disappears. Kaplow (2012) shows that any environmental reform can be made distribution-neutral by a type-by-type adjustment of the income tax, so a move to Pigouvian pricing is Pareto-improving whatever the schedule. Jacobs and Mooij (2015) show that, when goods and environmental quality are weakly separable from labor, the optimal pollution tax equals marginal damages exactly and the marginal cost of public funds is one.

When separability fails, the tax follows the logic established in Corlett and Hague (1953). A good is taxed above the Pigouvian level for a leisure complement, the case West and Williams (2007) substantiate for gasoline, and below it for a leisure substitute. Saez (2002) extends the benchmark to preferences that vary with ability. In a linear-tax setting, Jacobs and Ploeg (2019) show that the pollution tax stays Pigouvian when Engel curves are linear and slips below only when the poor spend disproportionately on the dirty good under nonlinear Engel curves. Every result in this literature is static, with no savings and no lifecycle dynamics. Therefore, it cannot speak to the age profile or the time path of a carbon tax.

The tools for this dynamic treatment come from the New Dynamic Public Finance literature. Golosov et al. (2003) derive a rationale for positive capital taxation in a dynamic Mirrlees economy, since it is harder to elicit future work from households that save privately. This overturns the classic zero-capital-tax benchmark common in the Ramsey tradition (Judd 1985; Chamley 1986). Moreover, the authors find that the Atkinson-Stiglitz uniform-commodity result carries over into the dynamic setting. Kocherlakota (2005) shows how to implement the capital wedge into a wealth-tax and Albanesi and Sleet (2006) derive a tractable recursive form of the problem. Kapička (2013) extends the first-order approach to persistent shocks, Pavan et al. (2014) give the general envelope characterization, Farhi and Werning (2013) characterize the labor wedge over the life cycle, and Golosov et al. (2016) extend the ABC form of the labor wedge derived in the static frameworks of Mirrlees (1971) and Saez (2001) into a dynamic ABCD form. The optimal wedges the dynamic Mirrlees model delivers vary with age, connecting to the age-dependent-tax literature (Weinzierl 2011; Bastani et al. 2013; Erosa and Gervais 2002), which prices age-varying labor rather than a commodity. None of this machinery has been aimed at commodity taxation under non-separable preferences or corrective taxation.

I extend the dynamic Mirrleesian apparatus from one consumption good to two, letting the second pollute, and characterize the optimal carbon tax alongside the labor and savings wedges over the life cycle. To my knowledge this is the first dynamic Mirrleesian derivation of the optimal carbon tax. The heterogeneous-agent Ramsey results and this paper share the Jacobs and Mooij (2015) and Kaplow (2012) irrelevance result, which both approaches carry into a climate economy by different routes. Douenne et al. (2026) carry it with an optimally-set linear transfer, showing analytically that the marginal cost of public funds averages to one across a balanced-growth path and equals one period by period when preferences are logarithmic. Douenne et al. (2025) find that this result still exists when agents face uninsurable labor risk. I consider uninsurable labor risk with a fully nonlinear income tax and obtain a carbon wedge equal to the social cost of carbon at every history under weak separability and any preference satisfying mild conditions, alongside a Mirrleesian labor wedge and a non-zero savings wedge following the inverse Euler condition. The two approaches practically converge in their policy conclusions however: pricing carbon at the social cost of carbon and redistributing through the income tax captures almost all the gains of a non-linear carbon wedge.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity, and consider a single country as well as a single generation living for $T+1$ periods.

Each household consumes a clean good c_t and a polluting good q_t, supplies labor l_t, and is exposed to the aggregate pollution stock S_t. Expected lifetime utility is

$$\mathbb{E}_0\left[\sum_{t=0}^{T}\beta^t U(c_t, q_t, l_t, S_t)\right]$$

$\beta \in (0,1)$ is the discount factor and utility is non-separable between its arguments. The clean good c is the numéraire, with prices normalized to 1. A unit of output transforms one-for-one into either good, so the producer price of the dirty good is also 1. The clean good is emission-free, whereas each unit of the dirty good emits a unit of carbon, so that date-t emissions equal aggregate dirty-good consumption Q_t. Emissions accumulate into a persistent pollution stock S_t, a function of current and past aggregate emissions.

$$S_t = S\left(Q_0, Q_1, \dots, Q_t\right)$$

where $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\, dF_0(\theta)$ is aggregate consumption of the dirty good at date t. I assume that S is differentiable and non-decreasing in each argument, but otherwise leave it unrestricted. The linear special case $S_t = \sum_{j=0}^{t}(1-d_j)\, Q_{t-j}$ in which each past emission flow leaves an additive trace $(1-d_j)$ in the current stock, is the carbon-cycle representation of

Golosov et al. (2014). The general form above nests it and allows nonlinear accumulation. A convenient leading example keeps a fraction δ of the stock from one period to the next.

$$S_t = \sum_{j=0}^{t} \delta^j Q_{t-j}$$

This is the recursion $S_t = \delta S_{t-1} + Q_t$ from an initial pollution stock. To keep the state finite, I assume emissions stop affecting the pollution stock after s periods.

$$S_t = S\left(Q_{t-s}, ..., Q_t\right)$$

Aside from the polluting good, the model follows the standard dynamic Mirrlees setup. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t, each agent knows their skill draws up to time t, which I denote by a skill history vector $\theta^t = (\theta_0, \ldots, \theta_t)$. Θ^t is the set of all possible skill histories.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable[1]. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \to \Theta^t$ describe how an agent with skill history θ^t reports it, and let Σ^t be the set of all possible reporting strategies. Capital transfers resources across periods through a linear technology with exogenous constant gross return $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta)\, \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a Pareto weight on the welfare of an agent with initial skill θ, normalized so that $\int_0^\infty \alpha(\theta)\, dF_0(\theta) = 1$. A utilitarian planner sets $\alpha(\theta) = 1$ for all θ. Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect to labor. Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U\left(c, q, \frac{y}{\theta}, S\right)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ.

Assumption 3.1 (preferences).
U is twice continuously differentiable with $U_c, U_q > 0$, $U_l, U_S < 0$, $U_{cc}, U_{qq}, U_{ll}, U_{SS} \leq 0$, strictly

[1] This is a plausible assumption for some goods such as electricity and air travel. For goods whose consumption cannot be observed, the planner is further constrained and the unrestricted wedges of Sections 4 and 5 no longer apply. The uniform tax of Section 6 survives: an anonymous linear price requires observing total expenditure, not its composition.

quasi-concave in (c, q) with $2U_cU_qU_{cq} - U_c^2U_{qq} - U_q^2U_{cc} > 0$ (regular strict quasi-concavity: a strictly convex, non-degenerate indifference map), and satisfies Inada conditions in (c, q). Single crossing holds at every S.

$$\frac{\partial}{\partial\theta}\frac{U_l(c, q, y/\theta)}{U_c(c, q, y/\theta, S)} \geq 0 \qquad \text{for all } (c, q, y, \theta)$$

At the optimal allocation, $1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) > 0$, where $\epsilon_t, \gamma_{c,t}$ are defined in (1) below.

Assumption 3.2 (shock process)**.**
For all $t \geq 1$, f_t is differentiable in both arguments with $f_t' = \partial f_t/\partial\theta$ and $f_{2,t} = \partial f_t/\partial\theta_{t-1}$. For all θ_{t-1}, the ratio

$$\frac{\theta_{t-1}\int_\theta^\infty f_{2,t}(x|\theta_{t-1})\,dx}{\theta f_t(\theta|\theta_{t-1})}$$

is bounded, and $\lim_{\theta\to\infty}\frac{1-F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})}$ is finite.

Any admissible allocation must be incentive compatible: the agent must weakly prefer truthful reporting to any other strategy. The pollution stock S_t is an aggregate, so no single agent's deviation moves it and it enters both sides of the constraint identically. For every initial type θ and every reporting strategy,

$$\begin{aligned}\mathbb{E}_0\left[\sum_{t=0}^{T}\beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)\middle|\theta\right]\\ \geq \mathbb{E}_0\left[\sum_{t=0}^{T}\beta^t U\left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t\right)\middle|\theta\right]\end{aligned}$$

Over a finite horizon, global incentive compatibility is equivalent to the absence of a profitable one-shot deviation after every history (Fernandes and Phelan 2000; Kapička 2013). I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing these constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_\theta\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right) + \beta\, w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1})\, \partial_{\theta_t} f_{t+1}(\theta_{t+1}|\theta_t)\, d\theta_{t+1}$ is the marginal continuation value of type θ^t, and the dot denotes a derivative with respect to the current draw θ_t. The continuation term is the future information rent of the current draw: with i.i.d. shocks, $\partial_{\theta_t} f_{t+1} = 0$ and it vanishes, since today's skill then carries no information about tomorrow's. Under Assumptions 3.1 and 3.2, this local condition is necessary for incentive compatibility (Kapička 2013). The monotonicity requirements on the allocation belong to the sufficiency direction, not to necessity. The local conditions define a relaxed problem and are necessary but not sufficient for global incentive compatibility. Indeed, no general primitive restriction guarantees sufficiency in dynamic models with multiple goods (Golosov et al. 2016). I solve the relaxed problem throughout and verify global incentive compatibility numerically in Section 8.

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t)|\theta]\, dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t)|\theta]\, dF_0(\theta) + RK_t$$

where K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t, summing over t, and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0\left[\sum_{t=0}^{T} \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t}\middle|\theta\right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0\left[\sum_{t=0}^{T} \frac{y_t(\theta^t)}{R^t}\middle|\theta\right] dF_0(\theta) + RK_0$$

The planner chooses allocations to maximize welfare subject to the local incentive constraint, the intertemporal resource constraint, and the pollutant law of motion.

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t,\ t=0,\dots,T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0\left[\sum_{t=0}^{T} \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)\middle|\theta\right]\right) dF_0(\theta)$$

Three wedges, the implicit marginal taxes on the labor, savings, and commodity margins, describe the distortions the optimal allocation imposes.

$$1 - \tau_t^y(\theta^t) \equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\theta_t\, U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}$$

$$1 - \tau_t^s(\theta^t) \equiv \frac{1}{\beta R}\,\frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1}\right)\right]}$$

$$1 + \tau_t^q(\theta^t) \equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}$$

I define six preference statistics used throughout. Two are the standard own-curvature elasticities, namely the inverse Frisch elasticity and the inverse intertemporal elasticity of substitution.

$$\epsilon_t \equiv \frac{U_{ll,t}\, l_t}{U_{l,t}} \qquad \sigma_t \equiv -\frac{U_{cc,t}\, c_t}{U_{c,t}} \tag{1}$$

Four are cross-partials that measure how the marginal utilities of the two goods and of the pollutant move with labor as well as with each other. In order, they represent the labor complementarity with the clean good, the labor complementarity with the dirty good, the labor complementarity of environmental quality, and the cross-elasticity between the two goods.

$$\gamma_{c,t} \equiv \frac{U_{cl,t}\, l_t}{U_{c,t}} \qquad \gamma_{q,t} \equiv \frac{U_{ql,t}\, l_t}{U_{q,t}} \qquad \xi_t \equiv \frac{U_{Sl,t}\, l_t}{U_{S,t}} \qquad \sigma_{cq,t} \equiv -\frac{U_{cq,t}\, q_t}{U_{c,t}}$$

Note that complementarity is measured with respect to labor rather than leisure here, so a positive γ signals a good whose marginal utility rises as the agent works more. To get the leisure complementarity interpretation, it suffices to flip the sign. Most of the commodity-wedge results hinge on the gap $\gamma_{c,t} - \gamma_{q,t}$ between the clean and dirty goods' complementarities with labor. If the gap is positive, the clean good is the relative labor complement, meaning an agent who works more wants more of it. Likewise, the results pertaining to the social cost of carbon hinge on the gap $\xi_t - \gamma_{c,t}$, which measures how the marginal damage of pollution varies with labor supply. If the gap is positive, an agent who works more suffers more from pollution and a cleaner environment is a relative labor complement.

4 Commodity wedge

With the introduction of a second good, the planner now has access to a third distortion margin through the commodity wedge in addition to the labor and savings wedges. The planner solves the usual recursive problem, following Kapička (2013)[2].

$$V_t(\hat{w}, \hat{w}_2, \theta_-) = \min_{c,q,y,u,w,w_2} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w(\theta)$$

In words, the planner's goal is to minimize resource cost while offering an agent expected lifetime utility $\hat{w}$ with marginal continuation value $\hat{w}_2$. $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ. It equals the primitive Pareto weight in the initial period, $\bar{\alpha}_0(\theta) = \alpha(\theta)$, and $\bar{\alpha}_t(\theta) = 1$ for $t > 0$. As in Golosov et al. (2016), the lifetime Pareto weights enter only at date 0. From date 1 on, efficiency weights each type by its marginal utility of the numéraire through the promise-keeping constraint, so the continuation weight collapses to one. Solving the problem yields the three wedges below, the labor and savings wedges written in the marginal utility of the numéraire.

Proposition 4.1 (Optimal wedges)**.**
The labor wedge is

$$\frac{\tau^y}{1-\tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \, \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} \, D_t(\theta)$$

[2] Kapicka's consumption set is an arbitrary convex and compact subset of $\mathbb{R}^N$, so consumption can be a vector of multiple goods rather than a scalar.

$$
\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right] (1 - \lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)) \frac{f_t(x)dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}
\end{aligned}
$$

Define $\widehat{U}_{c,t} \equiv U_{c,t}/(1 - \chi_t)$, with $\chi_t \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}} \frac{\tau_t^y}{1-\tau_t^y}$ as the complementarity-weighted labor wedge. Savings satisfy the generalized inverse Euler equation

$$
\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]
$$

The commodity wedge is

$$
\frac{\tau^q}{1 + \tau^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau^y}{1 - \tau^y}
$$

Proof. See Appendices A.1, A.2, and A.3. □

Corollary 4.1 (dynamic Atkinson-Stiglitz)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the commodity wedge vanishes at every history, $\tau_t^q = 0$.

Proof. Immediate from the commodity wedge of Proposition 4.1 at $\gamma_{c,t} = \gamma_{q,t}$. □

The commodity wedge can be expressed as the labor wedge scaled by the ratio of the labor complementarity gap between the two goods. Its sign follows a Corlett-Hague logic: the planner taxes the good that is the weaker complement of labor. Unlike the commodity wedge in a static Mirrlees framework however, the dynamic wedge is history-dependent whenever preferences are nonseparable and its shape is proportional to that of the labor wedge. When preferences are weakly separable, the commodity wedge vanishes. This is the dynamic Atkinson-Stiglitz uniform-commodity-taxation result of Golosov et al. (2003); the contribution here is the nonseparable case, where the wedge is a scaled labor wedge and inherits its entire history dependence.

The labor wedge is Golosov et al. (2016)'s decomposition term for term, now written in the marginal utility of the numéraire. All terms are unchanged from the one-good case except C_t, the social value of the resources extracted from types above θ. Each unit of goods taken from types $x > \theta$ is discounted by its welfare cost on x. The term in the exponent is the income effect on labor supply of those types, which now runs through the cross-curvature between the numéraire and the second good. The second good thus adds no new force to the labor wedge, only a reweighting of its income effect, which is signable. With the dirty good rising in type, the cross term scales C_t up when the goods are Edgeworth substitutes ($\sigma_{cq,t} > 0$) and down when they are complements. Read at a given allocation, Edgeworth substitutability between the goods thus raises the labor wedge and complementarity lowers it.

The savings wedge follows a generalized inverse Euler condition, but unlike the labor wedge is not directly affected by the presence of the second good. As such, it is unchanged from the one-good savings wedge. When preferences are non-separable, the cost of delivering utility is weighted by $1 - \chi_t$. When the numéraire complements labor, a unit of consumption makes work less painful and thus shrinks the information rent that truth-telling commands, rebating a share χ_t of the delivery cost. This expression of the wedge is comparable to Hellwig (2021), though it is stated pointwise rather than as a change of probability measure.

So far the dirty good is dirty in name only. Once its consumption pollutes, the planner acquires a corrective motive alongside the redistributive one, and the question is whether the two tangle.

5 The optimal carbon wedge

Let the dirty good pollute, so that its consumption feeds a pollution stock S_t that harms utility. As before, the incentive constraints reduce to a recursive problem, following Kapička (2013). Appendix B.1 shows the externality does not disturb the reduction. The argument is short. No report moves the aggregate pollution stock, so the agent's incentive problem is that of Section 4 with utility dated by the pollutant. Aggregates are deterministic, so internalizing them appends one new state $\mathbf{S}_-$, the vector of lagged aggregate emissions feeding current and future pollution. The continuation problem receives the updated state, with the current aggregate Q_t appended.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, \mathbf{S}) \right] f_t(\theta|\theta_-) d\theta$$

s.t.

$$
\begin{aligned}
\dot{u}(\theta) &= U_\theta\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta)\bar{\alpha}_t(\theta) f_t(\theta|\theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta \\
u(\theta) &= U\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right) + \beta w(\theta) \\
S &= S\left(\cdots, \int q(\theta) f(\theta) d\theta\right)
\end{aligned}
$$

$\mathbf{S}$ is the updated emission state ($\mathbf{S}_-$ with Q appended). Solving the problem leaves the income-side wedges untouched. The externality adds a single corrective term to the first-order condition for the dirty good, but does not otherwise change the optimality formulas for the other variables. Hence, the labor and savings wedges are those of Proposition 4.1. Only the carbon wedge is new and, under weak separability, it equals the social cost of carbon.

Proposition 5.1 (three-instrument separation)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrective and redistributive margins separate. The carbon wedge equals the social cost of carbon, the labor wedge keeps its Proposition 4.1 form, and the savings wedge keeps its inverse Euler form,

$$
\tau_t^q = \mathrm{SCC}_t \qquad \frac{\tau_t^y}{1-\tau_t^y} = A_t B_t C_t + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} D_t \qquad \frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]
$$

The carbon price carries no redistributive term; the income tax and the savings wedge do all the redistribution and insurance.

Proof. The first-order conditions are those of the no-pollution problem, so the labor and savings wedges are unchanged from Proposition 4.1. Proposition 5.2 gives $(\tau_t^q - \mathrm{SCC}_t)/(1+\tau_t^q) = (\gamma_{c,t} - \gamma_{q,t})[\,\cdot\,]$, which vanishes at $\gamma_{c,t} = \gamma_{q,t}$, leaving $\tau_t^q = \mathrm{SCC}_t$. □

Redistributing through an optimal nonlinear income tax fully separates externality correction from redistribution. The carbon price is set at the social cost of carbon, unlike the dynamic Ramsey planner in Barrage (2019) who prices carbon below the SCC. Douenne et al. (2026) recover an average near-Pigouvian price with linear instruments whereas it holds exactly here. This result generalizes the static Mirrlees irrelevance result in Jacobs and Mooij 2015 to a dynamic setting.

The SCC is itself the textbook Pigouvian price when abatement does not interact with the labor distortion, either because the pollution stock is as complementary with labor as the clean good at every date, $\xi_{t+j} = \gamma_{c,t+j}$, or because labor is undistorted.

$$\mathrm{SCC}_t = -\sum_{j=0}^{s} \frac{1}{R^j}\, \mathbb{E}_t \int \frac{U_{S,t+j}}{U_{c,t+j}}\, \Phi_j\, f_{t+j}\, d\theta \tag{2}$$

Here, $\Phi_j = \partial S_{t+j}/\partial Q_t$ is the pass-through of current emissions to the future pollution stock. Under weak separability and either of these conditions, the carbon wedge equals this Pigouvian price. Otherwise SCC_t carries a fiscal-adjustment factor, given with the corrective wedge below.

Proposition 5.2 (Optimal corrective wedge)**.**
The optimal corrective wedge between the clean good and the polluting good is

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

where

$$\mathrm{SCC}_t = -\sum_{j=0}^{s} \frac{1}{R^j}\, \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1-\tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1+\epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}}\, \Phi_j\, f_{t+j}\, d\theta$$

with $\Phi_j \equiv \partial S_{t+j}/\partial Q_t$ the effect of current aggregate emissions on the date-$(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by construction) and the expectation taken over future skill histories. The right-hand side is the no-pollution commodity wedge of Proposition 4.1 term for term, with A_t, B_t, C_t, D_t the labor-wedge terms. The externality changes only the left-hand side, replacing τ_t^q with the deviation $\tau_t^q - \mathrm{SCC}_t$.

Proof. See Appendix B.2. □

The corrective wedge decomposes into a Mirrleesian screening component and a corrective component built on the social cost of carbon SCC_t. The wedge now operates on the deviation $\tau_t^q - \mathrm{SCC}_t$ rather than on τ_t^q itself. The fiscal-adjustment factor on each damage term is what separates SCC_t from the Pigouvian benchmark (2). When $\xi_{t+j} > \gamma_{c,t+j}$, abatement relaxes the labor distortion and lends a fiscal bonus, so the SCC rises above the Pigouvian level. When $\xi_{t+j} < \gamma_{c,t+j}$, abatement tightens the distortion and the SCC falls below it. This is the "virtual subsidy" of Jacobs and Mooij (2015) carried to a dynamic setting. Their modified Samuelson rule scales each type's willingness to pay for environmental quality by one plus a term multiplying the labor wedge by $-\partial \ln(u_S/u_c)/\partial \ln l$, equivalent to $\xi - \gamma_c$ here. The planner over-abates relative to pure Pigouvian pricing when a cleaner environment complements work and under-abates when it complements leisure.

6 Uniform commodity wedge

The optimal commodity wedge is history dependent, tracking the labor wedge over one's lifetime, yet real-life commodity and carbon taxes are anonymous. In this section, I therefore consider the optimal tax system under the restriction of linear commodity wedges. A common consumer price $1 + \bar{\tau}_t$ means that the commodity wedge satisfies $U_q/U_c = 1 + \bar{\tau}_t$ for every type with the level $\bar{\tau}_t$ still chosen by the planner. This restriction can be expressed as a wedge constraint in the planner's problem, with Lagrange multiplier $\eta^q(\theta)\, f_t(\theta|\theta_-)$.

$$\frac{U_q\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}\right)}{U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}\right)} = 1 + \bar{\tau}_t \tag{3}$$

Although the planner assigns both goods to each type directly, we can view the constraint through the lens of demand. Any bundle satisfying (3) is the split an agent facing the common price would choose of the expenditure $x \equiv c + (1 + \bar{\tau}_t)q$, with Marshallian demands $c^d(x, l, \bar{\tau})$ and $q^d(x, l, \bar{\tau})$ pinned down by $U_q = (1 + \bar{\tau})U_c$ and the budget. Because each type faces the same price, this is equivalent to a one-good economy where the planner assigns each type total expenditure x, which the agent then splits between c and q.

Lemma 6.1 (implementation and first-order approach)**.**
An allocation satisfies (3) if and only if it assigns each type the private split of some pair (x, y) of expenditure and output at the common price. With indirect utility

$$\tilde{U}(x, l; \bar{\tau}) \equiv \max_{c,q \geq 0} U(c, q, l) \qquad \text{s.t.} \qquad c + (1 + \bar{\tau})q \leq x;$$

the split is unique since $\mathcal{D} < 0$. Incentive compatibility in the restricted economy is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{\tau}_t)$. The envelope condition is

$$\dot{u}(\theta) = -\frac{\tilde{U}_l\, l}{\theta} + \beta\, w_2(\theta) = -\frac{U_l\, l}{\theta} + \beta\, w_2(\theta)$$

Proof. See Appendix C.2. □

Before deriving constrained-optimal wedges, we need to define the Lagrange multiplier and a few relevant statistics. Let $\mathcal{D} \equiv U_{qq} - 2(1 + \bar{\tau})U_{cq} + (1 + \bar{\tau})^2 U_{cc} < 0$ be the curvature of utility along the budget line, negative at every price as a consequence of Assumption 3.1. Define $m_q \equiv (1 + \bar{\tau})\frac{\partial q^d}{\partial x}$ as the marginal budget share of the dirty good and $s_{qq} \equiv \frac{\partial q^d}{\partial(1+\bar{\tau})}\big|_u = \frac{U_c}{\mathcal{D}} < 0$ as its compensated own-price response.

Lemma 6.2 (multiplier)**.**
At every history,

$$\eta_t^q = (1+\bar{\tau}_t)\,|s_{qq,t}|\left[(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t} - \frac{\bar{\tau}_t}{1+\bar{\tau}_t}\right]$$

Proof. See Appendix C.4. □

The bracket is the gap between the commodity wedge the planner would set at the node if the instrument were free, $(\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f}$, and the wedge the uniform tax imposes. The compensated substitution response converts the gap into resources. The multiplier is positive where the uniform tax falls short of the node's Corlett-Hague level and negative where it overshoots. Substituting into the first-order conditions assembles the primitive statistics into composites, the curvatures a reader of the one-good reduction would compute from the indirect utility $\tilde{U}$.

Lemma 6.3 (composite statistics)**.**
Define

$$\tilde{\gamma} \equiv (1-m_q)\,\gamma_c + m_q\,\gamma_q \qquad \tilde{\epsilon} \equiv \epsilon - \frac{(1+\bar{\tau})^2\,U_c^2\,(\gamma_c-\gamma_q)^2}{l\,|U_l|\,|\mathcal{D}|} \;\le\; \epsilon \qquad \tilde{A} \equiv 1 + \tilde{\epsilon} - \tilde{\gamma}$$

Then $\tilde{\gamma} = \tilde{U}_{xl}\,l/\tilde{U}_x$ and $\tilde{\epsilon} = \tilde{U}_{ll}\,l/\tilde{U}_l$ for the indirect utility of Lemma 6.1. In weak separability, $\gamma_c = \gamma_q$, so $\tilde{\gamma} = \gamma_c = \gamma_q$, $\tilde{\epsilon} = \epsilon$, and the restricted problem is that of Section 4.

Proof. See Appendix C.7. □

The composite complementarity is the marginal-spending-weighted average of the two goods' complementarities: a marginal dollar of expenditure is split $(1-m_q, m_q)$ across the goods, so the marginal utility it buys co-moves with labor accordingly. The inequality on $\tilde{\epsilon}$ is a Le Chatelier property. The agent's freedom to re-optimize the split cushions labor shocks, so the effective disutility of labor is less convex and delivering expenditure is a weakly better hedge than delivering a rigid bundle. Both corrections are proportional to the same complementarity gap $\gamma_c - \gamma_q$ that drives the unconstrained wedge of Section 4. Everything the restriction does vanishes when the two goods are equally complementary with labor.

6.1 Constrained-optimal wedges

The three wedges follow from the first-order conditions, each carrying a fiscal-interaction term proportional to the multiplier that the unconstrained problem lacked.

Proposition 6.1 (labor wedge under a uniform commodity tax)**.**
At every history,

$$\frac{\tau^y_t}{1-\tau^y_t} = A_t\,\frac{\psi U_{c,t}}{\theta f_t} + \eta^q_t\,(1+\bar\tau_t)\left[\mathcal{I}_t - \frac{\gamma_{c,t}-\gamma_{q,t}}{(1-\tau^y_t)\,y_t}\right]$$

equivalently, in composite statistics,

$$\frac{\tau^y_t}{1-\tau^y_t} + \frac{\bar\tau_t}{1-\tau^y_t}\,\frac{dq^d_t}{dy}\bigg|_u = \tilde A_t\,\frac{\psi U_{c,t}}{\theta f_t} \qquad \frac{dq^d}{dy}\bigg|_u \equiv (1-\tau^y)\,\partial_x q^d + \frac{\partial_l q^d}{\theta}$$

where $dq^d/dy|_u$ is the utility-compensated response of dirty-good demand to earnings, the change in q^d when output rises by a unit and expenditure rises by $(1-\tau^y)$ units so as to hold utility fixed.

Proof. See Appendices C.5 and C.7. □

The restriction sets a comprehensive wedge, not the income-tax wedge alone, the dynamic counterpart of the effective labor wedge of Jacobs and Boadway (2014). When an agent earns a marginal unit of output on a compensated path, the planner collects τ^y_t through the income margin and $\bar\tau_t\,dq^d/dy|_u$ through the commodity margin. Their sum

$$\Omega_t \equiv \frac{\tau^y_t}{1-\tau^y_t} + \frac{\bar\tau_t}{1-\tau^y_t}\,\frac{dq^d_t}{dy}\bigg|_u$$

is what the screening condition pins down. The correction has two parts of opposite sign. The expenditure part $(1-\tau^y)\,\partial_x q^d$ is positive for a normal good. The labor part $\partial_l q^d/\theta$ is the statistic of Christiansen (1984), negative when the dirty good is a labor substitute. Working more at fixed expenditure shifts spending away from the good. When the labor-substitute part dominates, the measured income-tax wedge exceeds the naive $\tilde A_t\,\psi U_c/(\theta f)$. Discouraging labor then becomes fiscally cheaper: the agent who works less spends more on the taxed good, so part of the forgone output returns as commodity revenue. When the expenditure part dominates, the correction runs the other way, and the income-tax wedge sits below $\tilde A_t\,\psi U_c/(\theta f)$.

Proposition 6.2 (optimal uniform commodity tax)**.**
The optimal level of the uniform tax sets the population average of the multiplier to zero at every date, $\mathbb{E}_0[\eta^q_t] = 0$. Equivalently,

$$\bar\tau_t\;\mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\,l_t\,\partial_l q^d_t\right] \qquad\Longleftrightarrow\qquad \frac{\tau^q_t}{1+\tau^q_t} = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

where $\mathbb{E}_0$ is the unconditional population expectation over date-t histories θ^t. The uniform ad-valorem wedge is the substitution-weighted average of the unconstrained node-specific wedge

$(\gamma_c - \gamma_q)\,\psi U_c/(\theta f)$ of Section 4. This is the dynamic form of the optimal linear commodity tax of Jacobs and Boadway (2014).

Proof. See Appendix C.9. □

The single instrument averages the schedule the planner would set node by node if it could, weighing each history by the compensated substitutability $|s_{qq}|$ at that node. Nodes where the consumer can easily substitute away from the dirty good carry more weight, a deadweight-loss weighting in the Corlett-Hague tradition.[3] The condition is implicit: the weights and the node wedges are themselves evaluated at the restricted allocation, so $\bar{\tau}_t$ is characterized as a fixed point rather than in closed form. Two limits are immediate.

Corollary 6.1 (costless uniformity)**.**
If preferences are weakly separable, $\gamma_c = \gamma_q$ pointwise, then $\bar{\tau}_t = 0$ for every t. The multiplier vanishes at every node and the restricted optimum coincides with the unconstrained optimum. This is the dynamic Atkinson-Stiglitz benchmark of Corollary 4.1. More generally the restriction is costless at date t if and only if the unconstrained wedge $(\gamma_c - \gamma_q)\,\psi U_c/(\theta f)$ is constant across the date-t cross-section.

Corollary 6.2 (retirement)**.**
Suppose agents retire at a date $T_R \leq T$. From T_R on, there is no labor to report, so $\psi \equiv 0$. Hence, $\psi U_c/(\theta f) \equiv 0$ and $\bar{\tau}_t = 0$. The screening component of the uniform tax dies with the labor margin. With the externality below, the dirty-good price converges to pure Pigouvian pricing for retirees.

Proof. See Appendix C.12. □

Proposition 6.3 (savings wedge under a uniform commodity tax)**.**
Define the incentive-plus-fiscal adjustment

$$\chi_t^q \equiv \underbrace{\tilde{\gamma}_t \, \frac{\psi U_{c,t}}{\theta f_t}}_{\text{incentive}} + \underbrace{\varpi_t \, m_{q,t}}_{\text{fiscal}} \qquad \varpi_t \equiv \frac{\bar{\tau}_t}{1+\bar{\tau}_t}$$

equal to the primitive form $\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q\,(1+\bar{\tau}_t)\,\mathcal{I}_t$. The generalized inverse Euler equation holds in the adjusted marginal utility $U_{c,t}/(1-\chi_t^q)$,

$$\frac{1-\chi_t^q}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^q}{U_{c,t+1}}\right]$$

[3] The construction parallels the many-person Ramsey rule of Diamond (1975) in collapsing a heterogeneous population into a single linear-tax condition: there the optimal excise rule, here $\mathbb{E}_0[\eta_t^q] = 0$. The weighting differs in kind. Diamond aggregates through the distributional characteristic, the covariance of demand with the net social marginal utility of income; here the weights are pure compensated substitution responses, with the redistributive content carried inside the node wedges being averaged rather than in the weights.

and to second order $\tau_t^s \approx \mathrm{Var}_t(\ln U_{c,t+1}) + (\chi_t^q - \mathbb{E}_t\,\chi_{t+1}^q) + \mathrm{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1})$.

Proof. See Appendix C.6. □

The savings adjustment now has two parts. The incentive part $\tilde{\gamma}_t\,\psi U_c/(\theta f)$ is the Section 4 adjustment on the composite margin present when expenditure is complementary with labor. The fiscal part $\varpi_t m_{q,t}$ is new. Delivering a marginal unit of expenditure returns the fraction $\varpi_t m_{q,t}$ to the planner as commodity revenue, so its net resource cost is $1-\varpi_t m_{q,t}$ and the planner equalizes the present value of this net incentive-adjusted cost across dates and states. Through the second-order expression, the savings wedge picks up the term $\varpi_t m_{q,t} - \mathbb{E}_t[\varpi_{t+1} m_{q,t+1}]$, the expected change in the commodity-revenue content of spending. With a positive tax on a labor substitute, agents spend a larger share on the taxed good when leisure is high late in the working life, so this term tends to be negative. Deferring expenditure to high-leisure working years generates future commodity revenue and the planner credits saving for it. The credit stops at retirement, where Corollary 6.2 sets the screening tax and with it ϖ to zero.

The comprehensive wedge Ω_t inherits the full dynamic structure of the Section 4 labor wedge.

Proposition 6.4 (comprehensive-wedge decomposition)**.**
At every history,

$$\Omega_t = \tilde{A}_t(\theta)\,\tilde{B}_t(\theta)\,\tilde{C}_t(\theta) + \beta R\,\Omega_{t-1}\,\tilde{D}_t(\theta)$$

where $\tilde{B}_t = B_t$ is the hazard ratio of Proposition 4.1 and $\tilde{C}_t, \tilde{D}_t$ are its C_t and D_t with the integrating kernel built on the composite $\tilde{\gamma}$ and, in $\tilde{C}_t$, the redistributive weight carrying the revenue term $1 - \bar{\tau}_t\,\partial_x q^d - \lambda_{1,t}\bar{\alpha}_t U_{c,t}$.

Proof. See Appendix C.8. □

All of the qualitative structure survives the restriction. Proposition 6.1 then dictates, given $\bar{\tau}_t$, how Ω_t splits between the income-tax wedge and the commodity-revenue term.

6.2 What uniformity costs and where it binds

With the wedges set, what does the restriction cost? Because a single rate matches the personalized schedule only on average, the loss is second-order. It grows with the dispersion of the wedge the planner would otherwise set node by node, weighted at each history by how easily the consumer can substitute away from the dirty good.

Proposition 6.5 (cost of uniformity)**.**
Let $\varpi_t^*(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}$ denote the unconstrained node-specific ad-valorem wedge and let

$\mathrm{Var}_t^{|s|}$ denote the $|s_{qq,t}|$-weighted cross-sectional variance over date-t histories. To second order in the dispersion of ϖ_t^*, the date-t resource-equivalent loss from imposing a common tax rather than the node-specific schedule is

$$\mathrm{Loss}_t \approx \frac{1}{2}\,(1+\bar{\tau}_t)^3\ \mathbb{E}_0[|s_{qq,t}|]\ \mathrm{Var}_t^{|s|}(\varpi_t^*)$$

minimized at the uniform rate of Proposition 6.2.

Proof. See Appendix C.10. □

The cost of not personalizing the commodity tax is the substitution-weighted cross-sectional variance of the unconstrained wedge. It vanishes under weak separability, where $\varpi^* \equiv 0$, and it is small whenever the complementarity gap is small or the wedge $\varpi^* = (\gamma_c - \gamma_q)\,\psi U_c/(\theta f)$ is flat across the cross-section. This is the proportionality result of Section 4. Because the unconstrained wedge tracks the labor wedge, its dispersion is first-order in the dispersion of τ^y and what that tracking is worth is what uniformity forgoes. Every object in the formula is a demand-side or tax-side statistic, so it is a sufficient-statistics answer to the cost-of-mispricing question. The near-sufficiency of a flat carbon price documented in the quantitative section is precisely this variance being small.

Now reintroduce the externality of Section 5. Because the pollution stock is aggregate and deterministic, it is invariant to reports and acts as a dated preference shifter. The dirty good's first-order condition gains the shadow emission price $\nu_t = \mathrm{SCC}_t$ of Appendix B.2 and every condition of this section holds with the deviation $\bar{\tau}_t/(1+\bar{\tau}_t)$ replaced by $(\tau_t^q - \mathrm{SCC}_t)/(1+\tau_t^q)$, where $1+\tau_t^q$ is the common consumer price. The uniform per-unit tax net of marginal damages is the screening component $\bar{\tau}_t^{\mathrm{screen}} \equiv \tau_t^q - \mathrm{SCC}_t$, the revenue wedge net of the part that pays for damages.

Proposition 6.6 (additive split of the uniform price)**.**
At an interior optimum of the restricted economy with the externality, all of the preceding results hold with $\bar{\tau}_t = \tau_t^q - \mathrm{SCC}_t$ the screening tax and every statistic evaluated at the externality allocation. The optimal uniform consumer price splits additively,

$$1+\tau_t^q = 1+\mathrm{SCC}_t + \bar{\tau}_t^{\mathrm{screen}} \qquad \frac{\bar{\tau}_t^{\mathrm{screen}}}{1+\tau_t^q} = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

and the social cost of carbon keeps its discounted-damage form, its damage weight adding to the Section 5 fiscal factor a term proportional to the multiplier η_t^q (Appendix C.11).

Proof. See Appendix C.11. □

The uniformity restriction never binds the corrective component. The corrective part of the optimal dirty-good price was uniform already at the unconstrained optimum since a ton of emissions carries the same social cost whoever emits it and the emissions margin cannot screen. What uniformity constrains is only the Corlett-Hague screening part, forced from a history-dependent schedule down to the substitution-weighted scalar of Proposition 6.2. Hence, a legislated uniform carbon price sacrifices nothing on the corrective margin: the corrective wedge is a scalar the uniform price implements exactly, so the whole cost of uniformity falls on screening, which Proposition 6.5 measures. The level of the social cost of carbon is nonetheless co-determined with the constrained allocation, its damage weight carrying the multiplier node by node (Appendix C.11); this reprices carbon in the constrained second best but is not a loss on the corrective margin, which has no cross-sectional dispersion to forgo. This is the separation of Section 5 restated for the instrument governments actually use.

7 Third-best wedges

The separation of Proposition 5.1 is a property of the jointly optimal tax system, but real tax systems are often revised one instrument at a time. Hence, one might ask what happens if the tax system is currently set at a suboptimal setting and the planner can only adjust one instrument. In our setting, this is equivalent to a planner solving for the optimal allocation when one of the wedges is fixed at an exogenous schedule. This is a third-best optimum in the sense that the wedges are not optimized jointly.

7.1 Constrained labor wedge

Suppose the planner cannot adjust the labor wedge, such that the planner solves the problem of Section 5 with the labor wedge held to an exogenous schedule $\bar{\tau}_t^y(\theta) < 1$.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, \mathbf{S}) \right] f_t(\theta|\theta_-) d\theta$$

subject to the constraints of Section 5 and, at every type, the wedge constraint

$$\Gamma(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta\, U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0. \tag{4}$$

Let $\eta(\theta)\, f_t(\theta|\theta_-)$ be the multiplier on (4), so that η is the resource value per agent of type θ of a marginal increase in the frozen net-of-tax rate. Two statistics organize the results, alongside the normalized shadow price $\psi\, U_{c,t}/(\theta f_t)$ of the local incentive constraint of Section 6, all evaluated at the constrained allocation with $y = \theta l$:

$$\Gamma_{c,t} \equiv (1-\bar{\tau}_t^y)\,\frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{y_t} \qquad \Delta_t \equiv (1-\bar{\tau}_t^y)\,\frac{\sigma_t}{c_t} + \frac{\epsilon_t - 2\gamma_{c,t}}{y_t}$$

$\Gamma_{c,t} = \partial\Gamma/\partial c$ is the response of the implied net-of-tax rate to the numéraire. It is positive when leisure is a normal good, so that delivering consumption depresses labor supply (Appendix D). Δ_t is the same response along a utility-compensated increase in consumption and labor. It is proportional to the second-order condition of the agent's hours choice at the frozen rate, so $\Delta_t > 0$ wherever the schedule decentralizes an interior labor choice, which I assume throughout (Appendix D.5).

Lemma 7.1 (frozen schedule price)**.**
At every history,

$$\eta_t\,\Delta_t = \frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} - A_t\,\frac{\psi U_{c,t}}{\theta f_t}$$

with $A_t = 1 + \epsilon_t - \gamma_{c,t}$. The multiplier is proportional to the gap between the frozen wedge and the virtual wedge $A_t \frac{\psi U_c}{\theta f}$, the labor wedge the third-best costate would decentralize if the labor margin were free: $\eta_t > 0$ where the schedule overtaxes labor, $\eta_t < 0$ where it undertaxes, and $\eta_t = 0$ where it crosses the virtual optimum.

Proof. See Appendix D. □

The lemma is the section's engine. In the unconstrained problem, combining the first-order conditions for the numéraire and labor delivers the labor wedge, $\tau_t^y/(1-\tau_t^y) = A_t \frac{\psi U_c}{\theta f}$. Here, the same combination cannot determine the wedge, which is frozen. It determines instead the shadow price of freezing it. The virtual wedge is not the Section 4 wedge: ψ solves the same costate equation with an integrating factor and a source term modified by the constraint (Appendix D.3), so $\psi U_c/(\theta f)$ carries the hazard shape and the persistence of the ABCD decomposition evaluated in the third best. Setting $\eta_t \equiv 0$, which requires the schedule to coincide with the virtual wedge at every node returns every formula of Section 5.

Proposition 7.1 (third-best savings wedge)**.**
Define $\chi_t^\eta \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t\,\Gamma_{c,t}$ and the adjusted marginal utility $\widehat{U}_{c,t}^\eta \equiv U_{c,t}/(1-\chi_t^\eta)$. The generalized inverse Euler equation holds in $\widehat{U}_c^\eta$,

$$\frac{1}{\widehat{U}_{c,t}^\eta} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}^\eta}\right]$$

Proof. See Appendix D.1. □

The delivery-cost logic of Section 4 extends by one term. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and, as before, earns the incentive credit $\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t}$ when the numéraire is complementary with labor. The new term prices the income effect. The same unit of consumption moves the implied net-of-tax rate by $\Gamma_{c,t}$ and, with the rate frozen, labor supply must fall to restore (4). The planner values that induced fall at η_t. Where the schedule overtaxes labor ($\eta_t > 0$), consumption delivery drags labor down at a node where labor is already too low, so its net cost rises and χ_t^η falls.

Proposition 7.2 (third-best carbon wedge)**.**
Let $\mathcal{I}_t$ be the income-effect gap of Section 6, positive exactly when the dirty good is normal, evaluated here at the endogenous price $1+\tau_t^q$. At every history,

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t})\left(\frac{\psi U_{c,t}}{\theta f_t} + \frac{\eta_t}{y_t}\right) - \eta_t\,(1-\bar{\tau}_t^y)\,\mathcal{I}_t$$

where the social cost of carbon keeps its discounted-damage form with a third-best fiscal factor,

$$\mathrm{SCC}_t = -\sum_{j=0}^{s}\frac{1}{R^j}\,\mathbb{E}_t\int\left[1 + (\xi_{t+j} - \gamma_{c,t+j})\left(\frac{\psi U_{c,t+j}}{\theta f_{t+j}} + \frac{\eta_{t+j}}{y_{t+j}}\right) + \eta_{t+j}\,(1-\bar{\tau}_{t+j}^y)\,\frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}}\right]\frac{U_{S,t+j}}{U_{c,t+j}}\,\Phi_j\,f_{t+j}\,d\theta$$

with $\xi_{c,t} \equiv U_{Sc,t}\,c_t/U_{S,t}$ the complementarity of the pollutant with the numéraire.

Proof. See Appendices D.2 and D.4. □

Corollary 7.1 (third best separation failure)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = -\,\eta_t\,(1-\bar{\tau}_t^y)\,\mathcal{I}_t$$

so when the dirty good is normal, $\mathrm{sign}\,(\tau_t^q - \mathrm{SCC}_t) = -\,\mathrm{sign}\,(\eta_t)$. The optimum prices carbon below the social cost of carbon where the frozen schedule overtaxes labor and above it where the schedule undertaxes labor. The carbon wedge equals the social cost of carbon at every history only if the schedule coincides with the virtual wedge, $\eta_t \equiv 0$.

Delivering the numéraire depresses labor supply through its income effect. Where the schedule overtaxes labor, deliveries that depress labor are penalized at the shadow price η_t such that the planner tilts the assigned bundle toward the good that carries less drag and the relative price of the dirty good falls below its corrective level. Where the schedule undertaxes, the tilt reverses. The carbon price is bent to patch the labor distortion the frozen schedule leaves behind.

The corollary sharpens the reading of Kaplow (2012). His argument frees Pigouvian pricing from any optimality requirement on the income tax. Whatever the schedule, an environmental reform can be packaged with a type-by-type income-tax adjustment that keeps the distribution whole, so moving to marginal-damage pricing is a Pareto improvement. The third best forbids this package. With the labor wedge frozen node by node, the compensating adjustment is unavailable and the deviation $-\eta_t(1-\bar{\tau}^y_t)\mathcal{I}_t$ prices the missing degree of freedom. The same rigidity reaches the social cost of carbon itself: its fiscal factor now carries the schedule's error and even the $U = V(h(c,q,S),l)$ benchmark, which made the second-best SCC purely Pigouvian in Section 5, no longer does so unless $\eta_t = 0$. In the single-cohort reading of Section 8 where marginal damage is a constant resource cost, $\mathrm{SCC}_t = e$ and the third-best deviations act on $\tau^q_t - e$ undiluted. The uniform-tax exercise of Section 6 also has a labor counterpart: freeze the schedule flat and choose its level.

Proposition 7.3 (the optimal flat labor tax)**.**
Let the schedule be flat, $\bar{\tau}^y_t(\theta) = \bar{\tau}^y_t$, with its level chosen optimally at every working date. The optimal level sets the population average of the multiplier to zero, $\mathbb{E}_0[\eta_t] = 0$; equivalently,

$$\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t} = \frac{\mathbb{E}_0\left[s_{yy,t}\, A_t\, \frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[s_{yy,t}]} \qquad s_{yy,t} \equiv \left.\frac{\partial y}{\partial\,(1-\bar{\tau}^y_t)}\right|_u = \frac{1}{(1-\bar{\tau}^y_t)\,\Delta_t}$$

with $s_{yy,t}$ the compensated response of earnings to the net-of-tax rate.

Proof. See Appendix D.5. □

The optimal flat rate is the compensated-earnings-response-weighted average of the virtual Mirrleesian wedges, mirroring Proposition 6.2 with s_{yy} in place of $|s_{qq}|$. A single rate averages the schedule the planner would set node by node, weighting each node by how elastically its earnings respond. This is the optimal linear income tax of the Ramsey tradition rendered in mechanism-design statistics with the caveat that the constraint fixes the marginal rate while the mechanism still assigns consumption such that the implied instrument is a flat rate with history-dependent intercepts rather than a single linear budget line.

7.2 Constrained savings wedge

Now freeze the intertemporal margin instead such that the savings wedge must follow an exogenous schedule $\bar{\tau}^s_t(\theta) < 1$.

$$U_{c,t}(\theta^t) = (1-\bar{\tau}^s_t(\theta_t))\,\beta R\ \mathbb{E}_t\big[U_{c,t+1}(\theta^{t+1})\big]\,. \tag{5}$$

Unlike the labor constraint, (5) is intertemporal and cannot enter the period Hamiltonian pointwise. It fits the recursive problem through one more state, a marginal-utility promise, carried the same way the promises $\hat{w}$ and $\hat{w}_2$ are. Appendix E.1 verifies that this one additional state suffices. The date-t problem passes each successor the required expectation $M(\theta) \equiv U_{c,t}(\theta)/((1-\bar{\tau}_t^s(\theta))\,\beta R)$ and inherits its own requirement $\hat{M}$,

$$V_t(\hat{w},\hat{w}_2,\hat{M},\theta_-,\mathbf{S}_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta)+q(\theta)-y(\theta)+\frac{1}{R}V_{t+1}(w(\theta),w_2(\theta),M(\theta),\theta,\mathbf{S})\right] f_t(\theta|\theta_-)\,d\theta$$

subject to the constraints of Section 5 and the inherited requirement

$$\hat{M} = \int_0^\infty U_c\left(c(\theta),q(\theta),\frac{y(\theta)}{\theta},S\right) f_t(\theta|\theta_-)\,d\theta, \tag{6}$$

Raising the marginal utility of the numéraire at a node now has a price of its own. It serves the inherited requirement, worth ϑ_t per util, and raises the requirement passed forward, costing $\vartheta_{t+1}(\theta^t)/((1-\bar{\tau}_t^s)\,\beta R^2)$. This subsection's multiplier is as follows.

$$\eta_t^s(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1-\bar{\tau}_t^s)\,\beta R^2}\right] U_{c,t}(\theta^t)$$

Every first-order condition gains one term, η_t^s times the log-derivative of $U_{c,t}$ in the corresponding direction (Appendix E.2). The counterpart of Lemma 7.1 is now a law of motion rather than a static condition.

Lemma 7.2 (the multiplier's law of motion)**.**
Define $\chi_t^s \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s\,\sigma_t/c_t$. The planner's intertemporal condition keeps the inverse-Euler form in $U_{c,t}/(1-\chi_t^s)$,

$$\frac{1-\chi_t^s}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right]$$

Combining it with (5) pins the multiplier's drift. When the date-$(t+1)$ draw is degenerate as in retirement,

$$\eta_t^s\,\frac{\sigma_t}{c_t} - (1-\bar{\tau}_t^s)\,\eta_{t+1}^s\,\frac{\sigma_{t+1}}{c_{t+1}} = \left(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}\right)(\bar{\tau}_t^s - \tau_t^{s,*})$$

where the virtual savings wedge is defined by $1-\tau_t^{s,*} \equiv (1-\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t})/(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}})$, with initial condition $\vartheta_0 = 0$ and terminal value $\eta_T^s = \vartheta_T\,U_{c,T}$.

Proof. See Appendix E.3. □

The frozen wedge does not break the planner's inverse-Euler logic. What changes relative to Section 7.1 is what the error pins down. A frozen labor wedge is a static restriction, so its gap sets the multiplier's level node by node. A frozen savings wedge restricts a ratio of marginal utilities across dates, so its gap injects drift into η_t^s and the level is tied down at the two ends of life, $\vartheta_0 = 0$ at birth and $\eta_T^s = \vartheta_T U_{c,T}$ at the close. Setting $\bar{\tau}_t^s = \tau_t^{s,*}$ at every date solves the law of motion with $\eta^s \equiv 0$ and returns every formula of Section 5.

Proposition 7.4 (wedges under a frozen savings wedge)**.**
At every history,

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^s \left[\frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{\left(1-\tau_t^y\right) y_t}\right] \qquad \frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = \left(\gamma_{c,t} - \gamma_{q,t}\right) \frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s \mathcal{I}_t$$

with $\mathcal{I}_t$ the income-effect gap of Section 6. The SCC keeps its discounted-damage form with the following damage weight at every future node.

$$1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} - \eta^s \frac{\sigma + \xi_c}{c}$$

Proof. See Appendices E.4 through E.7. □

Corollary 7.2 (Frozen savings wedge tilts the carbon price)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = \eta_t^s \mathcal{I}_t$$

The deviations cannot all share one sign. Either $\eta_t^s \geq 0$ at every node or $\eta_t^s \leq 0$ at every node forces $\vartheta \equiv 0$ and $\eta^s \equiv 0$ (Appendix E.3). If the schedule is not the virtual optimum and the dirty good is normal, the carbon price lies above the social cost of carbon at some nodes and below it at others.

A two-period example signs the tilt. Let preferences separate the goods from labor and from each other and let the schedule overtax savings. The frozen Euler equation front-loads consumption. The numéraire is the only good whose delivery moves its own marginal utility, so the planner overdelivers it early and underdelivers it late. The dirty good absorbs the mirror image (Appendix E.3), so the carbon price starts above the social cost of carbon and ends below it. A savings subsidy tilts the profile the other way and the labor wedge tilts alongside through the first formula of Proposition 7.4. However badly the savings schedule is misset, it cannot justify pricing carbon uniformly below or above marginal damages and only reshuffles the carbon price over the life cycle.

The contrast between the two exercises is instructive. A missing labor optimum moves the carbon price node by node in one direction per node because the labor constraint misallocates the numéraire across types within a date. A missing savings optimum on the other hand tilts the carbon price across dates in both directions because the savings constraint misallocates the numéraire across dates within a life. The same two statistics price the leak in both cases, $\mathcal{I}_t$ on the commodity margin and $(\sigma + \xi_c)/c$ inside the social cost of carbon. Retirement separates them most sharply: the labor-frozen distortions die with the labor margin while the savings margin operates through retirement where the virtual wedge is zero, so a nonzero frozen wedge keeps injecting error and retirees' carbon price deviates from the social cost of carbon. The Pigouvian-in-retirement result of Corollary 6.2 survives a frozen labor wedge and fails against a frozen savings wedge. The intertemporal margin also admits the optimal-flat-level exercise.

Proposition 7.5 (optimal flat savings tax)**.**
Let the schedule be flat, $\bar{\tau}^s_t(\theta) = \bar{\tau}^s_t$, with its level chosen optimally at every date. The optimal level zeroes the U_c-weighted average of the forward multipliers,

$$\mathbb{E}_0[\vartheta_{t+1}\, U_{c,t}] = 0 \qquad \Longleftrightarrow \qquad \mathbb{E}_0[\eta^s_t] = \mathbb{E}_0[\vartheta_t\, U_{c,t}]$$

At $t = 0$, where nothing is inherited, $\mathbb{E}_0[\eta^s_0] = 0$.

Proof. See Appendix E.8. $\square$

The condition prices only what the date-t rate controls. The rate moves the marginal-utility requirements passed forward, valued at ϑ_{t+1} and weighted by how hard the constraint binds through $U_{c,t}$. The multiplier averages to zero only at birth and, afterward, its target drifts with the inherited term $\vartheta_t U_{c,t}$. Even at the optimal flat path, the multiplier is generally nonzero node by node, so the carbon-price deviations of Corollary 7.2 do not vanish. The optimal level disciplines their average, not their profile. Flatness costs here what uniformity cost in Proposition 6.5. A single rate cannot track the cross-section and the residual dispersion lands on the free margins.

7.3 Constrained commodity wedge

The last margin to freeze is the one governments actually freeze, often to zero in the case of carbon pricing. Section 5 asked what the carbon price should be. This subsection takes the carbon price as given and asks how the labor and savings wedges should adjust. The planner solves the problem of Section 5 with the commodity wedge held to an exogenous schedule $\bar{\tau}^q_t(\theta)$.

$$\frac{U_q\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} = 1 + \bar{\tau}_t^q(\theta) \equiv \bar{p}_t(\theta), \tag{7}$$

with multiplier $\eta^q(\theta) f_t(\theta|\theta_-)$. The leading case is a flat legislated price, $\bar{\tau}_t^q(\theta) = \bar{\tau}_t^q$, but the derivation allows for any schedule. Like the labor constraint and unlike the savings one, (7) is static, so the multiplier is pinned by a static condition and the regularity it needs is the second-order condition $\mathcal{D}_t < 0$ of Section 6, guaranteed by Assumption 3.1.

Lemma 7.3 (frozen commodity wedge price)**.**
At every history,

$$\eta_t^q = \bar{p}_t \, |s_{qq,t}| \left[(\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} - \frac{\bar{\tau}_t^q - \mathrm{SCC}_t}{1 + \bar{\tau}_t^q} \right]$$

where SCC_t is the shadow social cost of carbon of Proposition 7.6 below. The bracket is the virtual deviation the planner would set, $(\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f}$ by Proposition 5.2, minus the frozen one. We have $\eta_t^q > 0$ where the schedule underprices the dirty good and $\eta_t^q < 0$ where it overprices it. The compensated substitution response converts the pricing error into the multiplier.

Proof. See Appendix C.4. □

The planner can no longer set the carbon price, but marginal damages do not leave the problem. The shadow SCC_t inside the Lemma is what routes the damages the frozen price fails to collect into the remaining wedges.

Proposition 7.6 (wedges under a frozen commodity wedge)**.**
At every history, the labor wedge and the savings adjustment are

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t^q \, \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1 - \tau_t^y)\, y_t} \right] \qquad \chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \, \bar{p}_t \, \mathcal{I}_t$$

with the generalized inverse Euler equation holding in $U_{c,t}/(1 - \chi_t^q)$. The social cost of carbon keeps its discounted-damage form with the damage weight

$$1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q \left[\bar{p} \, \frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q} \right]$$

at each future node, where $\xi_{q,t} \equiv U_{Sq,t}\, q_t / U_{S,t}$ is the complementarity of the pollutant with the dirty good.

Proof. See Appendices C.5, C.6, and C.11. $\square$

Corollary 7.3 (the income tax collects the carbon the carbon price misses)**.**
Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrections collapse to the marginal budget share $m_{q,t}$ of Section 6 times the uncollected carbon price:

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} + m_{q,t} \frac{\mathrm{SCC}_t - \bar{\tau}_t^q}{1+\bar{\tau}_t^q} \qquad \chi_t^q = \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - m_{q,t} \frac{\mathrm{SCC}_t - \bar{\tau}_t^q}{1+\bar{\tau}_t^q}$$

A carbon price frozen at the social cost of carbon is harmless: $\bar{\tau}_t^q = \mathrm{SCC}_t$ gives $\eta_t^q = 0$ and returns every wedge to its Section 5 form. A carbon price frozen below it raises the labor wedge above the virtual level; one frozen above it lowers the labor wedge.

Proof. See Appendix C.5. $\square$

The mechanism is embedded carbon. A marginal compensated dollar of earnings is spent, a fraction $m_{q,t}$ of it on the dirty good, and each of those units carries uncollected marginal damage $(\mathrm{SCC}_t - \bar{\tau}_t^q)/(1 + \bar{\tau}_t^q)$. The income tax adds that charge, taxing the carbon content of marginal spending that the carbon price misses. The savings side applies the same charge intertemporally. Delivering consumption at date $t+1$ rather than t carries the uncollected damage of the extra dirty consumption it drags along, so the planner taxes savings where underpricing worsens over the life cycle and subsidizes it where pricing improves. A legislated price that stays flat while the social cost of carbon grows is the worsening case. Both corrections persist into retirement since the pricing error is a within-period object that never expires. Like the frozen savings wedge and unlike the frozen labor wedge, a mispriced carbon tax distorts retirees' margins. This is the Mirrleesian counterpart of Hänsel et al. (2022), who find that a mandated uniform carbon tax reshapes the optimal income tax. Here, the reshaping has a closed form, and it is the marginal budget share times the pricing error.

8 Quantitative analysis

How large is the Mirrleesian correction to the carbon price, and how much welfare does it deliver over a flat carbon tax? I answer both by solving the model's optimal mechanism globally, calibrated to US data, rather than evaluating the wedge formula at borrowed sufficient statistics. The solver extends the dual-costate recursion of Farhi and Werning (2013) and Golosov et al. (2016) to two consumption goods, a clean numéraire and a polluting energy good, and prices the externality as a constant marginal damage e entering the planner's resource constraint, so the social producer price of energy is $1+e$. This is the single-cohort reading of the social cost of carbon in Section 5: for a cohort too small to move the pollution stock, the discounted damage weights Φ_j collapse to one constant per-unit cost, no pollution-stock state is carried, and the Pigouvian benchmark is $\tau^q = \text{SCC} = e$. The pollution-stock dynamics are a separate exercise. What the solved mechanism settles is the object a formula evaluation cannot: the *level* and the *incidence* of the optimal commodity wedge, and the welfare cost of implementing it with an anonymous instrument.

8.1 Model and calibration

The life-cycle block is Farhi and Werning (2013)'s, replicated and validated before the two-good extension. Agents live sixty years, work for forty and retire for twenty; log-skills follow a geometric random walk with innovation variance $\hat{\sigma}^2 = 0.0095$; $\beta = 1/R = 0.95$, so $\beta R = 1$; skills are private information and the incentive constraints are handled by the first-order approach. The planner is utilitarian. Following Farhi and Werning (2013), newborns are ex ante identical, so the period-one cross-section is the first innovation of the random walk: the initial skill dispersion is already the $\hat{\sigma}^2 = 0.0095$ above, there is no separate initial-skill variance to set, and the common initial promised value is pinned by the economy's resource constraint. Each period the agent consumes a clean numéraire and an energy good (motor fuel, electricity, gas, fuel oil), both with producer price one. Period utility is

$$U(c, q, l) = \frac{1}{\rho} \log\left[(1 - b_q)\, c^{\rho} + b_q\, e^{-\varkappa l^{\alpha}}\, (q - \bar{q})^{\rho}\right] - \frac{l^{\alpha}}{\alpha} \qquad \rho \equiv 1 - \frac{1}{\varsigma}$$

a CES aggregate over the clean good and energy net of subsistence $\bar{q}$, inside the log shell that preserves Farhi and Werning (2013)'s balanced-growth homogeneity, with isoelastic labor disutility of curvature $\alpha = 3$ (Frisch elasticity one half). ς is the elasticity of substitution between the goods, b_q the energy weight, and $\bar{q} = 0$ in the homothetic benchmark; $\bar{q} > 0$ is the Stone-Geary necessity variant. The twist $e^{-\varkappa l^{\alpha}}$ makes energy a relative leisure complement: its utility weight falls with labor effort, so $U_{ql} \neq 0$ with the sign, in the notation of Section 4, that makes the complementarity gap $\gamma_c - \gamma_q$ positive. This is the empirically relevant case (West and Williams (2007); Christiansen (1984)), and the one that pushes the optimal carbon tax above Pigouvian.

Two solvers deliver the results. The *second best* assigns full consumption bundles, so the commodity wedge is unrestricted; this solver requires homothetic preferences. *Linear commodity taxation* layers an anonymous rate τ^q (flat or age-dependent) on the optimal nonlinear income and savings mechanism, and reduces exactly to a one-good problem with reshaped labor disutility, valid with subsistence and hence usable for the necessity calibration. The two solvers agree where they must: at $\gamma_c = \gamma_q$ the second best sets $\tau^q = e$ for every history and every age, the dynamic Atkinson-Stiglitz result, and it coincides with a flat Pigouvian tax to solver precision, returning the one-good initial condition $\hat{\lambda}_1 = 0.8797$. The incentive compatibility of the first-order-approach solution is verified ex post on the simulated benchmark.

The two-good calibration is disciplined by four US moments (Table 1). The energy expenditure share is 7% (Consumer Expenditure Survey, 2023). The income elasticity of energy is 0.45, below one because energy is a necessity, from residential-energy demand studies; the homothetic benchmark raises it to one, and the Stone-Geary variant keeps it at 0.45. The own-price elasticity is 0.4, from meta-analyses of energy demand; at a 7% share it pins the substitution elasticity $\varsigma \approx 0.4$. Both are long-run elasticities, the horizon a forty-year model asks for; short-run energy elasticities are roughly half as large, and the welfare numbers scale with the long-run value. The share moment pins the small CES energy weight b_q, and under Stone-Geary the income-elasticity moment implies a subsistence level $\bar{q} \approx 0.04$, about 57% of mean energy spending, so more than half of energy consumption is price-inelastic at the mean. The leisure complementarity $\varkappa$ is set so that, with the externality off, the second-best energy wedge at mid-career is about 10% ad valorem: West and Williams (2007)'s zero-externality gasoline tax, tempered for the household-energy composite, whose labor cross-effects are weaker than motor fuel's. The marginal damage is $e = 0.494$: the EPA's central social cost of carbon of \$190 per ton (**epaSCGHG2023**) times the carbon intensity of household energy spending, 0.0026 tons per producer dollar. Because $p_2 = 1$, the Pigouvian wedge $\tau^q = e$ is a 49% ad valorem tax on energy.

Table 1: Calibration

parameter	value	moment or source
life-cycle block (Farhi and Werning 2013)		
discount factor $\beta = 1/R$	0.95	FW13 ($\beta R = 1$)
labor-disutility curvature α	3	Frisch elasticity 0.5
log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages, via FW13
working / retirement periods	40/20	FW13
two-good block		
energy expenditure share	0.07	Consumer Expenditure Survey 2023
income elasticity of energy	0.45	residential-energy demand (necessity)
own-price elasticity of energy	0.4	energy-demand meta-analyses
leisure complementarity $\varkappa$	0.25	mid-career zero-externality wedge $\approx$ 10% (West and Williams 2007)
marginal damage e	0.494	\$190/tCO_2 (**epaSCGHG2023**) $\times$ 0.0026 tCO_2/\$

8.2 The optimal carbon wedge over the life cycle

The optimal wedge is the social cost of carbon plus a small, rising premium (Table 2, Figure 1). At labor-market entry the labor wedge is near zero, there is no incentive motive yet, and the carbon wedge sits at the Pigouvian level $e = 0.494$. It then rises concavely to about 0.63 near retirement. The non-Pigouvian premium over the Pigouvian gross price tops out around 9%. So the optimal carbon tax exceeds the social cost of carbon, in the direction a leisure complement predicts, but by less than a tenth over most of the life cycle.

The premium is a scaled labor wedge. Its first-order value is $\varkappa\, l^{\alpha}\, \tau^{y}/(1-\tau^{y})$, and this identity tracks the solved premium within about 10% at every age (last two rows of Table 2). The non-Pigouvian part of the carbon wedge therefore inherits the labor wedge's level and its life-cycle rise, which is the proportionality result of Section 4 made quantitative. The wedge is also genuinely history-dependent: its cross-sectional standard deviation fans out from 0.1% early in life to 5.5% near retirement, mirroring the spreading of the labor wedge as skill histories diverge.

The income-side wedges are essentially undisturbed by the second good. The labor wedge runs from 0.02 to 0.36 over the working life, against 0.02 to 0.37 in the one-good economy, and the savings wedge falls from 0.56% to 0.02%, the Farhi and Werning (2013) pattern intact. The commodity margin adds a corrective and screening instrument without reshaping the labor and savings margins, which is the separation showing up in the solved policies. Under the wedge the mean energy share falls from the calibrated 7% to about 6% at producer prices, the households priced off energy at the calibrated elasticity.

Table 2: Cross-sectional mean wedges over the working life (benchmark, second best)

working year t	1	10	20	30	40
labor wedge τ^{y}	0.021	0.167	0.272	0.336	0.363
carbon wedge τ^{q}	0.504	0.569	0.606	0.624	0.632
s.d. across histories	0.001	0.005	0.014	0.029	0.055
non-Pigouvian premium	0.007	0.050	0.075	0.087	0.092
identity $\varkappa l^{\alpha}\tau^{y}/(1-\tau^{y})$	0.006	0.047	0.069	0.079	0.083

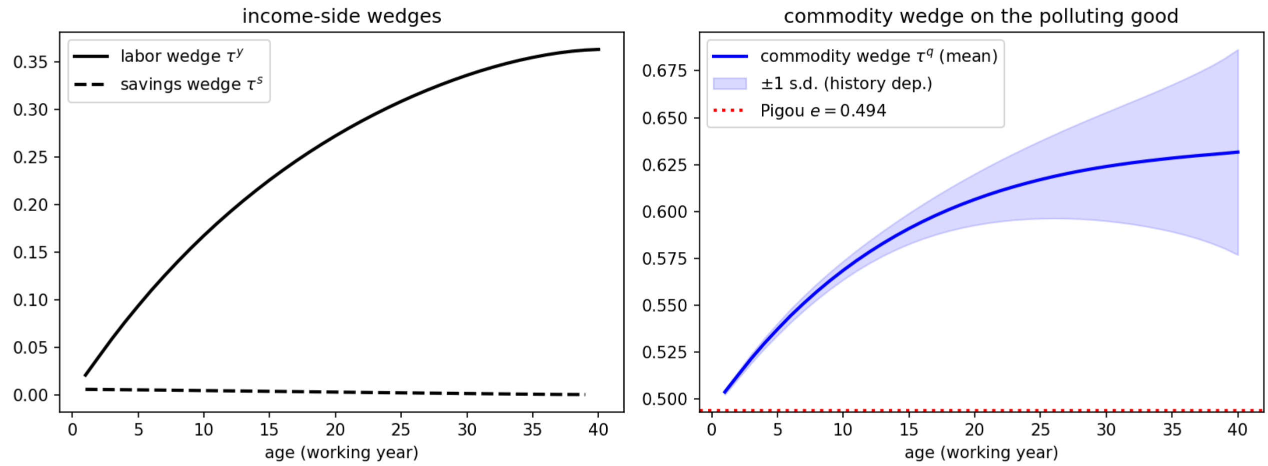

Figure 1: Income-side wedges (left) and the commodity wedge on energy (right), benchmark calibration. The band is ± 1 cross-sectional standard deviation across histories; the dotted line is the Pigouvian level $\tau^q = e = 0.494$.

8.3 The optimal wedge is progressive in skill

The optimal carbon wedge is progressive in skill (Figure 2). Binned by current productivity, the wedge is indistinguishable across quintiles early in life, when every history looks alike, and it fans out from mid-career: by the last working year it reaches about 70% for the top skill quintile against 57% for the bottom, a 13-point spread. High current productivity signals a history whose incentive constraints are expensive, and the leisure-complement margin is the planner's lever on exactly those histories, so the screening value of the commodity instrument concentrates on high-skill agents. Figure 2 is a schedule of marginal wedges by skill, not a map of the burden each household bears. That burden is set jointly by the nonlinear income tax and transfers alongside the wedge, so the schedule's progressivity speaks to the corrective instrument itself, not to the full incidence of carbon pricing.

Two readings follow. The optimal carbon wedge is not regressive: the planner taxes the energy of high-skill households more, not less, so the redistributive worry about carbon pricing is misplaced at the level of the optimal type-contingent wedge. The regressivity of real carbon taxes comes from elsewhere, from the flat-rate restriction, since an anonymous point-of-sale tax cannot condition on skill, and, in the necessity calibration, from energy's falling budget share. That is also why the flat rungs of the welfare ladder below leave money on the table.

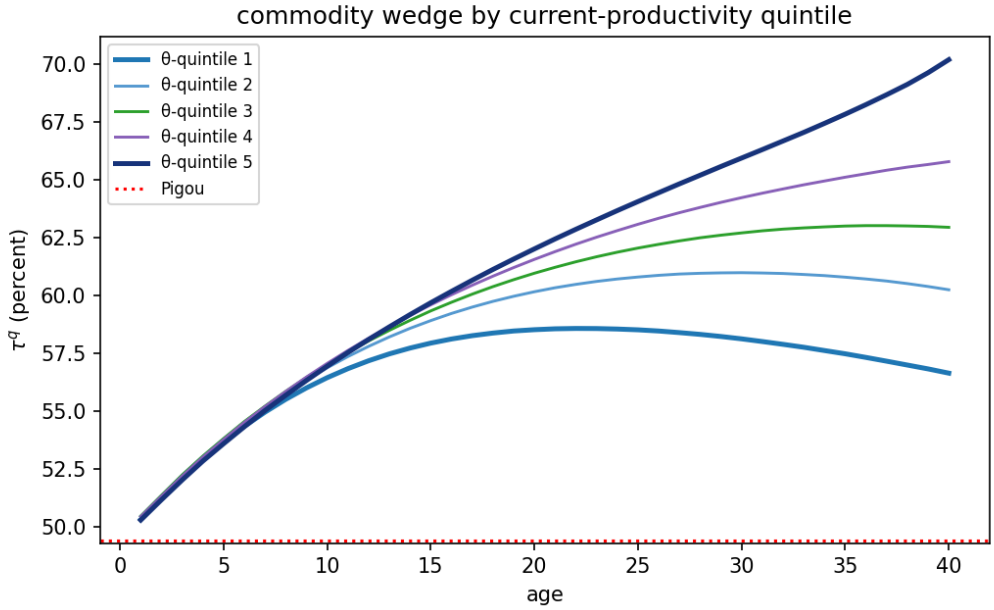

Figure 2: The optimal commodity wedge τ^q (%) by current-skill quintile and age, benchmark calibration.

8.4 The welfare value of carbon pricing and the cost of simpler rules

Table 3 is the welfare accounting: consumption-equivalent gains over having no carbon price, under the optimal nonlinear income and savings mechanism throughout, computed with common random numbers so that cross-regime differences net out simulation noise. Every regime pays for its own emissions in the planner's budget, so no rung is flattered by unpriced damages; the numerical resolution is about 0.002 percentage points, and differences below it are noise. Six findings, in order of size.

The value of a carbon price is overwhelmingly Pigouvian. At the EPA social cost of carbon, having any carbon price is worth 0.31% of consumption, and a flat rate at the social cost of carbon alone captures 98% of the full second-best value while cutting emissions by 13.5%. For scale, the entire insurance value of Farhi and Werning (2013)'s optimal income-tax apparatus over laissez-faire is 1.56% of consumption: the carbon instrument is worth about a fifth of that at this social cost of carbon.

The optimal flat rate exceeds the social cost of carbon by about 14%, $\tau^{q*} = 0.56$ against $e = 0.494$, the flat reflection of the Corlett-Hague premium. But the welfare gain from that uplift over pricing at the social cost of carbon is 0.003%: the objective is very flat in the rate, so "price at the social cost of carbon" is a near-optimal rule of thumb even with nonseparable preferences.

Sophistication buys almost nothing beyond the level. Making the rate age-dependent adds about 0.001 percentage points over the optimal flat rate, and full history dependence about 0.001 more, both at the numerical resolution. The premium varies over life and across histories by a few points of a 6% budget share with a substitution elasticity of 0.4, so the deadweight loss from mistracking it is second-order squared.

The pure Corlett-Hague motive is tiny. With the externality off, the entire commodity-wedge apparatus of the second best is worth 0.0036% of consumption, and its optimal flat rate is 5%, materially positive as in West and Williams (2007) but negligible in welfare. The flat rate sits below the 10% mid-career calibration target because it averages a premium that starts the working life near zero. Non-separability matters for the *level* of the optimal carbon tax, the 14% uplift, not as a stand-alone reason to tax energy.

Necessity halves the value and the abatement. Under the Stone-Geary calibration, in which energy's budget share falls with income, the value of the flat carbon tax falls to 0.15% and its abatement to 6%, because the subsistence part of energy does not respond to price. The optimal flat rate is essentially unchanged. This is the calibration in which a flat carbon tax is regressive, yet the mechanism absorbs the incidence through the income side, which is why the flat instrument loses so little welfare despite its incidence.

The separation placebo. With the leisure complementarity switched off ($\gamma_c = \gamma_q$), the second best and the flat Pigouvian tax coincide to five decimal places in welfare and in emissions: dynamic Atkinson-Stiglitz, and the exact knife-edge of Proposition 5.1, reproduced through two structurally different solvers.

Table 3: Welfare value of carbon pricing under progressively simpler rules (consumption-equivalent gain over $\tau^q = 0$, % of consumption)

	benchmark ($e = 0.494$)	Stone-Geary (necessity)	$\gamma_c = \gamma_q$ (placebo)	no externality ($e = 0$)
second best (nonlinear τ^q)	0.315	—	0.252	0.0036
age-dependent linear $\tau^q(t)$	0.314	—	—	0.0038
flat τ^{q*} (optimized)	0.313	0.153	—	0.0026
optimal flat rate τ^{q*}	0.56	0.56	—	0.051
flat Pigouvian $\tau^q = e$	0.309	0.151	0.252	—
emissions vs $\tau^q = 0$: second best	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−6.3%	−13.5%	−1.8%

Note: welfare entries are in % of consumption; the optimal-flat-rate row is an ad-valorem rate, and the final two rows report emissions changes relative to $\tau^q = 0$. Differences below the 0.002-point numerical resolution, including the apparent ranking of the age-dependent rate above the second best in the no-externality column, are simulation noise. Dashes mark three cases. *Not computed*: the second best requires homothetic preferences, so it and the age-dependent rung are blank under the non-homothetic Stone-Geary calibration. *Degenerate*: under the $\gamma_c = \gamma_q$ placebo the second best already coincides with the flat Pigouvian tax, so the intermediate rungs add nothing. *Not applicable*: with no externality ($e = 0$) there is no Pigouvian benchmark.

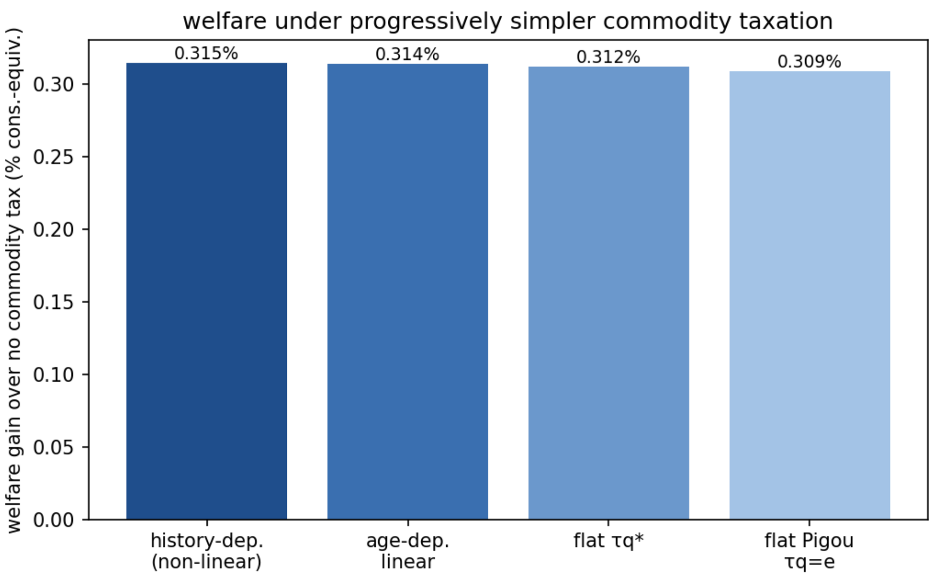

Figure 3: Consumption-equivalent welfare gain over no carbon price, benchmark calibration, under progressively simpler carbon-pricing rules.

8.5 Reading and limitations

The practical hierarchy for carbon policy in a Mirrleesian world is clean. Get the carbon price to the social cost of carbon, worth 0.31% of consumption; nudge it up by the premium for leisure complementarity, worth a further 0.003%; and spend the sophistication budget on the income tax, where Farhi and Werning (2013)'s own accounting shows the returns to age- and history-dependence are two orders of magnitude larger. The redistributive value embedded in the carbon tax is small, which is the separation principle in welfare terms: the income tax and the climate dividend, not the carbon price, should carry the distribution. This agrees with Douenne et al. (2026)'s near-Pigouvian conclusion, reached here from the Mirrleesian rather than the Ramsey side, and it answers the distributional concern that opens the paper. Regressive carbon pricing and the backlash it provokes are best met by adjusting the income tax, not by tilting, means-testing, or weakening the carbon price.

Four limitations bound the numbers. The externality is a constant marginal damage, so the pollution-stock dynamics of Section 5, which move the *level* of the tax rather than the wedge structure measured here, are a separate exercise. The second best requires homothetic preferences, so the necessity calibration is read off the exact linear-tax reduction rather than the full nonlinear mechanism. The log consumption index is required by the balanced-growth homogeneity the solver relies on. And the energy good carries a single carbon intensity, so within-energy heterogeneity is outside the model. None of these four bears on the paper's qualitative results, though the results rest on different footings. The premium's proportionality to the labor wedge is structural, the identity $\varkappa l^{\alpha} \tau^{y}/(1-\tau^{y})$, as is the near-sufficiency of a flat Pigouvian price, which follows from the objective's flatness in the rate. The premium's sign, size, and progressivity in skill are calibration-contingent: they rest on energy being a leisure complement ($\varkappa > 0$), the empirical input from West and Williams (2007). The premium scales roughly linearly in $\varkappa$, so the gasoline-only upper bound $\varkappa = 0.45$ about doubles the mid-career premium to 17–20% and $\varkappa = 0$ collapses it to the Atkinson-Stiglitz benchmark; the welfare stakes stay small across this range.

9 Conclusion

A pollution externality placed inside a dynamic Mirrlees economy leaves the corrective and redistributive tasks separate. When preferences are weakly separable, the carbon wedge is exactly the social cost of carbon and the labor and savings wedges are unchanged, so the below-Pigouvian result that the Ramsey tradition blames on fiscal distortion does not survive an optimal nonlinear income tax. Only nonseparability bends the carbon wedge away from the social cost of carbon, and it does so in the direction the good's complementarity with labor dictates.

Real carbon taxes are anonymous linear prices, not the history-dependent wedge the theory prescribes. Under that restriction the two-good problem collapses to a one-good problem in expenditure; the optimal uniform rate is a substitution-weighted average of the wedges it replaces, and its welfare cost is their dispersion, which vanishes under weak separability. With the externality the uniform price splits additively into the social cost of carbon and a screening term, so the restriction binds only the redistributive part and never the corrective one.

Carbon pricing is resisted as a burden on the poor, and governments answer by capping it, tilting it, or outright repealing it. This paper suggests a different response: once the income tax is set optimally, the carbon price has no real redistributive role and should simply be set at the social cost of carbon.

This is a claim about optimal design, not about political durability. Canada paired its consumer carbon tax with a per-household dividend and repealed it anyway, so whether households perceive the compensation and trust it to last can matter as much as whether it exists. The model takes the income tax and the dividend as credible and delivered; making them so is a question of salience and commitment that it does not address.

This reading carries one caveat. The separation is a property of the jointly optimal tax system, and presumes the labor and savings wedges are set optimally alongside the carbon price. When the labor wedge is held above or below its optimum, part of the redistribution it fails to carry falls back on the carbon price: Section 7 signs the deviation, below the social cost of carbon where the frozen schedule overtaxes labor and above it where the schedule undertaxes. A frozen savings wedge works differently: its error can only tilt the carbon-price profile, above the social cost of carbon at some dates and below it at others. And the flow runs both ways: a carbon price legislated below the social cost of carbon raises the optimal labor wedge by the marginal budget share of the dirty good times the uncollected carbon price, the income tax collecting the carbon the carbon price misses. The optimal linear instruments of the Ramsey tradition emerge inside the same family, as the levels at which the constraint multipliers average to zero.

Three extensions stand out. First, the paper characterizes wedges, not their implementation; how to replicate the optimal commodity wedge with real instruments is open. Second, the model follows a single generation, but climate change is intergenerational. The natural next step is an overlapping-generations extension, in which the single-generation model becomes the component problem of an infinite-horizon planner. The fixed point between the carbon-price path and the per-cohort mechanism-design problems would then endogenize the savings-wedge paths that Section 8 takes from calibrated optima. Third, outside the environmental context, the optimal commodity wedge is a time-varying object, which raises the question of age-dependent commodity taxation. Age-dependent commodity taxes may sound fanciful, yet age-based pricing of some goods and services, and discounts for low-income consumers, already implements a time-varying, income-based commodity wedge.

References

Albanesi, Stefania and Christopher Sleet (2006). "Dynamic Optimal Taxation with Private Information". In: *The Review of Economic Studies* 73.1, pp. 1–30. ISSN: 00346527, 1467937X.

Atkinson, A.B. and J.E. Stiglitz (July 1976). "The Design of Tax Structure: Direct versus Indirect Taxation". In: *Journal of Public Economics* 6.1, pp. 55–75. ISSN: 0047-2727. DOI: 10.1016/0047-2727(76)90041-4.

Barrage, Lint (2019). "Optimal Dynamic Carbon Taxes in a Climate–Economy Model with Distortionary Fiscal Policy". In: *The Review of Economic Studies* 87.1, pp. 1–39. ISSN: 0034-6527. DOI: 10.1093/restud/rdz055.

Bastani, Spencer, Soren Blomquist, and Luca Micheletto (Nov. 2013). "The Welfare Gains of Age-related Optimal Income Taxation". In: *International Economic Review* 54.4. Publisher: John Wiley & Sons, Ltd, pp. 1219–1249. ISSN: 0020-6598. DOI: 10.1111/iere.12034. URL: https://doi.org/10.1111/iere.12034 (visited on 07/22/2026).

Bovenberg, A. Lans and Ruud A. de Mooij (1994). "Environmental Levies and Distortionary Taxation". In: *The American Economic Review* 84.4. Publisher: American Economic Association JSTOR: 2118046, pp. 1085–1089. ISSN: 00028282. (Visited on 03/14/2023).

Bovenberg, A. Lans and Sjak A. Smulders (1996). "Transitional Impacts of Environmental Policy in an Endogenous Growth Model". In: *International Economic Review* 37.4. Publisher: [Economics Department of the University of Pennsylvania, Wiley, Institute of Social and Economic Research, Osaka University], pp. 861–893. ISSN: 00206598, 14682354. DOI: 10.2307/2527315. URL: http://www.jstor.org/stable/2527315 (visited on 07/21/2026).

Chamley, Christophe (1986). "Optimal Taxation of Capital Income in General Equilibrium with Infinite Lives". In: *Econometrica* 54.3. Publisher: [Wiley, Econometric Society], pp. 607–622. ISSN: 00129682, 14680262. DOI: 10.2307/1911310. URL: http://www.jstor.org/stable/1911310 (visited on 07/21/2026).

Chancel, Lucas (Nov. 2022). "Global carbon inequality over 1990–2019". In: *Nature Sustainability* 5.11, pp. 931–938. ISSN: 2398-9629. DOI: 10.1038/s41893-022-00955-z. URL: https://doi.org/10.1038/s41893-022-00955-z.

Christiansen, Vidar (July 1984). "Which commodity taxes should supplement the income tax?" In: *Journal of Public Economics* 24.2, pp. 195–220. ISSN: 0047-2727. DOI: 10.1016/0047-2727(84)90024-0. URL: https://www.sciencedirect.com/science/article/pii/0047272784900240.

Corlett, W. J. and D. C. Hague (Jan. 1953). "Complementarity and the Excess Burden of Taxation". In: *The Review of Economic Studies* 21.1, pp. 21–30. ISSN: 0034-6527. DOI: 10.2307/2296257. URL: https://doi.org/10.2307/2296257 (visited on 07/23/2026).

Cremer, Helmuth, Firouz Gahvari, and Norbert Ladoux (Dec. 1998). "Externalities and Optimal Taxation". In: *Journal of Public Economics* 70.3, pp. 343–364. ISSN: 0047-2727. DOI: 10.1016/S0047-2727(98)00039-5.

Cremer, Helmuth, Firouz Gahvari, and Norbert Ladoux (2001). "Second-Best Pollution Taxes and the Structure of Preferences". In: *Southern Economic Journal* 68.2, pp. 258–280. ISSN: 00384038. DOI: 10.2307/1061594.

— (Jan. 2010). "Environmental tax design with endogenous earning abilities (with applications to France)". In: *Journal of Environmental Economics and Management* 59.1, pp. 82–93. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2009.04.009. URL: https://www.sciencedirect.com/science/article/pii/S0095069609000850.

Dennig, Francis et al. (Dec. 2015). "Inequality, Climate Impacts on the Future Poor, and Carbon Prices". In: *Proceedings of the National Academy of Sciences* 112.52. Publisher: Proceedings of the National Academy of Sciences, pp. 15827–15832. DOI: 10.1073/pnas.1513967112. (Visited on 08/31/2023).

Diamond, P.A. (Nov. 1975). "A many-person Ramsey tax rule". In: *Journal of Public Economics* 4.4, pp. 335–342. ISSN: 0047-2727. DOI: 10.1016/0047-2727(75)90009-2. URL: https://www.sciencedirect.com/science/article/pii/0047272775900092.

Dietsch, Peter (Dec. 2024). "A fairer and more effective carbon tax". In: *Nature Sustainability* 7.12, pp. 1584–1591. ISSN: 2398-9629. DOI: 10.1038/s41893-024-01429-0. URL: https://doi.org/10.1038/s41893-024-01429-0.

Douenne, Thomas, Albert Jan Hummel, and Marcelo Pedroni (Jan. 2026). "Optimal Fiscal Policy in a Climate-Economy Model with Heterogeneous Households*". In: *The Economic Journal*, ueag006. ISSN: 0013-0133. DOI: 10.1093/ej/ueag006. URL: https://doi.org/10.1093/ej/ueag006 (visited on 04/19/2026).

Douenne, Thomas et al. (2025). *Optimal Climate Policy with Incomplete Markets*. Tech. rep. Issue: tecipa-807. University of Toronto, Department of Economics. URL: https://ideas.repec.org/p/tor/tecipa/tecipa-807.html.

Environment and Climate Change Canada (2016). *Technical update to Environment and Climate Change Canada's social cost of greenhouse gas estimates*. Tech. rep. Environment and Climate Change Canada.

Erosa, Andrés and Martin Gervais (Aug. 2002). "Optimal Taxation in Life-Cycle Economies". In: *Journal of Economic Theory* 105.2, pp. 338–369. ISSN: 0022-0531. DOI: 10.1006/jeth.2001.2877. URL: https://www.sciencedirect.com/science/article/pii/S0022053101928777.

Farhi, Emmanuel and Iván Werning (2013). "Insurance and Taxation over the Life Cycle". In: *The Review of Economic Studies* 80.2 (283), pp. 596–635. ISSN: 00346527, 1467937X.

Fernandes, Ana and Christopher Phelan (Apr. 2000). "A Recursive Formulation for Repeated Agency with History Dependence". In: *Journal of Economic Theory* 91.2, pp. 223–247. ISSN: 0022-0531. DOI: 10.1006/jeth.1999.2619.

Golosov, Mikhail, Narayana Kocherlakota, and Aleh Tsyvinski (2003). "Optimal Indirect and Capital Taxation". In: *Review of Economic Studies* 70.3, pp. 569–587. ISSN: 0034-6527. DOI: 10.1111/1467-937x.00256.

Golosov, Mikhail, Maxim Troshkin, and Aleh Tsyvinski (2016). "Redistribution and Social Insurance". In: *American Economic Review* 106.2, pp. 359–86. DOI: 10.1257/aer.20111550.

Golosov, Mikhail et al. (Jan. 2014). "Optimal Taxes on Fossil Fuel in General Equilibrium". In: *Econometrica* 82.1, pp. 41–88. ISSN: 0012-9682. DOI: 10.3982/ECTA10217.

Goulder, Lawrence H. et al. (July 2019). "Impacts of a Carbon Tax across US Household Income Groups: What Are the Equity-Efficiency Trade-Offs?" In: *Journal of Public Economics* 175, pp. 44–64. ISSN: 0047-2727. DOI: 10.1016/j.jpubeco.2019.04.002.

Hänsel, Martin C. et al. (Oct. 2022). "Optimal carbon taxation and horizontal equity: A welfare-theoretic approach with application to German household data". In: *Journal of Environmental Economics and Management* 116, p. 102730. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2022.102730. URL: https://www.sciencedirect.com/science/article/pii/S0095069622000833.

Hassett, Kevin A., Aparna Mathur, and Gilbert E. Metcalf (Apr. 2009). "The Incidence of a U.S. Carbon Tax: A Lifetime and Regional Analysis". In: *The Energy Journal* 30.2. Publisher: SAGE Publications, pp. 155–178. ISSN: 0195-6574. DOI: 10.5547/ISSN0195-6574-EJ-Vol30-No2-8. URL: https://doi.org/10.5547/ISSN0195-6574-EJ-Vol30-No2-8 (visited on 07/22/2026).

Hellwig, Christian (June 2021). "Static and Dynamic Mirrleesian Taxation with Non-separable Preferences: A Unified Approach". Working paper. Toulouse School of Economics.

Interagency Working Group on Social Cost of Greenhouse Gases (Feb. 2021). *Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide Interim Estimates under Executive Order 13990*. Tech. rep. Interagency Working Group on Social Cost of Greenhouse Gases.

Jacobs, Bas and Robin Boadway (Sept. 2014). "Optimal Linear Commodity Taxation under Optimal Non-Linear Income Taxation". In: *Journal of Public Economics* 117, pp. 201–210. ISSN: 0047-2727. DOI: 10.1016/j.jpubeco.2014.04.012.

Jacobs, Bas and Ruud A. de Mooij (May 2015). "Pigou Meets Mirrlees: On the Irrelevance of Tax Distortions for the Second-Best Pigouvian Tax". In: *Journal of Environmental Economics and Management* 71, pp. 90–108. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2015.01.003.

Jacobs, Bas and Frederick van der Ploeg (May 2019). "Redistribution and pollution taxes with non-linear Engel curves". In: *Journal of Environmental Economics and Management* 95, pp. 198–226. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2019.01.008. URL: https://www.sciencedirect.com/science/article/pii/S0095069617308549.

Judd, Kenneth L. (Oct. 1985). "Redistributive taxation in a simple perfect foresight model". In: *Journal of Public Economics* 28.1, pp. 59–83. ISSN: 0047-2727. DOI: 10.1016/0047-2727(85)90020-9. URL: https://www.sciencedirect.com/science/article/pii/0047272785900209.

Kapička, Marek (2013). "Efficient Allocations in Dynamic Private Information Economies with Persistent Shocks: A First-Order Approach". In: *The Review of Economic Studies* 80.3, pp. 1027–1054. ISSN: 0034-6527. DOI: 10.1093/restud/rds045.

Kaplow, Louis (May 2012). "Optimal Control of Externalities in the Presence of Income Taxation". In: *International Economic Review* 53.2. Publisher: John Wiley & Sons, Ltd, pp. 487–509. ISSN: 0020-6598. DOI: 10.1111/j.1468-2354.2012.00689.x. URL: https://doi.org/10.1111/j.1468-2354.2012.00689.x (visited on 04/20/2026).

Klenert, David et al. (Nov. 2018). "Environmental Taxation, Inequality and Engel's Law: The Double Dividend of Redistribution". In: *Environmental and Resource Economics* 71.3, pp. 605–624. ISSN: 1573-1502. DOI: 10.1007/s10640-016-0070-y.

Kocherlakota, Narayana R. (2005). "Zero Expected Wealth Taxes: A Mirrlees Approach to Dynamic Optimal Taxation". In: *Econometrica* 73.5, pp. 1587–1621. ISSN: 00129682, 14680262.

Kornek, Ulrike et al. (May 2021). "The Social Cost of Carbon and Inequality: When Local Redistribution Shapes Global Carbon Prices". In: *Journal of Environmental Economics and Management* 107, p. 102450. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2021.102450.

Mirrlees, J. A. (1971). "An Exploration in the Theory of Optimum Income Taxation". In: *The Review of Economic Studies* 38.2. Publisher: [Oxford University Press, Review of Economic Studies, Ltd.] JSTOR: 2296779, pp. 175–208. ISSN: 00346527, 1467937X. DOI: 10.2307/2296779. (Visited on 07/22/2023).

Pavan, Alessandro, Ilya Segal, and Juuso Toikka (Mar. 2014). "Dynamic Mechanism Design: A Myersonian Approach". In: *Econometrica* 82.2. Publisher: John Wiley & Sons, Ltd, pp. 601–653. ISSN: 0012-9682. DOI: 10.3982/ECTA10269. (Visited on 08/13/2023).

Ploeg, Frederick van der, Rezai Armon, and Reaños Miguel Tovar (July 2025). "Third-Best Carbon Taxation: Trading Off Emission Cuts, Equity, and Efficiency". In: *Journal of the Association of Environmental and Resource Economists* 12.4. Publisher: The University of Chicago Press, pp. 1023–1058. DOI: 10.1086/732802. URL: https://doi.org/10.1086/732802 (visited on 07/21/2026).

Poterba, James M. (1989). "Lifetime Incidence and the Distributional Burden of Excise Taxes". In: *The American Economic Review* 79.2. Publisher: American Economic Association, pp. 325–330. ISSN: 00028282. URL: http://www.jstor.org/stable/1827779 (visited on 07/22/2026).

Saez, Emmanuel (Jan. 2001). "Using Elasticities to Derive Optimal Income Tax Rates". In: *The Review of Economic Studies* 68.1, pp. 205–229. ISSN: 0034-6527. DOI: 10.1111/1467-937X.00166. (Visited on 07/20/2023).

— (Feb. 2002). "The desirability of commodity taxation under non-linear income taxation and heterogeneous tastes". In: *Journal of Public Economics* 83.2, pp. 217–230. ISSN: 0047-2727. DOI: 10.1016/S0047-2727(00)00159-6. URL: https://www.sciencedirect.com/science/article/pii/S0047272700001596.

Sandmo, Agnar (1975). “Optimal Taxation in the Presence of Externalities”. In: *The Swedish Journal of Economics* 77.1. Publisher: [Scandinavian Journal of Economics, Wiley] JSTOR: 3439329, pp. 86–98. ISSN: 00397318. DOI: 10.2307/3439329. (Visited on 02/18/2023).

Weinzierl, Matthew (2011). “The Surprising Power of Age-Dependent Taxes”. In: *The Review of Economic Studies* 78.4, pp. 1490–1518. ISSN: 00346527, 1467937X.

Werning, Iván (2007). “Optimal Fiscal Policy with Redistribution”. In: *The Quarterly Journal of Economics* 122.3. Publisher: Oxford University Press JSTOR: 25098865, pp. 925–967. ISSN: 00335533, 15314650. (Visited on 11/26/2023).

West, Sarah E. and Roberton C. Williams (Apr. 2007). “Optimal taxation and cross-price effects on labor supply: Estimates of the optimal gas tax”. In: *Journal of Public Economics* 91.3, pp. 593–617. ISSN: 0047-2727. DOI: 10.1016/j.jpubeco.2006.08.007. URL: https://www.sciencedirect.com/science/article/pii/S0047272706001162.

A Optimal wedge derivation without externalities

Because $U(c, q, l)$ can be written as $U(c, q, y/\theta)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian of the recursive problem is

$$
\begin{aligned}
\mathcal{H} = &\left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta)\right] f \\
&+ \psi \left[-U_l(c, q, y/\theta)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta) \bar{\alpha}_t f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi \left[u - U(c, q, y/\theta) - \beta w\right]
\end{aligned}
$$

Differentiating yields the following first-order conditions

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta
\end{aligned}
$$

The three wedges can be derived from here. The labor wedge is derived in Section A.1, the savings wedge in Section A.2, and the commodity wedge in Section A.3.

A.1 Labor wedge

Dividing the first-order condition for c by U_c and using $\gamma_c \equiv U_{cl} l/U_c$,

$$
\phi = \frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c. \tag{8}
$$

From the first-order condition for l and since $U_{ll}l + U_l = U_l(1+\epsilon)$ with $\epsilon \equiv U_{ll}l/U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta} U_l(1+\epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon)$$

Equating the two expressions for ϕ and collecting the terms in ψ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta}(1+\epsilon-\gamma_c)$$

The labor wedge is $1-\tau^y = -U_l/(\theta U_c)$, so $\theta U_c/U_l = -1/(1-\tau^y)$, making the left-hand side $\frac{f}{U_c}\left(1-\frac{1}{1-\tau^y}\right) = -\frac{f}{U_c}\frac{\tau^y}{1-\tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{\psi U_c}{\theta f} = A_t(\theta)\,\frac{\psi U_c}{\theta f} \tag{9}$$

Next, substitute ϕ from (8) into $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta}\psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}$$

This is a linear first-order ordinary differential equation. Multiplying by the integrating factor $\exp\left[-\int^\theta \gamma_c(\tilde{x})\,d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with the boundary condition $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx. \tag{10}$$

The remaining boundary condition $\psi(0) = 0$ pins down λ_1,

$$\lambda_{1,t} = \frac{\int_0^\infty \exp\left[-\int_0^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{1}{U_{c,t}(x)}f_t(x) + \lambda_{2,t}f_{2,t}(x)\right)dx}{\int_0^\infty \exp\left[-\int_0^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\bar{\alpha}_t(x)f_t(x)dx}. \tag{11}$$

Substituting (10) into (9),

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{U_c(\theta)}{\theta f(\theta)}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx$$

I split the integral into the $[f/U_c - \lambda_1 \bar{\alpha}_t f]$ part, which yields the intratemporal term $A_t B_t C_t$, and the $\lambda_2 f_2$ part, which yields the intertemporal term. For the first part, carry $U_c(\theta)/U_c(x)$ into the integrand. Along the optimal allocation $U_{c,t}$ depends on the type through $c(\theta)$, $q(\theta)$, and $l(\theta) = y(\theta)/\theta$. Hence,

$$\frac{d}{d\theta} \ln U_{c,t} = \frac{U_{cc}\dot{c} + U_{cq}\dot{q} + U_{cl}\dot{l}}{U_c} = -\sigma_t \frac{\dot{c}}{c} - \sigma_{cq,t}\frac{\dot{q}}{q} + \gamma_c \frac{\dot{l}}{l}$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the following:

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] = \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right], \quad (12)$$

Here, the $\gamma_c/\tilde{x}$ term from $\dot{l}/l$ cancels the integrating factor. The cross-curvature $\sigma_{cq,t}$ enters because, under nonseparability, extracting resources from a higher type shifts the marginal utility of the numéraire through the adjustment of both goods. The first part therefore equals $A_t B_t C_t$. For the second part, two preliminaries pin down $\lambda_{2,t}$. The first-order condition for w_2 rearranges to $\partial V_{t+1}/\partial w_2 = -\beta R\,\psi/f$, and the envelope theorem applied to the period-$(t+1)$ problem gives $\partial V_{t+1}/\partial \hat{w}_2 = -\lambda_{2,t+1}$, the multiplier with which threat keeping enters the Lagrangian. Equating the two expressions and shifting the date back one period,

$$\lambda_{2,t}(\theta^{t-1}) = \beta R \frac{\psi_{t-1}}{f_{t-1}}$$

Next, evaluating (9) at $t-1$ and solving for the costate,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} \frac{\theta_{t-1}}{U_{c,t-1}\,A_{t-1}}$$

The $\lambda_2 f_2$ block of the wedge is $\lambda_{2,t}$, a constant in the current type, times the same outer structure as the first part,

$$A_t(\theta)\frac{U_{c,t}(\theta)}{\theta f_t(\theta)}\lambda_{2,t}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx$$

Substituting the two preliminaries and collecting the ratios,

$$\beta R \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\, dx}{\theta f_t(\theta)} = \beta R \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} D_t(\theta)$$

Collecting both parts,

$$\frac{\tau^y}{1-\tau^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}} D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta)$$
$$B_t(\theta) = \frac{1-F_t(\theta)}{\theta f_t(\theta)}$$
$$C_t(\theta) = \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right] (1 - \lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)) \frac{f_t(x)dx}{1-F_t(\theta)}$$
$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}$$

This is Golosov et al. (2016)'s labor-wedge decomposition, with the cross-curvature $\sigma_{cq,t}$ in C_t.

A.2 Savings wedge

The savings wedge follows from the same first-order conditions. I define the incentive credit χ_t, the share of the cost of delivering utility through the numéraire that the planner recoups because consumption complements labor, relaxing the incentive constraint:

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}} \frac{\tau^y_t}{1-\tau^y_t}$$

Using $\frac{\tau^y_t}{1-\tau^y_t} = \frac{\psi U_c}{\theta f}(1+\epsilon_t-\gamma_{c,t})$, $\phi = f/U_c - \psi\gamma_c/\theta$ from (8) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c}\left(1-\gamma_c\psi U_c/(\theta f)\right) = \frac{1-\chi_t}{U_{c,t}}. \tag{13}$$

$\chi_t > 0$ when the numéraire is complementary with labor and labor is taxed and zero when preferences are separable in the numéraire against labor or labor is undistorted. The first-order condition for w and the envelope condition $\partial V_t/\partial\hat{w} = \lambda_{1,t}$ one period ahead give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1-\chi_t}{U_{c,t}}. \tag{14}$$

A second expression for $\lambda_{1,t}$ comes from integrating the costate equation over the whole type space. With $\psi(0) = \psi(\infty) = 0$,

$$0 = \int_0^\infty \dot{\psi}\, d\theta = \lambda_{1,t} \int_0^\infty \bar{\alpha}_t f\, d\theta - \lambda_{2,t} \int_0^\infty f_2\, d\theta - \int_0^\infty \phi\, d\theta$$

The Pareto weights are normalized so that $\int \bar{\alpha}_t f\, d\theta = 1$, and $\int f_t(\theta|\theta_-)\, d\theta = 1$ for every θ_- implies $\int f_{2,t}\, d\theta = 0$. Hence $\lambda_{1,t} = \int_0^\infty \phi\, d\theta$, which by (13) is

$$\lambda_{1,t} = \mathbb{E}_{t-1}\left[\frac{1-\chi_t}{U_{c,t}}\right], \tag{15}$$

where $\mathbb{E}_{t-1}$ integrates over θ_t conditional on θ_{t-1}. The multiplier on promise keeping is set before the current draw is known. Evaluating (15) at $t+1$ and equating with (14),

$$\frac{1-\chi_t}{U_{c,t}} = \frac{1}{\beta R}\, \mathbb{E}_t\left[\frac{1-\chi_{t+1}}{U_{c,t+1}}\right].$$

Defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\, \mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]$$

This is the inverse Euler equation in the adjusted marginal utility $\widehat{U}_c$. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and, when $U_{cl} > 0$, the unit lowers the information rent $-U_l\, l/\theta$, an incentive gain worth $\chi_t/U_{c,t}$.

A.3 Commodity wedge

The commodity wedge uses the first-order conditions for c and q of the preamble. There are different equivalent ways to express it, but I will seek the familiar labor wedge $ABCD$ form. First, because the first-order conditions for each good share the same ϕ,

$$\frac{f_t}{U_q} - \psi \frac{U_{ql}}{U_q}\frac{l}{\theta} = \frac{f_t}{U_c} - \psi \frac{U_{cl}}{U_c}\frac{l}{\theta}$$

Cross-multiplying by U_q and U_c and using $\gamma_c \equiv \frac{U_{cl}\, l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\, l}{U_q}$,

$$U_c\left(f - \frac{\psi}{\theta}\gamma_q U_q\right) = U_q\left(f - \frac{\psi}{\theta}\gamma_c U_c\right)$$

Dividing by $U_q f$ and rearranging,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f}(\gamma_q - \gamma_c)$$

Note that, by definition, $1 + \tau^q = \frac{U_q}{U_c}$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

The prefactor $\psi U_c/(\theta f)$ is the same object that drives the labor wedge. Solving (9) for it and substituting the $ABCD$ decomposition of Appendix A.1,

$$\frac{\psi U_c}{\theta f} = \frac{1}{A_t}\frac{\tau^y}{1-\tau^y} = B_t C_t + \beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\frac{D_t}{A_t}$$

Multiplying by the complementarity gap $\gamma_{c,t} - \gamma_{q,t}$,

$$\frac{\tau^q}{1+\tau^q} = (\gamma_{c,t} - \gamma_{q,t})\left[B_t C_t + \beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\frac{D_t}{A_t}\right]$$

with A_t, B_t, C_t, D_t as in Appendix A.1. Alternatively, multiplying $\frac{\psi U_c}{\theta f} = \frac{1}{A_t}\frac{\tau^y}{1-\tau^y}$ by the gap,

$$\frac{\tau^q}{1+\tau^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1+\epsilon_t - \gamma_{c,t}}\frac{\tau^y_t}{1-\tau^y_t}$$

The commodity wedge inherits every dynamic feature of the labor wedge. Its shape is that of the labor wedge, scaled by the elasticity term.

B Optimal wedge derivation with externalities

B.1 Validity of the recursive formulation

Section 4 takes its recursive form from Kapička (2013), whose consumption set is an arbitrary convex and compact set, so his reduction covers the consumption vector (c, q). The externality adds two ingredients his environment lacks: the pollution stock enters preferences and the planner internalizes its accumulation from aggregate emissions. This subsection shows that neither disturbs the validity of the recursive approach.

An allocation is a collection of measurable functions $c_t(\theta^t)$, $q_t(\theta^t)$, and $y_t(\theta^t)$ for $t = 0, \ldots, T$. Aggregate emissions at an allocation are

$$Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] \, dF_0(\theta) \qquad t = 0, \ldots, T$$

and the implied pollution stocks are $S_t = S(Q_{t-s}, \ldots, Q_t)$, with pre-period emissions given. As in the main text, $\Phi_j \equiv \partial S_{t+j}/\partial Q_t$, with $\Phi_j = 0$ for $j > s$ by the truncation of the carbon cycle. Multipliers dated beyond T are zero.

Lemma B.1 (incentive constraints with an aggregate pollution stock)**.**
Fix a deterministic emission sequence $(Q_0, \ldots, Q_T)$ and let $S_t = S(Q_{t-s}, \ldots, Q_t)$. Define the dated utility

$$\tilde{U}_t(c, q, l) \equiv U(c, q, l, S_t)$$

Given this sequence, (i) an allocation is incentive compatible in the economy with pollution if and only if it is incentive compatible in the no-externality economy of Section 4 with period utility $\tilde{U}_t$, and (ii) the local incentive constraint is

$$\dot{u}_t(\theta^t) = U_\theta\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right) + \beta\, w_{2,t}(\theta^t)$$

with no term in the pollution stock.

Proof. Step 1: deviations leave the pollution stock unchanged. Incentive compatibility compares one agent's payoffs under truthful reporting and under a reporting strategy σ, holding the allocation and all other reports fixed. Aggregate emissions integrate the allocation over the population and a single agent has measure zero in that integral, so Q_t and with it S_t take the same values under truth-telling and under the deviation. Every payoff entering the incentive constraint is therefore evaluated at the same pollution-stock sequence.

Step 2: the incentive constraints are those of a dated no-externality economy. By Step 1, the incentive constraint of Section 3 reads for every initial type θ and every reporting strategy σ,

$$\begin{aligned}\mathbb{E}_0 &\left[\sum_{t=0}^{T} \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right) \middle| \theta\right] \\ &\geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U\left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t\right) \middle| \theta\right]\end{aligned}$$

with the same fixed S_t on both sides. Substituting the definition of $\tilde{U}_t$ on both sides,

$$\begin{aligned}\mathbb{E}_0 &\left[\sum_{t=0}^{T} \beta^t \tilde{U}_t\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right) \middle| \theta\right] \\ &\geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t \tilde{U}_t\left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}\right) \middle| \theta\right]\end{aligned}$$

The two displays are the same statement, so an allocation satisfies one if and only if it satisfies the other. The second is the incentive constraint of the no-externality economy with period utility $\tilde{U}_t$, which proves (i).

Step 3: the local constraint. Part (i) places the reporting problem in the environment of Kapička (2013) with period utility $\tilde{U}_t$. His requirements bear on the period utility date by date and $\tilde{U}_t$ inherits them from Assumption 3.1 because S_t enters U as a fixed parameter: $\tilde{U}_t$ is twice continuously differentiable, increasing in both goods, jointly concave in (c, q, l), and differentiable in the type. The dating of $\tilde{U}_t$ adds nothing beyond the date dependence Section 4 already carries in f_t and $\bar{\alpha}_t$ and the horizon is finite, so the one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) apply date by date:

$$\dot{u}_t(\theta^t) = \frac{\partial}{\partial \theta_t} \tilde{U}_t\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right) + \beta\, w_{2,t}(\theta^t)$$

The derivative passes through the pollution stock, which does not vary with the agent's type:

$$\frac{\partial}{\partial \theta_t} \tilde{U}_t \left(c_t, q_t, \frac{y_t}{\theta_t} \right) = U_\theta \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right) + U_S \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right) \frac{\partial S_t}{\partial \theta_t} = U_\theta \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right)$$

since $\partial S_t / \partial \theta_t = 0$. This proves (ii). □

The lemma is the formal version of the observation in Section 3 that the pollution stock enters both sides of the incentive constraint identically. To the agent, the pollution stock is a date-specific preference parameter, not a state she controls. The planner is different because it moves the pollution stock. The next proposition shows what internalizing it costs.

Proposition B.1 (recursive formulation with an externality).
Let the recursive formulation of Section 4 be valid for the no-externality economy. The relaxed planner problem with pollution then admits the recursive formulation of Section 5 with the emission window $\mathbf{S}_- = (Q_{t-s}, \ldots, Q_{t-1})$ as the only new state, the accumulation constraint as the only new restriction, and the first-order system of the recursive problem as that of the sequence problem. In that system, (i) given the emission price ν_t, every date-t history solves the node problem of Section 4 with period utility $\tilde{U}_t$ and the resource cost of the dirty good raised from 1 to $1 + \nu_t$; (ii) the pollution-stock multiplier μ_t is common to all date-t histories and equals the damage integral over the date-t cross-section; and (iii) the emission price collects the discounted pollution-stock multipliers, $\nu_t = \sum_{j=0}^{s} R^{-j} \mu_{t+j} \Phi_j$.

Proof. Step 1: the aggregates are deterministic and internalizing them is a finite-dimensional addition. Skills are the only shocks and they are idiosyncratic, allocations are measurable functions of skill histories, and agents form a continuum, so the cross-sectional distribution of histories at each date coincides with the ex-ante law of a single agent's history. Aggregate emissions Q_t are therefore numbers known at date 0 and so are the pollution stocks S_t. The relaxed planner problem is then equivalent to a joint choice of the allocation and two vectors $(Q_0, \ldots, Q_T)$ and $(S_0, \ldots, S_T)$, subject to the consistency constraint $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\, dF_0(\theta)$ and the accumulation constraint $S_t = S(Q_{t-s}, \ldots, Q_t)$ at each date. Substituting the two constraints back into the objective recovers the original problem, so the two problems select the same allocations.

Step 2: Lagrangian. Attach the multiplier ν_t in present value ν_t / R^t to the date-t consistency constraint and μ_t in present value μ_t / R^t to the date-t accumulation constraint, with the sign convention of the Hamiltonian in Appendix B.2. The Lagrangian of the relaxed problem is

$$\mathcal{L} = \mathcal{L}^{alloc}\left(\{c_t, q_t, y_t\}; \{S_t\}\right) + \sum_{t=0}^{T} \frac{\nu_t}{R^t}\left(\int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta]\, dF_0(\theta) - Q_t\right)$$
$$- \sum_{t=0}^{T} \frac{\mu_t}{R^t}\left[S_t - S(Q_{t-s}, \ldots, Q_t)\right]$$

where $\mathcal{L}^{alloc}$ collects every term of the Section 4 Lagrangian: the resource cost, the local incentive constraints, and the promise- and threat-keeping constraints, all with period utility $\tilde{U}_t$, as Lemma B.1 requires. The pollution stocks enter $\mathcal{L}^{alloc}$ only as the dates of $\tilde{U}_t$. The aggregates enter only the two constraint blocks.

Step 3: given the aggregates, the allocation problem is the Section 4 problem, re-dated and re-priced. Fix $(Q_t, S_t, \nu_t)_{t=0}^T$. Merging the consistency block with the resource cost turns the date-t flow cost of a history into

$$c_t(\theta^t) + (1 + \nu_t)\, q_t(\theta^t) - y_t(\theta^t)$$

plus the term $-\nu_t Q_t$, a constant with respect to the allocation. The allocation problem is therefore the no-externality problem with two changes: period utility $\tilde{U}_t$ and dirty-good resource cost $1 + \nu_t$. Both stay inside the class the Section 4 reduction covers. Dating the utility adds nothing beyond the date dependence already present in f_t and $\bar{\alpha}_t$, and over the finite horizon the recursion is a backward induction from $V_{T+1} \equiv 0$, so no stationarity is used. Re-pricing keeps the period cost linear, hence continuous and convex, and increasing in q for $\nu_t > -1$. Since ν_t will turn out to be the social cost of carbon, this asks only that the carbon price not subsidize the dirty good by more than its unit resource cost. The factorization of the incentive-constraint set into current controls and continuation promises never involves the objective, so the re-pricing leaves it untouched. Given the aggregates, the allocation problem is therefore recursive in the Section 4 states $(\hat{w}, \hat{w}_2, \theta_-)$ and its value functions solve the Bellman equation of that section with $\tilde{U}_t$ and the re-priced dirty good. This is (i).

Step 4: first-order conditions for the aggregates. The pollution stock S_t appears in $\mathcal{L}$ in two places: the accumulation block, with derivative $-\mu_t/R^t$, and the date-t utility terms of $\mathcal{L}^{alloc}$, at every date-t history. Within a history, utility enters two constraints: the local incentive constraint with costate ψ and the definition of u with multiplier ϕ, exactly as in the Hamiltonian of Appendix B.2. Differentiating each with respect to S,

$$\frac{\partial}{\partial S}\psi\left[-U_l\left(c, q, \frac{y}{\theta}, S\right)\frac{l}{\theta} + \beta w_2\right] = -\psi\, U_{Sl}\frac{l}{\theta} \qquad \frac{\partial}{\partial S}\phi\left[u - U\left(c, q, \frac{y}{\theta}, S\right) - \beta w\right] = -\phi\, U_S$$

The pollution stock is common to all date-t histories, so stationarity of $\mathcal{L}$ in S_t sums these contributions across the date-t cross-section,

$$\mu_t = \int_0^\infty \left(-\psi \, U_{Sl} \frac{l}{\theta} - \phi \, U_S \right) d\theta$$

where the integral runs over the date-t population of skill histories. This is the first-order condition for S in Appendix B.2 whose type integral is to be read over that cross-section: at $t = 0$ the cross-section is F_0 and the reading is literal and at later dates the social-cost-of-carbon formula applies it by writing the date-$(t + j)$ damage integrals as population averages, the $\mathbb{E}_t \int (\cdot) \, f_{t+j} \, d\theta$ terms of Proposition 5.2. Because there is no aggregate risk, these averages are deterministic: $\mathbb{E}_t$ denotes a cross-section, the same for every date-t history, not a forecast specific to the current node. In particular, μ_t is one number per date, which is (ii).

Current aggregate emissions Q_t appear in the consistency block with derivative $-\nu_t / R^t$ and, in the accumulation blocks of dates t, through $t + s$ since $S_\tau = S(Q_{\tau-s}, \ldots, Q_\tau)$ carries Q_t as an argument exactly when $t \leq \tau \leq t + s$. Stationarity of $\mathcal{L}$ in Q_t gives

$$-\frac{\nu_t}{R^t} + \sum_{\tau=t}^{t+s} \frac{\mu_\tau}{R^\tau} \frac{\partial S_\tau}{\partial Q_t} = 0 \quad \iff \quad \nu_t = \sum_{j=0}^{s} \frac{1}{R^j} \, \mu_{t+j} \, \Phi_j$$

which is (iii).

Step 5: recursive problem of Section 5 encodes the same system. Two checks remain: the state and the first-order conditions. From date t on, the aggregates matter only through the dated utilities $\tilde{U}_\tau$ for $\tau \geq t$, hence through $S_\tau = S(Q_{\tau-s}, \ldots, Q_\tau)$ for $\tau \geq t$, and each of these depends on past aggregates only through the window $\mathbf{S}_- = (Q_{t-s}, \ldots, Q_{t-1})$, the rest being current and future choices. Appending $\mathbf{S}_-$ to the Section 4 state is therefore sufficient and the window updates recursively, appending Q_t and dropping Q_{t-s}. The conditions: differentiate the Hamiltonian of Appendix B.2. The condition for q carries the extra term $\left[\mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q} \right] f$, so the dirty good's effective resource cost is $1 + \nu_t$ with $\nu_t = \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q_t}$, and the envelope chain of that appendix resolves $\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s} R^{-(j-1)} \mu_{t+j} \Phi_j$, so that

$$\nu_t = \mu_t \, \Phi_0 + \frac{1}{R} \sum_{j=1}^{s} \frac{1}{R^{j-1}} \, \mu_{t+j} \, \Phi_j = \sum_{j=0}^{s} \frac{1}{R^j} \, \mu_{t+j} \, \Phi_j$$

The condition for S is the population damage integral of Step 4. The conditions for (c, l, u, w, w_2) contain no aggregate term and are those of the Section 4 problem with $\tilde{U}_t$, matching Step 3.

The two first-order systems coincide condition by condition. $\square$

The argument concerns the relaxed problem, as everywhere in the paper: the envelope condition stands in for global incentive compatibility, and the sufficiency caveat of Section 3 is unchanged since Lemma B.1 maps the externality economy into the class for which that caveat was stated. Section 8 verifies global incentive compatibility numerically at the computed optima.

B.2 Commodity wedge

The Hamiltonian of this problem is as follows:

$$
\begin{aligned}
\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, \mathbf{S})\right] f \\
& + \psi\left[-U_l(c, q, y/\theta, S)\frac{l}{\theta} + \beta w_2\right] \\
& - \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi\left[u - U(c, q, y/\theta, S) - \beta w\right] \\
& - \mu\left[S - S(\ldots, Q)\right]
\end{aligned}
$$

This yields the following first-order conditions.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} + \left[\mu S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}\right] f = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta}\psi\left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S\right) d\theta = \mu
\end{aligned}
$$

Here $S_q \equiv \partial S/\partial Q$ is the marginal effect of current aggregate emissions on the current pollution stock and the density f in the first-order condition for q comes from differentiating the aggregate $Q = \int q(\theta)f(\theta)\,d\theta$ which enters both the current accumulation constraint and, as the newest entry of the emission state $\mathbf{S}$, the continuation value through $\partial V_{t+1}/\partial Q \equiv \partial V_{t+1}/\partial Q_t$. The scalar current stock S is not itself an argument of $V_{t+1}(\cdot, \mathbf{S})$; the continuation value sees current emissions only through $\mathbf{S}$, via $\partial V_{t+1}/\partial Q$, which is why no V term appears in the first-order condition for S. Define the full shadow price of current emissions,

$$\nu_t \equiv \mu_t S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q}. \tag{16}$$

As in the previous proof,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Multiplying by U_q and U_c,

$$U_c\left(f - \psi U_{ql}\frac{l}{\theta} + \nu_t f\right) = U_q\left(f - \psi U_{cl}\frac{l}{\theta}\right)$$

Using $\gamma_c \equiv \frac{U_{cl}\,l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\,l}{U_q}$, or, equivalently, $\gamma_c U_c = U_{cl}\,l$ and $\gamma_q U_q = U_{ql}\,l$,

$$U_c\left(f - \frac{\psi}{\theta}\gamma_q U_q + \nu_t f\right) = U_q\left(f - \frac{\psi}{\theta}\gamma_c U_c\right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \nu_t U_c f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f}(\gamma_q - \gamma_c) - \nu_t \frac{U_c}{U_q}$$

Note that, by definition, $1 + \tau^q = U_q/U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \nu_t \frac{U_c}{U_q}$$

Since $\frac{U_c}{U_q} = \frac{1}{1+\tau^q}$,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \frac{\nu_t}{1+\tau^q}$$

Rearranging,

$$\frac{\tau^q - \nu_t}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

The right-hand side is exactly that of the no-pollution primitive condition $\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$ of Appendix A.3, with τ^q replaced by the deviation $\tau^q - \nu_t$. The first-order conditions for c and l and the costate equation for ψ are untouched by the externality, so the integrating-factor steps there carry over verbatim and give

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

with A_t, B_t, C_t, D_t the labor-wedge terms of Appendix A.1. The externality leaves this block untouched and enters only by replacing τ^q with the deviation $\tau^q - \nu_t$ on the left.

Now consider ν_t. Its first piece is the date-t pollution-stock multiplier μ_t, the marginal damage of a unit of the current pollution stock. Substituting ϕ from the first-order condition for c into the first-order condition for S, then dividing and multiplying by f,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}\right] U_S \right) d\theta = \int_0^\infty \left(-\frac{\psi}{\theta f} U_{Sl} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l\right] \frac{U_S}{U_c} \right) f\, d\theta$$

Rewrite $\psi/\theta f$ by noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{U_c}\frac{1}{1+\epsilon-\gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{Sl} l}{U_S}$,

$$\begin{aligned}
\mu_t &= \int \left(-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{1+\epsilon-\gamma_c}\frac{U_{Sl} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{1+\epsilon-\gamma_c}\frac{U_{cl} l}{U_c}\right]\frac{U_S}{U_c} \right) f\, d\theta \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{1}{1+\epsilon-\gamma_c}\frac{U_{Sl} l}{U_c}\frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma_c}{1+\epsilon-\gamma_c}\right]\frac{U_S}{U_c} \right) f\, d\theta \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\xi}{1+\epsilon-\gamma_c}\frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma_c}{1+\epsilon-\gamma_c}\right]\frac{U_S}{U_c} \right) f\, d\theta \\
&= -\int \left[1 + \left(\frac{\tau^y}{1-\tau^y}\right)\frac{\xi-\gamma_c}{1+\epsilon-\gamma_c}\right]\frac{U_S}{U_c} f\, d\theta
\end{aligned}$$

The second piece of ν_t prices the echo of current emissions in future pollution stocks. By the envelope theorem applied to the period-$(t+1)$ problem, Q_t enters continuation costs only through the accumulation constraints of the next s periods: $S_{t+j} = S(Q_{t+j-s}, ..., Q_{t+j})$ carries Q_t as an argument for $j \leq s$ and drops it afterward. Each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \mu_{t+1}\,\Phi_1 + \frac{1}{R}\frac{\partial V_{t+2}}{\partial Q_t} \qquad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t}$$

and iterating until Q_t drops out ($\Phi_j = 0$ for $j > s$ by construction),

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s} \frac{1}{R^{j-1}}\,\mu_{t+j}\,\Phi_j$$

Substituting into (16), with $\Phi_0 = S_q$,

$$\nu_t = \mu_t\,\Phi_0 + \frac{1}{R}\sum_{j=1}^{s}\frac{1}{R^{j-1}}\,\mu_{t+j}\,\Phi_j = \sum_{j=0}^{s}\frac{1}{R^j}\,\mu_{t+j}\,\Phi_j$$

Substituting the damage block above for each μ_{t+j} and writing the date-$(t+j)$ population integrals as expectations over future skill histories,

$$\mathrm{SCC}_t \equiv \nu_t = -\sum_{j=0}^{s}\frac{1}{R^j}\,\mathbb{E}_t\int\left[1+\left(\frac{\tau^y_{t+j}}{1-\tau^y_{t+j}}\right)\frac{\xi_{t+j}-\gamma_{c,t+j}}{1+\epsilon_{t+j}-\gamma_{c,t+j}}\right]\frac{U_{S,t+j}}{U_{c,t+j}}\,\Phi_j\,f_{t+j}\,d\theta$$

Here, s is the number of periods emissions affect the pollution stock.

C Constrained-optimal commodity wedge

This appendix derives the results of Sections 6 and 7.3 in one pass. The commodity wedge is held to a schedule $\bar p_t(\theta) \equiv 1+\bar\tau^q_t(\theta)$: Section 6 reads every condition at a flat schedule, $\bar p_t(\theta) = 1+\bar\tau_t$, whose level is chosen optimally, and Section 7.3 at an arbitrary exogenous schedule. The shadow emission price ν_t of (16) is carried throughout; the no-externality economy of Section 6's first read sets $\nu_t = 0$. Multiplier conventions follow Appendix A.1, and $\eta^q(\theta)f_t(\theta|\theta_-)$ is the multiplier on the wedge constraint. Two shorthands recur: the ad-valorem deviation and its virtual counterpart,

$$\varpi_t(\theta) \equiv \frac{\bar\tau^q_t(\theta)-\nu_t}{1+\bar\tau^q_t(\theta)} \qquad\qquad \varpi^*_t(\theta^t) \equiv (\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}$$

which at $\nu_t = 0$ and a flat schedule are the ϖ_t and ϖ^*_t of Section 6.

C.1 The problem and first-order conditions

The Hamiltonian is that of Appendix B.2 plus the constraint term $\eta^q\,[U_q/U_c - \bar p]\,f$. Write $\Gamma^q \equiv U_q/U_c - \bar p$ and Γ^q_z for its partial derivatives; each follows from the quotient rule,

$$\Gamma^q_z = \frac{U_{qz}}{U_c} - \frac{U_q}{U_c}\frac{U_{cz}}{U_c} = \frac{U_{qz}}{U_c} - \bar p\,\frac{U_{cz}}{U_c}$$

evaluated at the constrained allocation, where $U_q/U_c = \bar p$. For the numéraire, with $U_{qc}/U_c = -\sigma_{cq}/q$ and $U_{cc}/U_c = -\sigma/c$,

$$\Gamma^q_c = -\frac{\sigma_{cq}}{q} + \bar p\,\frac{\sigma}{c} = \bar p\left[\frac{\sigma}{c} - \frac{\sigma_{cq}}{\bar p\, q}\right] = \bar p\,\mathcal{I}_t, \tag{17}$$

the income-effect gap of Section 6 evaluated at $1 + \tau^q = \bar p$. For the dirty good, define $\sigma_{q,t} \equiv -U_{qq}\, q_t/U_q$, the own-curvature of the dirty good's marginal utility, so that $U_{qq}/U_c = (U_{qq}q/U_q)(U_q/U_c)/q = -\bar p\,\sigma_q/q$ and

$$\Gamma^q_q = -\bar p\,\frac{\sigma_q}{q} + \bar p\,\frac{\sigma_{cq}}{q} = -\bar p\,\frac{\sigma_q - \sigma_{cq}}{q} \equiv -\bar p\,\mathcal{J}_t, \tag{18}$$

with $\mathcal{J}_t \equiv (\sigma_{q,t} - \sigma_{cq,t})/q_t$ the mirror image of $\mathcal{I}_t$. For labor, with $U_{ql}/U_c = (U_{ql}l/U_q)(U_q/U_c)/l = \bar p\,\gamma_q/l$ and $U_{cl}/U_c = \gamma_c/l$,

$$\Gamma^q_l = \bar p\,\frac{\gamma_q}{l} - \bar p\,\frac{\gamma_c}{l} = -\bar p\,\frac{\gamma_c - \gamma_q}{l}. \tag{19}$$

For the pollution stock, with $\xi_c \equiv U_{Sc}\, c/U_S$ and $\xi_q \equiv U_{Sq}\, q/U_S$ the complementarities of the pollutant with each good, $U_{qS}/U_c = \xi_q(U_S/U_c)/q$ and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so

$$\Gamma^q_S = \frac{U_S}{U_c}\left[\frac{\xi_q}{q} - \bar p\,\frac{\xi_c}{c}\right]. \tag{20}$$

The first-order conditions are those of Appendix B.2, each augmented by $\eta^q f\,\Gamma^q_z$:

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \eta^q f\,\Gamma^q_c = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \eta^q f\,\Gamma^q_q = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \eta^q f\,\Gamma^q_l = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta^q f\,\Gamma^q_S\right) d\theta = \mu
\end{aligned}$$

with the conditions for u, w, and w_2 unchanged, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the promise-keeping chain carry over. In the no-externality economy $\nu_t = 0$ and the pollution-stock condition is absent. When the level of a flat schedule is itself chosen, one more derivative is needed; it is taken in Appendix C.9.

C.2 Implementation and first-order approach

Fix a common price $\bar p = 1 + \bar\tau$ and consider the agent's split of expenditure x,

$$\tilde U(x, l; \bar\tau) = \max_{c,q\geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar p\, q = x$$

The first-order condition of the split is $U_q = \bar p\, U_c$. If we substitute c using the budget constraint, $U(x - \bar p\, q, q, l)$ and therefore $\tilde U_q = -U_c \bar p + U_q$ and $\tilde U_{qq} = U_{cc}\bar p^2 - 2U_{cq}\bar p + U_{qq}$. Assumption 3.1 implies that $\mathcal{D} < 0$ and that an interior split exists and is unique. A bundle (c, q) therefore satisfies the wedge constraint $U_q/U_c = \bar p$ if and only if it is the split of $x = c + \bar p q$.

For the envelope identities, differentiate $\tilde U = U(c^d, q^d, l)$. Differentiating the budget over x gives $\frac{\partial c^d}{\partial x} + \bar p\, \frac{\partial q^d}{\partial x} = 1$. By the first-order condition,

$$\tilde U_x = U_c \frac{\partial c^d}{\partial x} + U_q \frac{\partial q^d}{\partial x} = U_c \left(\frac{\partial c^d}{\partial x} + \bar p\, \frac{\partial q^d}{\partial x} \right) = U_c$$

Differentiating the budget in l at fixed x gives $\frac{\partial c^d}{\partial l} + \bar p\, \frac{\partial q^d}{\partial l} = 0$.

$$\tilde U_l = U_l + U_c \left(\frac{\partial c^d}{\partial l} + \bar p\, \frac{\partial q^d}{\partial l} \right) = U_l$$

A deviator who reports $\hat\theta$ receives the bundle assigned to $\hat\theta$, meaning the pair $(x(\hat\theta), y(\hat\theta))$. They split $x(\hat\theta)$ at the same common price, so the deviation space is that of a one-good economy in (x, y) with period utility $\tilde U(x, y/\theta; \bar\tau)$. The type enters $\tilde U$ only through $l = y/\theta$, so the envelope condition is $\dot u = -\tilde U_l\, l/\theta + \beta w_2 = -U_l\, l/\theta + \beta w_2$ and the first-order approach applies as in the one-good economy of Appendix A.1.

C.3 Demand identities

Differentiating the tangency $U_q = \bar p U_c$ together with the budget gives the demand derivatives. In x at fixed $l, \bar p$, the budget gives $\partial_x c^d = 1 - \bar p\, \partial_x q^d$. Substituting into $U_{qc}\partial_x c^d + U_{qq}\partial_x q^d = \bar p(U_{cc}\partial_x c^d + U_{cq}\partial_x q^d)$ and collecting,

$$\mathcal{D}\, \partial_x q^d = -\left(U_{cq} - \bar p U_{cc}\right) \qquad \Longrightarrow \qquad \partial_x q^d = \frac{U_{cq} - \bar p U_{cc}}{|\mathcal{D}|} = \frac{\bar p\, U_c\, \mathcal{I}_t}{|\mathcal{D}|}, \tag{21}$$

where the last step uses $U_{cq} - \bar p U_{cc} = U_c\left(\sigma \bar p/c - \sigma_{cq}/q\right) = \bar p\, U_c\, \mathcal{I}_t$ from the definitions of the statistics. Since $\partial_x q^d$ is the income expansion of dirty-good demand, $\mathcal{I}_t > 0$ exactly when the dirty good is normal. The budget then gives $\partial_x c^d = 1 - m_q$ with $m_q \equiv \bar{p}\,\partial_x q^d$, which satisfies $U_q \mathcal{J}_t = \bar{p}U_{cq} - U_{qq}$ and hence $\partial_x c^d = U_q \mathcal{J}_t / |\mathcal{D}|$ is positive when c is normal. In l at fixed $x, \bar{p}$, the budget gives $\partial_l c^d = -\bar{p}\,\partial_l q^d$ and the differentiated tangency yields $\mathcal{D}\,\partial_l q^d = \bar{p}U_{cl} - U_{ql}$, with $U_{cl} = \gamma_c U_c / l$ and $U_{ql} = \gamma_q \bar{p} U_c / l$,

$$\partial_l q^d = \frac{\bar{p}\,U_c\,(\gamma_c - \gamma_q)}{l\,\mathcal{D}} = \frac{\bar{p}\,U_c\,(\gamma_q - \gamma_c)}{l\,|\mathcal{D}|}. \tag{22}$$

Compensated own-price response: along an indifference curve $dc = -\bar{p}\,dq$ and differentiating the tangency in $\bar{p}$ at constant utility gives $\mathcal{D}\,dq = U_c\,d\bar{p}$. Hence,

$$s_{qq} \equiv \left.\frac{\partial q^d}{\partial \bar{p}}\right|_u = \frac{U_c}{\mathcal{D}} = -\frac{U_c}{|\mathcal{D}|} < 0 \qquad \frac{\partial q^d}{\partial \bar{p}} = s_{qq} - q^d\,\partial_x q^d \quad \text{(Slutsky)}$$

Three identities tie the constraint statistics to the demand objects. Combining (22) with s_{qq},

$$l\,\partial_l q^d = -\bar{p}\,|s_{qq}|(\gamma_c - \gamma_q)\,. \tag{23}$$

Adding $\bar{p}\,\mathcal{I}_t = (U_{cq} - \bar{p}U_{cc})/U_c$ and $\mathcal{J}_t = (\bar{p}U_{cq} - U_{qq})/U_q = (\bar{p}U_{cq} - U_{qq})/(\bar{p}U_c)$,

$$\Delta_t^q \equiv \bar{p}\,\mathcal{I}_t + \mathcal{J}_t = \frac{\bar{p}U_{cq} - \bar{p}^2 U_{cc} + \bar{p}U_{cq} - U_{qq}}{\bar{p}\,U_c} = \frac{-\mathcal{D}}{\bar{p}\,U_c} = \frac{1}{\bar{p}\,|s_{qq}|} > 0. \tag{24}$$

Combining (21) and (24), the marginal budget share is the numéraire-side curvature share,

$$m_q = \bar{p}\,\partial_x q^d = \frac{\bar{p}^2\,U_c\,\mathcal{I}_t}{|\mathcal{D}|} = \frac{\bar{p}\,\mathcal{I}_t}{\Delta_t^q}. \tag{25}$$

C.4 The multiplier

Dividing the first-order condition for c by $U_c f$, using $U_{cl} l = \gamma_c U_c$ and (17),

$$\frac{U_c\,\phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q\,\bar{p}\,\mathcal{I}_t. \tag{26}$$

Dividing the first-order condition for q by $U_q f$, multiplying by U_c, and substituting $U_c/U_q = 1/\bar{p}$ and (18),

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} + \eta^q\,\frac{\Gamma^q_q}{\bar{p}} = \frac{1+\nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} - \eta^q\,\mathcal{J}_t. \tag{27}$$

Equating (26) and (27) and collecting,

$$1 - \frac{1+\nu_t}{\bar{p}} = (\gamma_c - \gamma_q)\frac{\psi U_c}{\theta f} - \eta^q\left(\bar{p}\,\mathcal{I}_t + \mathcal{J}_t\right)$$

Since $1-(1+\nu_t)/\bar{p} = (\bar{\tau}^q_t - \nu_t)/(1+\bar{\tau}^q_t) = \varpi_t$ with Δ^q_t from (24),

$$\eta^q_t\,\Delta^q_t = \varpi^*_t - \varpi_t \qquad \Longleftrightarrow \qquad \eta^q_t = \bar{p}_t\,|s_{qq,t}|(\varpi^*_t - \varpi_t)\,. \tag{28}$$

At $\nu_t = 0$ and a flat schedule, this is Lemma 6.2. With the externality, it is Lemma 7.3 with $\mathrm{SCC}_t \equiv \nu_t$. At $\eta^q_t = 0$, (28) is the primitive corrective-wedge condition of Appendix B.2, so the multiplier vanishes where the frozen deviation equals the virtual one.

C.5 Labor wedge

From the first-order condition for l, using $U_{ll}l + U_l = U_l(1+\epsilon)$,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \frac{\eta^q f\,\Gamma^q_l}{U_l}$$

With $U_l = -(1-\tau^y_t)\theta U_c$, where τ^y_t is the endogenous labor wedge, and (19),

$$\frac{\Gamma^q_l}{U_l} = \frac{-\bar{p}(\gamma_c - \gamma_q)/l}{-(1-\tau^y_t)\theta U_c} = \frac{\bar{p}\,(\gamma_c - \gamma_q)}{(1-\tau^y_t)\,y\,U_c}$$

Multiplying by U_c/f,

$$\frac{U_c\,\phi}{f} = \frac{1}{1-\tau^y_t} - (1+\epsilon)\frac{\psi U_c}{\theta f} + \eta^q\,\frac{\bar{p}\,(\gamma_c - \gamma_q)}{(1-\tau^y_t)\,y} \tag{29}$$

Equating (26) and (29) and collecting, with $1/(1-\tau^y_t) - 1 = \tau^y_t/(1-\tau^y_t)$ and $A_t = 1+\epsilon-\gamma_c$,

$$\frac{\tau^y_t}{1-\tau^y_t} = A_t\frac{\psi U_c}{\theta f} + \eta^q_t\,\bar{p}_t\left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1-\tau^y_t)\,y_t}\right] \tag{30}$$

This is the primitive labor wedge of Propositions 6.1 and 7.6. The bracket is the response of the implied commodity wedge $\ln(U_q/U_c)$ to a utility-compensated unit of earnings. Along $(dc, dl) = (1-\tau_t^y, 1/\theta)$, we have $d\ln(U_q/U_c) = (1-\tau_t^y)\,\mathcal{I}_t + (\gamma_q - \gamma_c)/y$. Dividing by $1-\tau_t^y$ gives the bracket. Under weak separability the bracket is $\mathcal{I}_t$. Using (28) with $\varpi^* = 0$ and (25),

$$\eta_t^q\,\bar{p}_t\,\mathcal{I}_t = -\frac{\bar{p}_t\,\mathcal{I}_t}{\Delta_t^q}\,\varpi_t = -m_{q,t}\,\varpi_t = m_{q,t}\,\frac{\nu_t - \bar{\tau}_t^q}{1+\bar{\tau}_t^q}$$

which is Corollary 7.3.

C.6 Savings wedge

Define $\chi_t^q \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$, so that (26) reads

$$\frac{\phi}{f} = \frac{1-\chi_t^q}{U_{c,t}}.$$

The promise-keeping chain of Appendix A.2 is untouched, since the wedge constraint enters none of its steps. The first-order condition for w and the envelope condition $\partial V_{t+1}/\partial\hat{w} = \lambda_{1,t+1}$ give

$$\lambda_{1,t+1}(\theta^t) = \beta R\,\frac{\phi(\theta)}{f(\theta)} = \beta R\,\frac{1-\chi_t^q}{U_{c,t}}.$$

Integrating the costate equation over the type space with $\psi(0) = \psi(\infty) = 0$, $\int \bar{\alpha}_t f\,d\theta = 1$, and $\int f_2\,d\theta = 0$ gives the second expression

$$\lambda_{1,t} = \int_0^\infty \phi\,d\theta = \mathbb{E}_{t-1}\left[\frac{1-\chi_t^q}{U_{c,t}}\right].$$

Evaluating this at $t+1$ and equating with the previous display one period apart,

$$\frac{1-\chi_t^q}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^q}{U_{c,t+1}}\right]$$

Substituting the multiplier (28) and the curvature-share identity (25) assembles the adjustment into composite form,

$$\chi_t^q = \gamma_c\frac{\psi U_c}{\theta f} - m_q\,(\varpi_t^* - \varpi_t) = [\gamma_c - m_q(\gamma_c - \gamma_q)]\,\frac{\psi U_c}{\theta f} + m_q\,\varpi_t = \tilde{\gamma}_t\,\frac{\psi U_{c,t}}{\theta f_t} + m_{q,t}\,\varpi_t$$

with $\tilde{\gamma} \equiv (1-m_q)\gamma_c + m_q\gamma_q$ as in Lemma 6.3. At $\nu_t = 0$, this is the $\chi^q = \tilde{\gamma}\,\psi U_c/(\theta f) + \varpi m_q$ of Proposition 6.3 and $m_q\varpi = (\bar{\tau}^q - \nu)\,\partial_x q^d$ is the commodity revenue net of marginal damages, returned by a marginal dollar of expenditure. It remains to pass from the inverse Euler equation to the savings wedge. The savings wedge τ_t^s is defined by the agent's intertemporal condition $U_{c,t} = \beta R\,(1-\tau_t^s)\,\mathbb{E}_t[U_{c,t+1}]$, so

$$1-\tau_t^s = \frac{U_{c,t}}{\beta R\;\mathbb{E}_t[U_{c,t+1}]}.$$

The generalized inverse Euler equation gives $\beta R = U_{c,t}\;\mathbb{E}_t\big[(1-\chi_{t+1}^q)/U_{c,t+1}\big]\,/(1-\chi_t^q)$. Substituting,

$$1-\tau_t^s = \frac{1-\chi_t^q}{\mathbb{E}_t\big[(1-\chi_{t+1}^q)/U_{c,t+1}\big]\;\mathbb{E}_t[U_{c,t+1}]}, \tag{31}$$

the exact wedge. To approximate it, write $g_{t+1} \equiv \ln U_{c,t+1} = \bar g + \hat g$ with $\mathbb{E}_t\,\hat g = 0$, and treat the fluctuation $\hat g$ and the adjustment χ_{t+1}^q as small. Expanding $e^{\pm g_{t+1}}$ to second order in $\hat g$ and keeping first order in χ^q,

$$\begin{aligned}
\mathbb{E}_t[U_{c,t+1}] &= e^{\bar g}\left(1+\frac{1}{2}\operatorname{Var}_t\hat g\right),\\
\mathbb{E}_t\left[\frac{1-\chi_{t+1}^q}{U_{c,t+1}}\right] &= e^{-\bar g}\left(1-\mathbb{E}_t\,\chi_{t+1}^q+\operatorname{Cov}_t(\chi_{t+1}^q,\hat g)+\frac{1}{2}\operatorname{Var}_t\hat g\right),
\end{aligned}$$

where $\mathbb{E}_t\,\hat g = 0$ turns the linear term of the second expansion into the covariance. Their product is $1-\mathbb{E}_t\,\chi_{t+1}^q+\operatorname{Cov}_t(\chi_{t+1}^q,\hat g)+\operatorname{Var}_t\hat g$ to this order, so inverting (31),

$$1-\tau_t^s = (1-\chi_t^q)\left[1+\mathbb{E}_t\,\chi_{t+1}^q-\operatorname{Cov}_t(\chi_{t+1}^q,\hat g)-\operatorname{Var}_t\hat g\right] \approx 1-\chi_t^q+\mathbb{E}_t\,\chi_{t+1}^q-\operatorname{Cov}_t(\chi_{t+1}^q,\hat g)-\operatorname{Var}_t\hat g.$$

Since $\operatorname{Cov}_t(\chi_{t+1}^q,\hat g) = \operatorname{Cov}_t(\chi_{t+1}^q,\ln U_{c,t+1})$,

$$\tau_t^s \approx \operatorname{Var}_t(\ln U_{c,t+1}) + \big(\chi_t^q-\mathbb{E}_t\,\chi_{t+1}^q\big) + \operatorname{Cov}_t(\chi_{t+1}^q,\ln U_{c,t+1}),$$

the second-order form of Proposition 6.3. The baseline expansion of Appendix A.2 is the special case $\chi^q \mapsto \chi$, and the frozen-commodity expansion of Proposition 7.6 the same at $\nu_t = \mathrm{SCC}_t$.

C.7 Composite statistics and the comprehensive wedge

Substituting the multiplier (28) into the labor condition (30) yields the comprehensive form of Proposition 6.1. Using $\bar p/\Delta^q = \bar p^2|s_{qq}|$ from (24), the correction term expands into four pieces,

$$\eta^q\bar p\left[\mathcal{I}-\frac{\gamma_c-\gamma_q}{(1-\tau^y)y}\right] = \left[m_q-\frac{\bar p^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\right](\varpi^*-\varpi)$$

and, expanding with $\varpi^* = (\gamma_c-\gamma_q)\,\psi U_c/(\theta f)$,

$$
\begin{aligned}
m_q\,\varpi^* &= m_q(\gamma_c-\gamma_q)\frac{\psi U_c}{\theta f} = (\gamma_c-\tilde{\gamma})\frac{\psi U_c}{\theta f}\\
-\frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\,\varpi^* &= -\frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)^2}{(1-\tau^y)y}\,\frac{\psi U_c}{\theta f} = -\left(\epsilon-\tilde{\epsilon}\right)\frac{\psi U_c}{\theta f}\\
-m_q\,\varpi &= -\left(\bar{\tau}^q-\nu\right)\partial_x q^d\\
+\frac{\bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)}{(1-\tau^y)y}\,\varpi &= -\frac{\bar{\tau}^q-\nu}{1-\tau^y}\cdot\frac{\partial_l q^d}{\theta}
\end{aligned}
$$

where the second line uses $l|U_l| = (1-\tau^y)\theta l\,U_c = (1-\tau^y)y\,U_c$ to match Lemma 6.3's Le Chatelier correction, $\epsilon-\tilde{\epsilon} = \bar{p}^2U_c^2(\gamma_c-\gamma_q)^2/(l|U_l||\mathcal{D}|) = \bar{p}^2|s_{qq}|(\gamma_c-\gamma_q)^2/((1-\tau^y)y)$, and the fourth line uses $\partial_l q^d/\theta = -\bar{p}|s_{qq}|(\gamma_c-\gamma_q)/y$ from (23) and $\varpi\bar{p} = \bar{\tau}^q-\nu$. Moving the two fiscal pieces to the left-hand side of (30) and collecting the $\psi U_c/(\theta f)$ terms on the right,

$$
\frac{\tau_t^y}{1-\tau_t^y}+\frac{\bar{\tau}_t^q-\nu_t}{1-\tau_t^y}\left[(1-\tau_t^y)\,\partial_x q^d+\frac{\partial_l q^d}{\theta}\right] = \left[A_t+(\gamma_c-\tilde{\gamma})-(\epsilon-\tilde{\epsilon})\right]\frac{\psi U_c}{\theta f} = \tilde{A}_t\,\frac{\psi U_c}{\theta f}
$$

which is the comprehensive form, with $\bar{\tau}^q-\nu$ the screening tax and the bracket the compensated demand response $dq^d/dy|_u$. Along a perturbation dy with $dx = (1-\tau^y)dy$, utility is unchanged and $dq^d = [(1-\tau^y)\partial_x q^d + \partial_l q^d/\theta]dy$.

The composite statistics are also the curvatures of the indirect utility of Lemma 6.1, which proves the second claim of Lemma 6.3. Since $\tilde{U}_x = U_c(c^d,q^d,l)$ and $\partial_l c^d = -\bar{p}\,\partial_l q^d$,

$$
\tilde{U}_{xl} = U_{cl} + (U_{cq}-\bar{p}\,U_{cc})\,\partial_l q^d
$$

Dividing by $\tilde{U}_x = U_c$, multiplying by l, and substituting (21) and (22),

$$
\frac{\tilde{U}_{xl}\,l}{\tilde{U}_x} = \gamma_c + \frac{l\,(U_{cq}-\bar{p}U_{cc})}{U_c}\cdot\frac{\bar{p}U_c(\gamma_q-\gamma_c)}{l\,|\mathcal{D}|} = \gamma_c + m_q\,(\gamma_q-\gamma_c) = \tilde{\gamma}
$$

Similarly $\tilde{U}_l = U_l(c^d,q^d,l)$ gives, using $U_{lq}-\bar{p}U_{lc} = (\bar{p}U_c/l)(\gamma_q-\gamma_c)$,

$$
\tilde{U}_{ll} = U_{ll} + (U_{lq}-\bar{p}\,U_{lc})\,\partial_l q^d = U_{ll} + \frac{\bar{p}^2U_c^2(\gamma_c-\gamma_q)^2}{l^2\,|\mathcal{D}|}
$$

and dividing by $\tilde{U}_l = U_l < 0$ flips the sign of the positive correction,

$$\frac{\tilde{U}_{ll}\, l}{\tilde{U}_l} = \epsilon - \frac{\bar{p}^2\, U_c^2\, (\gamma_c - \gamma_q)^2}{l\, |U_l|\, |\mathcal{D}|} = \tilde{\epsilon} \leq \epsilon$$

Under weak separability, the corrections vanish and the problem is that of Appendix A.1.

C.8 The costate and the comprehensive-wedge decomposition

Substituting $\phi = f/U_c - \psi\gamma_c/\theta + \eta^q\bar{p}\,\mathcal{I}_t\, f/U_c$ from (26) into the costate equation and eliminating the multiplier with $\eta^q\bar{p}\mathcal{I}_t = m_q(\varpi^* - \varpi)$ and $m_q\varpi^*\, f/U_c = m_q(\gamma_c - \gamma_q)\,\psi/\theta$,

$$\dot{\psi} - \frac{\tilde{\gamma}}{\theta}\,\psi = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - [1 - m_q\,\varpi]\,\frac{f}{U_c}$$

Solving this ODE with the integrating factor $\exp[-\int \tilde{\gamma}\, d\tilde{x}/\tilde{x}]$ and $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \tilde{\gamma}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left([1 - m_q(x)\,\varpi(x)]\,\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right) dx$$

Here, $\psi(0) = 0$ pins $\lambda_{1,t}$ as in (11). To carry $U_c(\theta)$ into the integrand, multiply the identity (12) by $\exp[\int_\theta^x(\gamma_c - \tilde{\gamma})\, d\tilde{x}/\tilde{x}]$, with $\gamma_c - \tilde{\gamma} = m_q(\gamma_c - \gamma_q)$:

$$\frac{U_c(\theta)}{U_c(x)}\exp\left[-\int_\theta^x \tilde{\gamma}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] = \exp\left[\int_\theta^x \left(\sigma\frac{\dot{c}}{c} + \sigma_{cq}\frac{\dot{q}}{q} - \gamma_c\frac{\dot{y}}{y} + m_q\,(\gamma_c - \gamma_q)\,\frac{1}{\tilde{x}}\right) d\tilde{x}\right] \equiv \tilde{K}(\theta, x)$$

By Appendix C.7, $\Omega_t = \tilde{A}_t\,\psi U_c/(\theta f)$, so multiplying the costate solution by $\tilde{A}_t U_c/(\theta f)$ and splitting the integral as in Appendix A.1,

$$\Omega_t = \tilde{A}_t(\theta)\,\tilde{B}_t(\theta)\,\tilde{C}_t(\theta) + \beta R\,\Omega_{t-1}\,\tilde{D}_t(\theta)$$
$$\tilde{B}_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$
$$\tilde{C}_t(\theta) = \int_\theta^\infty \tilde{K}(\theta, x)\,(1 - m_q(x)\,\varpi(x) - \lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x))\,\frac{f_t(x)dx}{1 - F_t(\theta)}$$
$$\tilde{D}_t(\theta) = \frac{\tilde{A}_t(\theta)}{\tilde{A}_{t-1}}\,\frac{U_{c,t}(\theta)}{U_{c,t-1}}\,\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \tilde{\gamma}_t(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}$$

The persistence block uses $\lambda_{2,t} = \beta R\,\psi_{t-1}/f_{t-1}$ and, from $\Omega_{t-1} = \tilde{A}_{t-1}\,\psi_{t-1}U_{c,t-1}/(\theta_{t-1}f_{t-1})$,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1}\,\Omega_{t-1}}{\tilde{A}_{t-1}\,U_{c,t-1}}$$

Hence, the recursing object is the comprehensive wedge, which is Proposition 6.4. The revenue term $-m_q\varpi$ inside the $\tilde{C}_t$ weight is the only trace of the second good in the redistributive valuation.

C.9 The optimal uniform price

Let the schedule be flat, $\bar{\tau}^q_t(\theta) = \bar{\tau}_t$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date-t wedge constraints, each contributing $\eta^q f\,\partial[U_q/U_c - (1+\bar{\tau}_t)]/\partial\bar{\tau}_t = -\eta^q f$. The R^{-t} discounting and the transition densities convert the sum over date-t nodes into the unconditional population expectation, as in Step 4 of Proposition B.1. By the envelope theorem,

$$\frac{dW_0}{d\bar{\tau}_t} = -\frac{1}{R^t}\,\mathbb{E}_0[\eta^q_t]$$

Hence, the optimal level sets $\mathbb{E}_0[\eta^q_t] = 0$. In the reduction bookkeeping, the same condition emerges from Roy's identity offsetting the incentive value of compensation. The constraint formulation nets the two automatically and only the substitution distortion against the screening benefit survives. Substituting (28) with $\bar{p}_t$ and ϖ_t common across the date-t cross-section,

$$\mathbb{E}_0[\eta^q_t] = \bar{p}_t\,\{\mathbb{E}_0[|s_{qq,t}|\,\varpi^*_t] - \varpi_t\,\mathbb{E}_0[|s_{qq,t}|]\} = 0 \qquad \Longrightarrow \qquad \varpi_t = \frac{\mathbb{E}_0\left[|s_{qq,t}|\,(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Multiplying this by $\bar{p}_t\,\mathbb{E}_0[|s_{qq}|]$ and replacing $\bar{p}|s_{qq}|(\gamma_c-\gamma_q)\,\psi U_c/(\theta f) = -\,(\psi U_c/(\theta f))\,l\,\partial_l q^d$ and $\bar{p}\varpi_t = \bar{\tau}_t - \nu_t$,

$$(\bar{\tau}_t - \nu_t)\,\mathbb{E}_0[|s_{qq,t}|] = -\,\mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\,l_t\,\partial_l q^d_t\right] \qquad \Longleftrightarrow \qquad (\bar{\tau}_t - \nu_t)\,\mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t}\,l_t\,\partial_l q^d_t\right]$$

At node-specific prices the same condition without the integral gives $\varpi(\theta^t) = \varpi^*(\theta^t)$, the unconstrained wedge of Appendix A.3. The uniform tax is its $|s_{qq}|$-weighted average. Under weak separability, every node wedge is zero, so $\bar{\tau}_t = \nu_t$ (Corollaries 6.1 and 7.3). In retirement, $\psi \equiv 0$, so $\psi U_c/(\theta f) \equiv 0$ and $\bar{\tau}_t = \nu_t$ (Corollary 6.2).

C.10 The cost of uniformity

By the envelope theorem applied node by node, the date-t resource value of moving a single node's price is $-\eta^q(\theta)f(\theta)$ per unit of $\bar{\tau}^q(\theta)$. By (28),

$$-\eta^q\, f = \bar{p}\, |s_{qq}| (\varpi - \varpi^*)\, f$$

This value zeroes out at $\varpi = \varpi^*$, reproducing the node-specific optimum. The marginal resource cost of moving the node's ad-valorem wedge is $\bar{p}^2$ times this since $\varpi = 1 - (1+\nu)/\bar{p}$ gives $d\bar{\tau}^q = \bar{p}^2\, d\varpi/(1+\nu)$, which at $\nu = 0$ is $d\bar{\tau} = \bar{p}^2 d\varpi$. The loss from setting ϖ rather than ϖ^* at a node is then, to second order

$$\int_{\varpi^*}^{\varpi} \bar{p}^3\, |s_{qq}|\, (v - \varpi^*)\, dv = \frac{1}{2}\, \bar{p}^3\, |s_{qq}|\, (\varpi - \varpi^*)^2$$

Aggregating over the date-t cross-section, with $\bar{p}$ common because the price is uniform, and minimizing the resulting quadratic over the common ϖ yields the weighted mean $\varpi = \mathbb{E}_t^{|s|}[\varpi_t^*]$ of Appendix C.9 and, at that minimizer,

$$\text{Loss}_t \approx \frac{1}{2}\, (1+\bar{\tau}_t)^3\, \mathbb{E}_0[|s_{qq,t}|]\, \text{Var}_t^{|s|}(\varpi_t^*)$$

where $\mathbb{E}_t^{|s|}$ and $\text{Var}_t^{|s|}$ are the $|s_{qq,t}|$-weighted population mean and variance over date-t histories, which is Proposition 6.5.

C.11 The externality and the social cost of carbon

The conditions above already carry the shadow emission price: every deviation is measured net of ν_t, so all of Section 6's results hold with $\bar{\tau}_t = \tau_t^q - \nu_t$ the screening tax and every statistic evaluated at the externality allocation, and the optimal flat level of Appendix C.9 delivers the additive split of Proposition 6.6. What remains is the pollution-stock multiplier. Substitute the three pieces of the first-order condition for S,

$$-\psi U_{Sl}\frac{l}{\theta} = -\frac{\psi U_c}{\theta f}\, \xi\, \frac{U_S}{U_c}\, f \qquad -\phi U_S = -\,(1-\chi_t^q)\, \frac{U_S}{U_c}\, f \qquad \eta^q f\, \Gamma_S^q = \eta^q\, \frac{U_S}{U_c} \left[\frac{\xi_q}{q} - \bar{p}\, \frac{\xi_c}{c}\right] f$$

the second from (26) and the third from (20), to get

$$\mu_t = -\int_0^\infty \left[\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^q - \eta^q\left(\frac{\xi_q}{q} - \bar p\,\frac{\xi_c}{c}\right)\right]\frac{U_S}{U_c}\,f\,d\theta$$

Expanding $1-\chi_t^q = 1-\gamma_c\frac{\psi U_c}{\theta f} + \eta^q \bar p \mathcal{I}_t$ with $\bar p \mathcal{I}_t = \bar p\,\sigma/c - \sigma_{cq}/q$ and collecting,

$$\mu_t = -\int_0^\infty \left[1 + (\xi-\gamma_c)\,\frac{\psi U_c}{\theta f} + \eta^q\left(\bar p\,\frac{\sigma+\xi_c}{c} - \frac{\sigma_{cq}+\xi_q}{q}\right)\right]\frac{U_S}{U_c}\,f\,d\theta$$

The shadow emission price aggregates these date-$(t+j)$ damages exactly as in Appendix B.2. Current emissions enter the continuation value only through the accumulation constraints of the next s periods, so each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^{s}\frac{1}{R^{j-1}}\,\mu_{t+j}\,\Phi_j, \qquad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t},$$

and, with $\Phi_0 = S_q$, (16) gives $\nu_t = \sum_{j=0}^{s} R^{-j}\mu_{t+j}\Phi_j$. Substituting this expression for each μ_{t+j} and writing the date-$(t+j)$ population integrals as expectations over future skill histories,

$$\mathrm{SCC}_t \equiv \nu_t = -\sum_{j=0}^{s}\frac{1}{R^j}\,\mathbb{E}_t\int\left[1+(\xi_{t+j}-\gamma_{c,t+j})\,\frac{\psi U_{c,t+j}}{\theta f_{t+j}} + \eta^q_{t+j}\left(\bar p_{t+j}\,\frac{\sigma_{t+j}+\xi_{c,t+j}}{c_{t+j}} - \frac{\sigma_{cq,t+j}+\xi_{q,t+j}}{q_{t+j}}\right)\right]\frac{U_{S,t+j}}{U_{c,t+j}}\,\Phi_j\,f_{t+j}\,d\theta$$

whose bracket is the damage weight of Proposition 7.6 at each future node. The new piece is the response of the damage valuation to a budget-neutral clean-for-dirty swap: along $(dc, dq) = (\bar p, -1)$, which holds expenditure at consumer prices fixed, $d\ln(U_S/U_c) = \bar p(\sigma+\xi_c)/c - (\sigma_{cq}+\xi_q)/q$, using $\partial_c \ln(U_S/U_c) = (\sigma+\xi_c)/c$ and $\partial_q \ln(U_S/U_c) = (\sigma_{cq}+\xi_q)/q$. At $\eta^q \equiv 0$ the weight collapses to the fiscal factor of Appendix B.2; at the optimal flat level the multiplier averages to zero but is not zero node by node, so the weight carries the term even in Section 6, which is the sense in which Proposition 6.6 says the fiscal factor gains a term proportional to η^q. The corrective component of the price is nonetheless uniform across types and histories, because damages depend only on aggregate emissions, so the emissions margin carries no information about types and cannot screen; the uniformity restriction binds only the Corlett-Hague screening component and never the corrective one.

C.12 Retirement

This subsection proves Corollary 6.2 and the retirement claims of Section 7. Retirement dates are $t \geq T_R$. Output is zero and the period utility is $U(c, q, 0, S_t)$, so the current type enters neither preferences nor technology.

Lemma C.1 (retirement kills the costate)**.**
At every retirement date an optimal allocation is measurable with respect to the retirement-entry

history θ^{T_R-1}, the marginal promises vanish, $w_{2,t} \equiv 0$ for $t \geq T_R - 1$, the retirement component problem is deterministic node by node, and the costate on the local incentive constraint is $\psi_t \equiv 0$ for $t \geq T_R$. At the junction, the envelope condition of the last working date loses its future-rent term, $\dot{u}_{T_R-1} = U_\theta$.

Proof. Step 1: payoff-irrelevant reports cannot be separated. At a retirement date the report moves the assignment but neither preferences nor an output requirement, so the utility an agent obtains from reporting $\hat{\theta}$,

$$U\Big(c(\hat{\theta}), q(\hat{\theta}), 0, S_t\Big) + \beta\, w(\hat{\theta})$$

does not depend on the true type. Incentive compatibility then forces it to be constant in $\hat{\theta}$ on the support. Were it higher at some report, every type would prefer that report and truth-telling would fail for a positive mass of types. Hence, u_t is constant across the date-t draws given θ^{t-1} at every retirement date. The assignment can then be taken flat as well: every draw faces the same delivery problem, minimizing $c + q + \frac{1}{R}V_{t+1}$ over the bundles and continuation promises that deliver the common utility, so an optimal assignment common to all draws exists, and without loss the allocation from T_R on is measurable with respect to θ^{T_R-1}.

Step 2: the marginal promise dies at the junction. For $t \geq T_R - 1$, the marginal continuation value integrates a constant against the derivative of a density. By Step 1, $u_{t+1}(\theta^t, \theta')$ is constant in θ', equal to some $\bar{u}_{t+1}(\theta^t)$, so

$$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta')\, \partial_{\theta_t} f_{t+1}(\theta'|\theta_t)\, d\theta' = \bar{u}_{t+1}(\theta^t)\, \frac{\partial}{\partial \theta_t} \int_0^\infty f_{t+1}(\theta'|\theta_t)\, d\theta' = 0$$

since the density integrates to one at every θ_t. At the junction, $w_2 = 0$ enters the date-$(T_R - 1)$ problem as a feasibility restriction rather than a first-order condition: the retirement block can deliver no other value, so the choice of w_2 disappears, the last working date's costate is left free, and its envelope condition reads $\dot{u} = U_\theta$.

Step 3: the retirement block is deterministic and carries no costate. With the allocation measurable in θ^{T_R-1} and $w_2 \equiv 0$, the envelope condition at retirement dates reads $\dot{u} = U_\theta + \beta w_2 = 0$, which the flat profile satisfies identically. The type integral drops from the component problem, which becomes deterministic node by node,

$$V_t(\hat{w}, \mathbf{S}_-) = \min_{c,q,w,S}\; c + q + \frac{1}{R} V_{t+1}(w, \mathbf{S}) \qquad \text{s.t.} \qquad \hat{w} = U(c, q, 0, S) + \beta w$$

with the emission state and its aggregate constraint carried as in Section 5. There is no envelope constraint and no threat-keeping constraint, hence no costate and no threat multiplier: $\psi_t \equiv 0$ and $\lambda_{2,t} = 0$ for $t \geq T_R$. The working-age first-order system confirms the consistency of this reading: at $\lambda_2 = 0$, $\bar{\alpha}_t = 1$, and a flat allocation, the condition for the numéraire gives $\phi = f/U_c$, promise keeping gives $\lambda_{1,t} = \int \phi \, d\theta = 1/U_{c,t}$, and the costate equation integrates to

$$\dot{\psi} = \lambda_1 f - \phi = \frac{f}{U_{c,t}} - \frac{f}{U_{c,t}} = 0$$

which, with $\psi(0) = 0$, gives $\psi \equiv 0$.

Step 4: the relaxed problem alone would not deliver this. With persistent skills, retirement draws remain informative about working-life types and the relaxed problem, which keeps only the envelope condition, would tilt retirement utility against those draws to relax the last working date's incentive constraint through w_2. Step 1 rules the tilt out: nothing disciplines a payoff-irrelevant report, so no such tilt is globally incentive compatible. The measurability restriction of Step 1 is therefore a restriction to the implementable class, not an approximation, and it is how the life-cycle block of Section 8, following Farhi and Werning (2013), treats retirement.

Step 5: the formulas do not need the costate. In retirement, the formulas of interest are the commodity wedge, the savings wedge, and the social cost of carbon. In each of them, ψ enters multiplied by one of γ_c, γ_q, or ξ. Each of these statistics is proportional to l through its defining cross-partial, so each vanishes at $l = 0$. The retirement conclusions, a zero screening tax, the standard inverse Euler equation and Pigouvian carbon pricing therefore hold whatever the costate. □

Corollary 6.2 follows: $\psi \equiv 0$ gives $\psi U_c/(\theta f) \equiv 0$ at every retirement node, so the optimal uniform level of Appendix C.9 is $\bar{\tau}_t = \nu_t$, the pure Pigouvian price. The retirement statements of Sections 7.1 and 7.2 follow by the same steps. With no hours choice to decentralize, the frozen labor-wedge constraint is not imposed in retirement, so $\eta \equiv 0$ there while the frozen savings wedge keeps operating across retirement dates, which is why its distortions survive where the labor-frozen ones die.

D Third-best derivations: the frozen labor wedge

The modified problem's Hamiltonian is

$$\begin{aligned}
\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, \mathbf{S})\right] f \\
& + \psi \left[-U_l(c, q, y/\theta, S)\frac{l}{\theta} + \beta w_2\right] \\
& - \lambda_1 u(\theta)\bar{\alpha}_t f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi\left[u - U(c, q, y/\theta, S) - \beta w\right] \\
& - \mu\left[S - S(\ldots, Q)\right] \\
& + \eta\left[-\frac{U_l(c, q, y/\theta, S)}{\theta\, U_c(c, q, y/\theta, S)} - (1 - \bar{\tau}_t^y(\theta))\right] f
\end{aligned}$$

Write $\Gamma \equiv -U_l/(\theta U_c) - (1 - \bar{\tau}_t^y(\theta))$ for the constraint and $\Gamma_c, \Gamma_q, \Gamma_l, \Gamma_S$ for its partial derivatives. The first-order conditions are as follows:

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} + \eta f\,\Gamma_c = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \eta f\,\Gamma_q = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l + \eta f\,\Gamma_l = \frac{1}{\theta}\psi\left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1\bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta f\,\Gamma_S\right) d\theta = \mu
\end{aligned}$$

with $\nu_t \equiv \mu_t S_q + \frac{1}{R}\partial V_{t+1}/\partial Q$. Each derivative of Γ can be expressed as follows:

$$
\begin{aligned}
\Gamma_c &= (1-\bar{\tau}_t^y)\frac{\sigma}{c} - \frac{\gamma_c}{y} \\
\Gamma_q &= (1-\bar{\tau}_t^y)\frac{\sigma_{cq}}{q} - \frac{\gamma_q(1+\tau^q)}{y} \\
\Gamma_l &= (1-\bar{\tau}_t^y)\frac{\epsilon-\gamma_c}{l} \\
\Gamma_S &= -\frac{U_S}{U_c}\left[\frac{\xi}{y} + (1-\bar{\tau}_t^y)\frac{\xi_c}{c}\right]
\end{aligned}
$$

$\xi_c \equiv U_{Sc}c/U_S$ is the elasticity of the pollutant with respect to the numéraire consumption good.

Section 7.1 reads $\Gamma_c > 0$ as the normality of leisure; the equivalence is as follows. Consider the hours choice the frozen schedule decentralizes, as in Appendix D.5: the agent picks l at the net-of-tax rate $\omega \equiv 1-\bar{\tau}_t^y$, so the marginal wage is $\omega\theta$, with first-order condition $h \equiv \omega\theta\, U_c + U_l = 0$. Give the agent a marginal gift dm of the numéraire, holding the dirty-good assignment fixed, so that consumption moves by $dc = dm + \omega\theta\, dl$. Differentiating h,

$$
dh = \omega\theta\,[U_{cc}\,dc + U_{cl}\,dl] + U_{lc}\,dc + U_{ll}\,dl = (\omega\theta\,U_{cc} + U_{cl})\,dm + \left[\omega^2\theta^2 U_{cc} + 2\omega\theta\,U_{cl} + U_{ll}\right] dl = 0
$$

The bracket is the second-order form of the hours choice, computed in Appendix D.5 as $-\omega\theta^2 U_c\,\Delta_t$, negative at an interior maximum. For the coefficient on dm, substituting $U_{cc} = -\sigma U_c/c$ and $U_{cl} = \gamma_c U_c/l$, with $y = \theta l$,

$$
\omega\theta\,U_{cc} + U_{cl} = -\omega\theta\,\frac{\sigma}{c}\,U_c + \frac{\gamma_c}{l}\,U_c = -\theta U_c\left[\omega\,\frac{\sigma}{c} - \frac{\gamma_c}{y}\right] = -\theta U_c\,\Gamma_c
$$

Hence

$$
\frac{dl}{dm} = -\frac{\omega\theta\,U_{cc} + U_{cl}}{\omega^2\theta^2 U_{cc} + 2\omega\theta\,U_{cl} + U_{ll}} = -\frac{-\theta U_c\,\Gamma_c}{-\omega\theta^2 U_c\,\Delta_t} = -\frac{\Gamma_c}{\omega\theta\,\Delta_t}
$$

Since $\Delta_t > 0$ at an interior choice and $\omega\theta > 0$, $\operatorname{sign}(dl/dm) = -\operatorname{sign}(\Gamma_c)$: a gift of the numéraire depresses labor supply exactly when $\Gamma_c > 0$, which is the sense in which Section 7.1 calls leisure normal.

From the first-order condition for c,

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c + \eta\,\frac{\Gamma_c}{U_c} f \tag{32}$$

From the first-order condition for l,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \eta\frac{\Gamma_l}{U_l} f$$

Hence,

$$\frac{f}{U_c} - \frac{\psi}{\theta}\gamma_c + \eta\,\frac{\Gamma_c}{U_c} f = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \eta\frac{\Gamma_l}{U_l} f$$

After collecting terms,

$$\left(\frac{U_l}{\theta U_c} + 1\right)\theta f = -\frac{\psi U_l}{\theta}(1+\epsilon-\gamma_c) + \eta U_l \left[\frac{\Gamma_l}{U_l} - \frac{\Gamma_c}{U_c}\right] f$$

Because $\bar{\tau}_t^y = 1 + \frac{U_l}{\theta U_c}$,

$$\bar{\tau}_t^y \theta f = -\frac{\psi U_l}{\theta}(1+\epsilon-\gamma_c) + \eta U_l \left[\frac{\Gamma_l}{U_l} - \frac{\Gamma_c}{U_c}\right] f$$

Dividing by θf, multiplying the RHS by $\frac{U_c}{U_c}$, and using the wedge definition,

$$\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} = \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) - \eta\left[\Gamma_l \frac{U_c}{U_l} - \Gamma_c\right]$$

Substituting Γ_l by its definition,

$$\begin{aligned}
\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} &= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) - \eta\left[(1-\bar{\tau}_t^y)\frac{\epsilon-\gamma_c}{l}\frac{U_c}{U_l} - (1-\bar{\tau}_t^y)\frac{\sigma}{c} + \frac{\gamma_c}{y}\right] \\
&= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) - \eta\left[(1-\bar{\tau}_t^y)\frac{\epsilon-\gamma_c}{y}\frac{\theta U_c}{U_l} - (1-\bar{\tau}_t^y)\frac{\sigma}{c} + \frac{\gamma_c}{y}\right] \\
&= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) + \eta\left[\frac{\epsilon-\gamma_c}{y} + (1-\bar{\tau}_t^y)\frac{\sigma}{c} - \frac{\gamma_c}{y}\right] \\
&= \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma_c) + \eta\left[(1-\bar{\tau}_t^y)\frac{\sigma}{c} + \frac{\epsilon-2\gamma_c}{y}\right]
\end{aligned}$$

D.1 Savings wedge

Define $\chi_t^\eta \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t}$. Then (32) reads

$$\frac{\phi}{f} = \frac{1-\chi_t^\eta}{U_{c,t}} \tag{33}$$

The chain of Appendix A.2 is unchanged, because the constraint enters none of its steps. The first-order condition for w and the envelope condition $\partial V_{t+1}/\partial \hat{w} = \lambda_{1,t+1}$ give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1-\chi_t^\eta}{U_{c,t}}$$

Integrating the costate equation with $\psi(0) = \psi(\infty) = 0$, $\int \bar{\alpha}_t f \, d\theta = 1$, and $\int f_2 \, d\theta = 0$,

$$\lambda_{1,t} = \int_0^\infty \phi \, d\theta = \mathbb{E}_{t-1}\left[\frac{1-\chi_t^\eta}{U_{c,t}}\right]$$

Evaluating the second expression at $t+1$ and equating with the first,

$$\frac{1-\chi_t^\eta}{U_{c,t}} = \frac{1}{\beta R}\ \mathbb{E}_t\left[\frac{1-\chi_{t+1}^\eta}{U_{c,t+1}}\right]$$

This is the adjusted generalized inverse Euler equation.

D.2 Commodity wedge

Dividing the first-order condition for q by U_q, using $U_{ql}l = \gamma_q U_q$,

$$\phi = \frac{(1+\nu_t)f}{U_q} - \frac{\psi}{\theta}\gamma_q + \eta\,\frac{\Gamma_q}{U_q}f$$

Multiplying by $\frac{U_c}{f}$, with $U_c/U_q = 1/(1+\tau^q)$,

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{1+\tau^q} - \frac{\psi U_c}{\theta f}\gamma_q + \eta\,\frac{\Gamma_q}{1+\tau^q}$$

Substituting Γ_q by its definition,

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{1+\tau^q} - \frac{\psi U_c}{\theta f}\gamma_q + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} - \frac{\gamma_q}{y}\right]$$

The same operations on the FOC for c, (32), with Γ_c substituted by its definition yields

$$\frac{U_c\,\phi}{f} = 1 - \frac{\psi U_c}{\theta f}\gamma_c + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma}{c} - \frac{\gamma_c}{y}\right]$$

Hence,

$$1 - \frac{\psi U_c}{\theta f}\gamma_c + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma}{c} - \frac{\gamma_c}{y}\right] = \frac{1+\nu_t}{1+\tau^q} - \frac{\psi U_c}{\theta f}\gamma_q + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} - \frac{\gamma_q}{y}\right]$$

Moving every term except $\frac{1+\nu_t}{1+\tau^q}$ to the left,

$$1 - \frac{1+\nu_t}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \eta\left[(1-\bar{\tau}_t^y)\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} - \frac{\gamma_q}{y} - (1-\bar{\tau}_t^y)\,\frac{\sigma}{c} + \frac{\gamma_c}{y}\right]$$

Grouping the $1/y$ terms with the complementarity gap and the curvature terms with $1-\bar{\tau}_t^y$,

$$1-\frac{1+\nu_t}{1+\tau^q}=(\gamma_c-\gamma_q)\left[\frac{\psi U_c}{\theta f}+\frac{\eta}{y}\right]-\eta\,(1-\bar{\tau}_t^y)\left[\frac{\sigma}{c}-\frac{\sigma_{cq}}{(1+\tau^q)\,q}\right]$$

Because $1-\frac{1+\nu_t}{1+\tau^q}=\frac{\tau^q-\nu_t}{1+\tau^q}$ and $\mathcal{I}_t\equiv\frac{\sigma}{c}-\frac{\sigma_{cq}}{(1+\tau^q)\,q}$,

$$\frac{\tau_t^q-\nu_t}{1+\tau_t^q}=(\gamma_{c,t}-\gamma_{q,t})\left[\frac{\psi U_{c,t}}{\theta f_t}+\frac{\eta_t}{y_t}\right]-\eta_t\,(1-\bar{\tau}_t^y)\,\mathcal{I}_t$$

This is Proposition 7.2 with $\mathrm{SCC}_t\equiv\nu_t$ (Appendix D.4). At $\eta_t=0$ the formula reduces to the primitive condition $\frac{\tau^q-\nu_t}{1+\tau^q}=\frac{\psi U_c}{\theta f}(\gamma_c-\gamma_q)$ of Appendix B.2.

The sign of $\mathcal{I}_t$ is the normality of the dirty good. By the demand identity (21) of Appendix C.3 evaluated at the price $1+\tau_t^q$, the income expansion of dirty-good demand is $\partial_x q^d=(1+\tau^q)\,U_c\,\mathcal{I}_t/|\mathcal{D}|$, so $\mathcal{I}_t>0$ exactly when the dirty good is normal.

D.3 The costate

Substituting (32) into the costate equation $\dot{\psi}=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\phi$,

$$\dot{\psi}=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\frac{f}{U_c}+\frac{\psi}{\theta}\gamma_c-\eta\,\frac{\Gamma_c}{U_c}f$$

Moving the γ_c term to the left,

$$\dot{\psi}-\frac{\gamma_c}{\theta}\psi=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\frac{f}{U_c}-\eta\,\frac{\Gamma_c}{U_c}f$$

The multiplier is itself a function of the costate. Solving Lemma 7.1 for η,

$$\eta=\frac{1}{\Delta_t}\left[\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}-A_t\,\frac{\psi U_c}{\theta f}\right]$$

Hence,

$$\eta\,\frac{\Gamma_c}{U_c}f=\frac{\Gamma_c}{\Delta_t}\left[\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}-A_t\,\frac{\psi U_c}{\theta f}\right]\frac{f}{U_c}=\frac{\Gamma_c}{\Delta_t}\,\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}\,\frac{f}{U_c}-\frac{\Gamma_c A_t}{\Delta_t}\,\frac{\psi}{\theta}$$

Substituting into the ordinary differential equation, moving the new ψ term to the left, and defining the constraint-augmented complementarity $\gamma^{\eta}_{c,t} \equiv \gamma_{c,t} + A_t \, \Gamma_{c,t}/\Delta_t$,

$$\dot{\psi} - \frac{\gamma^{\eta}_c}{\theta}\psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \left[1 + \frac{\Gamma_c}{\Delta_t}\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t}\right]\frac{f}{U_c}$$

This is a linear first-order ordinary differential equation of the same form as in Appendix A.1, with γ^{η}_c in place of γ_c in the homogeneous part and the frozen schedule entering the source. Multiplying by the integrating factor $\exp\left[-\int^{\theta} \gamma^{\eta}_c(\tilde{x})\, d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with $\psi(\infty) = 0$,

$$\psi(\theta) = \int_{\theta}^{\infty} \exp\left[-\int_{\theta}^{x} \gamma^{\eta}_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] \left(\left[1 + \frac{\Gamma_c}{\Delta_t}\frac{\bar{\tau}^y_t}{1-\bar{\tau}^y_t}\right]\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x)\right) dx$$

The boundary condition $\psi(0) = 0$ pins down $\lambda_{1,t}$ exactly as in (11), with γ^{η}_c in the kernel and the bracketed source in place of 1. The threat-keeping multiplier is unchanged in form, $\lambda_{2,t} = \beta R\, \psi_{t-1}/f_{t-1}$. Because Lemma 7.1 at $t-1$ gives

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1}}{U_{c,t-1}}\,\frac{\psi_{t-1}\, U_{c,t-1}}{\theta_{t-1} f_{t-1}} = \frac{\theta_{t-1}}{A_{t-1}\, U_{c,t-1}}\left[\frac{\bar{\tau}^y_{t-1}}{1-\bar{\tau}^y_{t-1}} - \eta_{t-1}\Delta_{t-1}\right]$$

it reads

$$\lambda_{2,t} = \beta R\, \frac{\theta_{t-1}}{A_{t-1}\, U_{c,t-1}}\left[\frac{\bar{\tau}^y_{t-1}}{1-\bar{\tau}^y_{t-1}} - \eta_{t-1}\Delta_{t-1}\right]$$

The persistence channel loads on the frozen wedge net of the schedule's error, rather than on the optimal wedge. Through ψ, the ratio $\psi U_c/(\theta f)$ inherits the hazard and persistence structure of the ABCD decomposition, evaluated in the third best; the wedges of Section 7 then follow from it and Lemma 7.1 without further integration.

D.4 The social cost of carbon

The first-order condition for S determines the date-t pollution-stock multiplier,

$$\mu_t = \int_0^{\infty} \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \eta f\, \Gamma_S\right) d\theta$$

Each piece can be expressed in the marginal damage valuation U_S/U_c. Using $U_{Sl}l = \xi U_S$,

$$-\psi U_{Sl}\frac{l}{\theta} = -\frac{\psi}{\theta}\,\xi\,U_S = -\frac{\psi U_c}{\theta f}\,\xi\,\frac{U_S}{U_c}\,f$$

From (33),

$$-\phi U_S = -\left(1-\chi_t^{\eta}\right)\frac{U_S}{U_c}\,f$$

Substituting Γ_S by its definition,

$$\eta f\,\Gamma_S = -\eta\left[\frac{\xi}{y} + \left(1-\bar{\tau}_t^{y}\right)\frac{\xi_c}{c}\right]\frac{U_S}{U_c}\,f$$

Hence,

$$\mu_t = -\int_0^{\infty}\left[\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^{\eta} + \eta\,\frac{\xi}{y} + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\xi_c}{c}\right]\frac{U_S}{U_c}\,f\,d\theta$$

Expanding $1-\chi_t^{\eta} = 1 - \gamma_c\frac{\psi U_c}{\theta f} + \eta\,\Gamma_c$ with Γ_c written out, the bracket becomes

$$\begin{aligned}
&\frac{\psi U_c}{\theta f}\,\xi + 1 - \chi_t^{\eta} + \eta\,\frac{\xi}{y} + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\xi_c}{c} \\
&\quad = 1 + (\xi-\gamma_c)\frac{\psi U_c}{\theta f} + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\sigma}{c} - \eta\,\frac{\gamma_c}{y} + \eta\,\frac{\xi}{y} + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\xi_c}{c} \\
&\quad = 1 + (\xi-\gamma_c)\frac{\psi U_c}{\theta f} + \eta\,\frac{\xi-\gamma_c}{y} + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\sigma+\xi_c}{c} \\
&\quad = 1 + (\xi-\gamma_c)\left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y}\right) + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\sigma+\xi_c}{c}
\end{aligned}$$

so that

$$\mu_t = -\int_0^{\infty}\left[1 + (\xi-\gamma_c)\left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y}\right) + \eta\left(1-\bar{\tau}_t^{y}\right)\frac{\sigma+\xi_c}{c}\right]\frac{U_S}{U_c}\,f\,d\theta$$

The envelope iteration that assembles ν_t from the future pollution-stock multipliers is that of Appendix B.2 unchanged: the wedge constraints at future dates depend on S only through the utility function, and their effect on continuation costs is already internalized in the future multipliers μ_{t+j}, so $\nu_t = \sum_{j=0}^{s} R^{-j}\mu_{t+j}\Phi_j$. Substituting the damage block and writing the date-$(t+j)$ population integrals as expectations over future skill histories gives the SCC_t of Proposition 7.2. At $\eta \equiv 0$ the weight collapses to $1+(\xi-\gamma_c)\,\psi U_c/(\theta f) = 1+\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\xi-\gamma_c}{1+\epsilon-\gamma_c}$, the fiscal factor of Appendix B.2. The new piece is an income effect: $(\sigma+\xi_c)/c = \partial \ln(U_S/U_c)/\partial c$ is the sensitivity of the marginal damage valuation to the numéraire, and it is priced at $\eta(1-\bar{\tau}_t^y)$ exactly as $\mathcal{I}_t$ is in the commodity wedge. Because it does not vanish under $U = V(h(c,q,S),l)$, the benchmark that made the second-best SCC_t purely Pigouvian in Section 5 no longer does so in the third best unless $\eta_t = 0$.

D.5 The optimal flat schedule

Let the schedule be flat, $\bar{\tau}_t^y(\theta) = \bar{\tau}_t^y$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date-t wedge constraints, each contributing

$$\eta f \,\frac{\partial}{\partial \bar{\tau}_t^y}\left[-\frac{U_l}{\theta U_c} - (1-\bar{\tau}_t^y)\right] = \eta f$$

The R^{-t} discounting and the transition densities convert the sum over date-t nodes into the unconditional population expectation, as in Appendix C.9. By the envelope theorem,

$$\frac{dW_0}{d\bar{\tau}_t^y} = \frac{1}{R^t}\,\mathbb{E}_0[\eta_t]$$

so the optimal level sets $\mathbb{E}_0[\eta_t] = 0$. Solving Lemma 7.1 for η_t and taking the expectation,

$$\mathbb{E}_0[\eta_t] = \mathbb{E}_0\left[\frac{1}{\Delta_t}\left(\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y} - A_t\,\frac{\psi U_{c,t}}{\theta f_t}\right)\right] = 0$$

Because the flat rate is common across the cross-section,

$$\frac{\bar{\tau}_t^y}{1-\bar{\tau}_t^y}\,\mathbb{E}_0\left[\frac{1}{\Delta_t}\right] = \mathbb{E}_0\left[\frac{A_t}{\Delta_t}\,\frac{\psi U_{c,t}}{\theta f_t}\right]$$

The weight $1/\Delta_t$ is a compensated earnings response. Consider the hours problem the frozen schedule decentralizes: the agent chooses l at the net-of-tax rate $\omega \equiv 1-\bar{\tau}_t^y$, with first-order condition

$$h \equiv \omega\theta\, U_c + U_l = 0$$

Perturb ω holding utility fixed. The compensation is $dc_0 = -\theta l\, d\omega$, so consumption moves only through hours, $dc = \omega\theta\, dl$, and

$$dh = \theta U_c\, d\omega + \left[\omega^2\theta^2 U_{cc} + 2\omega\theta U_{cl} + U_{ll}\right] dl = 0$$

The bracket is the second-order condition of the hours choice, and it is proportional to Δ_t. Substituting $U_{cc} = -\sigma U_c/c$, $U_{cl} = \gamma_c U_c/l$, and $U_{ll} = \epsilon U_l/l = -\epsilon\, \omega\theta U_c/l$, which uses $U_l = -\omega\theta U_c$ at the constrained allocation,

$$\begin{aligned}\omega^2\theta^2 U_{cc} + 2\omega\theta U_{cl} + U_{ll} &= -\omega\theta U_c\left[\omega\theta\,\frac{\sigma}{c} - \frac{2\gamma_c}{l} + \frac{\epsilon}{l}\right] \\ &= -\omega\theta^2 U_c\left[\omega\,\frac{\sigma}{c} + \frac{\epsilon - 2\gamma_c}{y}\right] = -\omega\theta^2 U_c\,\Delta_t\end{aligned}$$

with $y = \theta l$, so $\Delta_t > 0$ wherever the hours choice is an interior maximum. Hence,

$$dh = \theta U_c\, d\omega - \omega\theta^2 U_c\,\Delta_t\, dl = 0 \qquad \Longrightarrow \qquad \left.\frac{dl}{d\omega}\right|_u = \frac{1}{\omega\theta\,\Delta_t}$$

With $y = \theta l$,

$$s_{yy,t} \equiv \left.\frac{\partial y}{\partial\left(1 - \bar\tau_t^y\right)}\right|_u = \theta\left.\frac{dl}{d\omega}\right|_u = \frac{1}{\left(1 - \bar\tau_t^y\right)\Delta_t} \qquad \Longrightarrow \qquad \frac{1}{\Delta_t} = \left(1 - \bar\tau_t^y\right) s_{yy,t}$$

Substituting into the optimality condition, with $1 - \bar\tau_t^y$ common,

$$\frac{\bar\tau_t^y}{1 - \bar\tau_t^y} = \frac{\mathbb{E}_0\left[s_{yy,t}\, A_t\,\frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0\left[s_{yy,t}\right]}$$

This is Proposition 7.3. At node-specific rates the same condition without the integral returns $\bar\tau^y/(1-\bar\tau^y) = A_t\frac{\psi U_c}{\theta f}$, the virtual wedge; the optimal flat rate is its compensated-response-weighted average, exactly as the optimal uniform commodity tax of Appendix C.9 averages the commodity wedges node by node.

E Third-best derivations: the frozen savings wedge

This appendix derives the results of Section 7.2 with every intermediate step shown. Multiplier conventions follow Appendix D; the wedge constraint is now the intertemporal condition (5), carried by the marginal-utility promise state that the first subsection validates.

E.1 Validity of the recursive formulation

The wedge constraints of Appendices C and D are pointwise: each involves only the date-t controls at its own node, so appending it to the Hamiltonian changes feasibility and nothing else. The frozen-savings constraint (5) is not: it ties the date-t allocation to the expected marginal utility of the date-$(t+1)$ allocation, so it cannot enter the period Hamiltonian pointwise, and Section 7.2 carries it instead through the marginal-utility promise state $\hat{M}$. This subsection shows that treatment is exact. The argument parallels Appendix B.1: a lemma splits the constraint into date-local pieces linked by one scalar per node, and a proposition matches the first-order system of the augmented recursive problem to that of the sequence problem.

Throughout, the frozen-savings constraint is imposed at every node θ^t with $t = 0, \ldots, T-1$, and $U_{c,t}(\theta^t)$ abbreviates the marginal utility of the numéraire at the node,

$$U_{c,t}(\theta^t) \equiv U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)$$

Lemma E.1 (promise form of the frozen-savings constraint)**.**
An allocation satisfies (5) at every node if and only if there exist functions $M_t : \Theta^t \to \mathbb{R}_{++}$, $t = 0, \ldots, T-1$, such that at every node

$$M_t(\theta^t) = \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}^s_t(\theta_t))\, \beta R} \tag{34}$$

and

$$\int_0^\infty U_{c,t+1}(\theta^t, \theta')\, f_{t+1}(\theta'|\theta_t)\, d\theta' = M_t(\theta^t). \tag{35}$$

The definition (34) involves only the date-t allocation at θ^t; the requirement (35) involves only the date-$(t+1)$ allocation at its successors.

Proof. Suppose the allocation satisfies (5) at every node, and define M_t by (34). At node θ^t, (5) reads

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}^s_t(\theta_t))\, \beta R \int_0^\infty U_{c,t+1}(\theta^t, \theta')\, f_{t+1}(\theta'|\theta_t)\, d\theta'$$

Dividing both sides by $(1 - \bar{\tau}^s_t)\, \beta R$, positive since $\bar{\tau}^s_t < 1$,

$$\int_0^\infty U_{c,t+1}(\theta^t, \theta')\, f_{t+1}(\theta'|\theta_t)\, d\theta' = \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}^s_t(\theta_t))\, \beta R} = M_t(\theta^t)$$

which is (35). Conversely, substituting (34) into (35) and multiplying by $(1 - \bar{\tau}^s_t)\, \beta R$ returns (5) at the node. For the locality claim: $U_{c,t}(\theta^t)$ is a function of the date-t bundle at θ^t and of S_t, a deterministic date-t object by Step 1 of Proposition B.1, and the integral in (35) runs over the date-$(t+1)$ bundles at the successors (θ^t, θ'). □

The lemma isolates the constraint's only intertemporal content: one number per node, computed at the parent from its own controls by (34) and delivered by the successors on average by (35). That is a promise, of exactly the kind the state already carries. The promise $\hat{w}$ fixes the successors' $\bar{\alpha}_t f_t$-weighted average utility, the threat $\hat{w}_2$ their $f_{2,t}$-weighted average, and $\hat{M}$ now fixes their f_t-weighted average marginal utility; nothing in the constraint reaches beyond one period.

Proposition E.1 (recursive formulation with a frozen savings wedge)**.**
Suppose the recursive formulation of Section 5 is valid (Proposition B.1). Then the relaxed planner problem with the frozen savings schedule admits the recursive formulation of Section 7.2: the promise $\hat{M}$ is the only new state, the inherited requirement (6) the only new constraint, the passed state $M(\theta)$ is pinned by (34) rather than free, and the first-order system of the recursive problem is that of the sequence problem. In that system $\partial V_t/\partial \hat{M} = -\vartheta_t$, and $\vartheta_{t+1}(\theta^t)$ is the sequence multiplier on the node-θ^t constraint, so each first-order condition carries the combination $[\vartheta_t - \vartheta_{t+1}/((1 - \bar{\tau}^s_t)\, \beta R^2)]\, U_{cz} = (\eta^s_t / U_{c,t})\, U_{cz}$.

Proof. Step 1: the constraint set factorizes with one more promise. By Lemma E.1, replace the frozen-savings constraints with the pairs (34) and (35). Read at the date-t component problem with state $(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-)$ and the number $\hat{M}$ appended: the parent's requirement (35) is the constraint (6) on the date-t allocation rule alone, and the definition (34) computes the number each successor inherits from the date-t controls, adding neither a constraint nor a choice at date t, since $M(\theta)$ is a function of $(c, q, y)(\theta)$ and the date. Neither piece involves $u(\theta)$, $w(\theta)$, $w_2(\theta)$, or the reporting margin: the frozen wedge restricts the planner's assignment so that the allocation is consistent with free saving at the frozen return, changing feasibility and not incentive compatibility, so the factorization of the incentive-constraint set (Lemma B.1 and Proposition B.1) is untouched. Backward induction then delivers the augmented factorization. The final period inherits a requirement and passes none, since (5) runs only to $T-1$. At earlier dates, an allocation rule is feasible for $(\hat{w}, \hat{w}_2, \hat{M}, \theta_-, \mathbf{S}_-)$ if and only if it satisfies the envelope condition, promise and threat keeping, and (6), with continuation triples $(w(\theta), w_2(\theta), M(\theta))$ feasible at date $t+1$ and $M(\theta)$ given by (34). At $t = 0$ nothing is inherited, and the convention $\vartheta_0 = 0$ makes the date-0 conditions read as the general ones. The Bellman of Section 7.2 is this factorization written as a value function.

Step 2: the constraint terms of the recursive form. Two pieces of Appendix E.2's derivation carry the constraint. The date-t Lagrangian holds the inherited requirement as $\vartheta_t \left[\int U_c\, f\, d\theta - \hat{M}\right]$, so $\hat{M}$ appears only there and the envelope theorem gives $\partial V_t / \partial \hat{M} = -\vartheta_t$. The passed state enters through the continuation value, and the chain rule through (34) gives, for each control $z \in \{c, q, l, S\}$,

$$\frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M}{\partial z} f = -\frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s)\, \beta R^2} U_{cz}\, f$$

so each first-order condition carries

$$\left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s)\, \beta R^2}\right] U_{cz}\, f = \frac{\eta_t^s}{U_{c,t}} U_{cz}\, f$$

the combination of Section 7.2.

Step 3: the sequence Lagrangian carries the same terms. Let $p_t(\theta^t)$ be the date-t path density, the product of the initial density and the transitions. Attach to the constraint of node θ^t, written as (35) minus (34), the multiplier $a_t(\theta^t)$, scaled by $R^{-(t+1)} p_t(\theta^t)$; the sequence Lagrangian gains

$$\sum_{t=0}^{T-1} \frac{1}{R^{t+1}} \int a_t(\theta^t) \left[\int_0^\infty U_{c,t+1}(\theta^t, \theta')\, f_{t+1}(\theta'|\theta_t)\, d\theta' - \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}_t^s(\theta_t))\, \beta R}\right] p_t(\theta^t)\, d\theta^t$$

A date-t control $z(\theta^t)$ meets this block twice. Through the parent's constraint, at θ^{t-1}, its contribution to the first-order condition is

$$\frac{1}{R^t} a_{t-1}(\theta^{t-1})\, U_{cz,t}\, p_{t-1}(\theta^{t-1})\, f_t(\theta_t|\theta_{t-1}) = \frac{1}{R^t} a_{t-1}(\theta^{t-1})\, U_{cz,t}\, p_t(\theta^t)$$

and through its own constraint,

$$-\frac{1}{R^{t+1}} \frac{a_t(\theta^t)}{(1 - \bar{\tau}_t^s(\theta_t))\, \beta R} U_{cz,t}\, p_t(\theta^t)$$

Now normalize: divide the sequence condition at node θ^t by $R^{-t}\, p_{t-1}(\theta^{t-1})$. The date-$t$ flow cost then carries the coefficient $f_t(\theta_t|\theta_{t-1}) = f$, the Hamiltonian's convention, and the two constraint contributions become

$$a_{t-1}(\theta^{t-1})\, U_{cz}\, f - \frac{a_t(\theta^t)}{(1-\bar{\tau}_t^s)\, \beta R^2}\, U_{cz}\, f$$

Comparing with Step 2 identifies $\vartheta_t = a_{t-1}(\theta^{t-1})$ and $\vartheta_{t+1}(\theta^t) = a_t(\theta^t)$: the inherited multiplier is the parent's, the forward multiplier the node's own, one number per component problem in both cases. The boundaries follow. At $t = 0$ there is no parent constraint, so the first term is absent and $\vartheta_0 = 0$; at $t = T$ there is no own constraint, so the second term is absent and the condition carries ϑ_T alone, which is $\eta_T^s = \vartheta_T U_{c,T}$. The terms in U_{cS} aggregate across the date-t cross-section inside the first-order condition for S, exactly as in Step 4 of Proposition B.1, with the η^s combination appearing node by node, which is the pollution-stock condition of Appendix E.2. The two first-order systems therefore coincide condition by condition. $\square$

E.2 The problem and first-order conditions

The constraint at date t, node θ^t, requires $\mathbb{E}_t[U_{c,t+1}] = U_{c,t}/((1-\bar{\tau}_t^s(\theta))\, \beta R)$. It involves the next period's controls, so it enters the recursive problem through the marginal-utility promise state of Section 7.2: the date-t problem inherits the requirement (6) with multiplier ϑ_t and passes each successor the state

$$M(\theta) = \frac{U_c\Big(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\Big)}{(1-\bar{\tau}_t^s(\theta))\, \beta R}$$

which is pinned by the current allocation and is not a free control. With the Lagrangian carrying $\vartheta_t \left[\int U_c f\, d\theta - \hat{M}\right]$, the envelope theorem gives $\partial V_t/\partial \hat{M} = -\vartheta_t$. At $t = 0$ there is no inherited requirement, so $\vartheta_0 = 0$; at $t = T$ no state is passed, since there is no date $T+1$. The Hamiltonian is

$$
\begin{aligned}
\mathcal{H} = &\left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c,q,l,S), \theta, \mathbf{S})\right] f \\
&+ \psi \left[-U_l(c,q,y/\theta,S)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta) \bar{\alpha}_t f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi \left[u - U(c,q,y/\theta,S) - \beta w\right] \\
&- \mu \left[S - S(\ldots, Q)\right] \\
&+ \vartheta_t \, U_c(c,q,y/\theta,S) \, f
\end{aligned}
$$

Differentiating the continuation term with respect to a control $z \in \{c, q, l, S\}$, by the chain rule through M and the envelope condition at $t+1$,

$$
\frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M}{\partial z} f = \frac{1}{R} \left(-\vartheta_{t+1}(\theta^t)\right) \frac{U_{cz}}{(1-\bar{\tau}_t^s)\,\beta R} f = -\frac{\vartheta_{t+1}(\theta^t)}{(1-\bar{\tau}_t^s)\,\beta R^2} U_{cz} \, f
$$

Adding the $\vartheta_t U_{cz} f$ term from the inherited requirement, every first-order condition gains the combination

$$
\left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1-\bar{\tau}_t^s)\,\beta R^2}\right] U_{cz} \, f = \frac{\eta_t^s}{U_{c,t}} U_{cz} \, f
$$

with η_t^s as defined in Section 7.2. The first-order conditions are as follows:

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: \; f - \psi U_{cl}\frac{l}{\theta} + \frac{\eta_t^s}{U_c} U_{cc}\, f = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: \; f - \psi U_{ql}\frac{l}{\theta} + \nu_t f + \frac{\eta_t^s}{U_c} U_{cq}\, f = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: \; -\theta f - \phi U_l + \frac{\eta_t^s}{U_c} U_{cl}\, f = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: \; \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\partial \mathcal{H}}{\partial w} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \; \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \; \int_0^\infty \left(-\psi U_{Sl}\frac{l}{\theta} - \phi U_S + \frac{\eta_t^s}{U_c} U_{cS}\, f\right) d\theta = \mu
\end{aligned}
$$

with ν_t the shadow price of current emissions of (16). In the statistics of Table **??**, $U_{cc}/U_c = -\sigma/c$, $U_{cq}/U_c = -\sigma_{cq}/q$, $U_{cl}/U_c = \gamma_c/l$, and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so the new term in each condition is η_t^s times the log-derivative of U_c in the corresponding direction. The constraint involves neither u, w, nor w_2, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the envelope identities for $\hat{w}$ and $\hat{w}_2$ are unchanged.

E.3 The multiplier's law of motion

Dividing the first-order condition for c by $U_c f$, using $U_{cl} l = \gamma_c U_c$ and $U_{cc}/U_c = -\sigma/c$,

$$
\frac{U_c\,\phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} \;\equiv\; 1 - \chi_t^s \tag{36}
$$

with $\chi_t^s \equiv \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s \sigma_t / c_t$. The chain of Appendix A.2 is unchanged, because the constraint enters none of its steps. The first-order condition for w and the envelope condition give

$$
\lambda_{1,t+1} = \beta R\,\frac{\phi}{f} = \beta R\,\frac{1 - \chi_t^s}{U_{c,t}}
$$

Integrating the costate equation over $(0, \infty)$ with $\psi(0) = \psi(\infty) = 0$,

$$
\lambda_{1,t} = \int_0^\infty \phi\, d\theta = \mathbb{E}_{t-1}\left[\frac{1 - \chi_t^s}{U_{c,t}}\right]
$$

Evaluating the second expression at $t+1$ and equating with the first,

$$\frac{1-\chi_t^s}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right] \tag{37}$$

This is the generalized inverse Euler equation of Lemma 7.2. Multiplying (37) by $U_{c,t}$ and substituting the frozen constraint $U_{c,t} = (1-\bar{\tau}_t^s)\beta R\,\mathbb{E}_t[U_{c,t+1}]$,

$$1-\chi_t^s = (1-\bar{\tau}_t^s)\,\mathbb{E}_t[U_{c,t+1}]\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right]$$

Because $\mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[XY] - \mathrm{Cov}(X,Y)$, with $X = U_{c,t+1}$ and $Y = (1-\chi_{t+1}^s)/U_{c,t+1}$,

$$1-\chi_t^s = (1-\bar{\tau}_t^s)\left\{\mathbb{E}_t\big[1-\chi_{t+1}^s\big] - \mathrm{Cov}_t\left(U_{c,t+1},\,\frac{1-\chi_{t+1}^s}{U_{c,t+1}}\right)\right\} \tag{38}$$

When the date-$(t+1)$ draw is degenerate the covariance vanishes and (38) reads

$$1-\chi_t^s = (1-\bar{\tau}_t^s)\left(1-\chi_{t+1}^s\right)$$

Writing $\chi^s = \gamma_c\,\psi U_c/(\theta f) + \eta^s\sigma/c$ on both sides,

$$\left(1-\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t}\right) - \eta_t^s\frac{\sigma_t}{c_t} = (1-\bar{\tau}_t^s)\left[\left(1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}\right) - \eta_{t+1}^s\frac{\sigma_{t+1}}{c_{t+1}}\right]$$

Dividing through by $1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}$, with the virtual savings wedge defined by

$$1-\tau_t^{s,*} \equiv \frac{1-\gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t}}{1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}}$$

gives

$$(1-\tau_t^{s,*}) - \frac{\eta_t^s\,\sigma_t/c_t}{1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}} = (1-\bar{\tau}_t^s) - \frac{(1-\bar{\tau}_t^s)\,\eta_{t+1}^s\,\sigma_{t+1}/c_{t+1}}{1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}}$$

Multiplying by $1-\gamma_{c,t+1}\frac{\psi U_{c,t+1}}{\theta f_{t+1}}$ and collecting the η^s terms on the left,

$$\eta_t^s \frac{\sigma_t}{c_t} - (1-\bar{\tau}_t^s)\,\eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \left(1 - \gamma_{c,t+1} \frac{\psi U_{c,t+1}}{\theta f_{t+1}}\right)(\bar{\tau}_t^s - \tau_t^{s,*}) \tag{39}$$

This is the law of motion of Lemma 7.2. Its boundary conditions come from the definition of η^s: at $t = 0$ there is no inherited requirement, so $\vartheta_0 = 0$ and $\eta_0^s = -\vartheta_1(\theta^0)\,U_{c,0}/((1-\bar{\tau}_0^s)\,\beta R^2)$; at $t = T$ no requirement is passed, so $\eta_T^s = \vartheta_T\,U_{c,T}$. The chain $\vartheta_{t+1}(\theta^t) = (1-\bar{\tau}_t^s)\,\beta R^2\,[\vartheta_t - \eta_t^s/U_{c,t}]$ recovers the raw multipliers from the η^s sequence.

Sign reversal. Suppose $\eta^s \geq 0$ at every node; the case $\eta^s \leq 0$ is symmetric. Write $\kappa_t \equiv (1-\bar{\tau}_t^s)\,\beta R^2 > 0$. Along any path: at $t = 0$, $\vartheta_0 = 0$ and $\eta_0^s \geq 0$ give $\vartheta_1 \leq 0$; at interior dates, $\eta_t^s \geq 0$ gives $\vartheta_{t+1} \leq \kappa_t\,\vartheta_t$, so $\vartheta_t \leq 0$ propagates forward and $\vartheta_t \leq 0$ for all $t \geq 1$. At the final date, $\eta_T^s = \vartheta_T U_{c,T} \geq 0$ forces $\vartheta_T = 0$. Walking back: $0 = \vartheta_T \leq \kappa_{T-1}\vartheta_{T-1}$ gives $\vartheta_{T-1} \geq 0$, hence $\vartheta_{T-1} = 0$, and by induction $\vartheta_t = 0$ at every node, so $\eta^s \equiv 0$. A nonzero multiplier profile therefore takes both signs, which is Corollary 7.2.

The two-period example. Let $T = 1$, preferences separable ($\gamma_c = \sigma_{cq} = 0$, so $\tau_0^{s,*} = 0$), and $\bar{\tau}_0^s > 0$. The boundary conditions give $\eta_0^s = -\vartheta_1 U_{c,0}/\kappa_0$ and $\eta_1^s = \vartheta_1 U_{c,1}$. At $\gamma_c = 0$ the right-hand side of (39) at $t = 0$ is $\bar{\tau}_0^s - \tau_0^{s,*} = \bar{\tau}_0^s$; substituting both boundary expressions into the left,

$$-\vartheta_1 \left[\frac{U_{c,0}\,\sigma_0}{\kappa_0\,c_0} + (1-\bar{\tau}_0^s)\,\frac{U_{c,1}\,\sigma_1}{c_1}\right] = \bar{\tau}_0^s \qquad \Longrightarrow \qquad \vartheta_1 = -\frac{\bar{\tau}_0^s}{\dfrac{U_{c,0}\,\sigma_0}{\kappa_0\,c_0} + (1-\bar{\tau}_0^s)\,\dfrac{U_{c,1}\,\sigma_1}{c_1}} \; < \; 0$$

so $\eta_0^s > 0 > \eta_1^s$: with savings overtaxed, the carbon price sits above the social cost of carbon at date 0 and below it at date 1.

E.4 Labor wedge

From the first-order condition for l, using $U_{ll}l + U_l = U_l(1+\epsilon)$,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon) + \frac{\eta_t^s}{U_c}\frac{U_{cl}}{U_l}\,f$$

Now $U_l = -(1-\tau_t^y)\theta U_c$, where τ_t^y is the endogenous labor wedge, so $-\theta f/U_l = f/((1-\tau_t^y)U_c)$, and with $U_{cl} = \gamma_c U_c/l$,

$$\frac{\eta_t^s}{U_c}\frac{U_{cl}}{U_l} = \frac{\eta_t^s}{U_c}\frac{\gamma_c U_c/l}{-(1-\tau_t^y)\theta U_c} = -\frac{\eta_t^s\,\gamma_c}{(1-\tau_t^y)\,y\,U_c}$$

Multiplying by $\frac{U_c}{f}$,

$$\frac{U_c\,\phi}{f} = \frac{1}{1-\tau_t^y} - (1+\epsilon)\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\gamma_c}{(1-\tau_t^y)\,y} \tag{40}$$

Equating (36) and (40),

$$1 - \gamma_c\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} = \frac{1}{1-\tau_t^y} - (1+\epsilon)\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\gamma_c}{(1-\tau_t^y)\,y}$$

Moving 1 to the right and the $\psi U_c/(\theta f)$ and η_t^s terms to the left, with $\frac{1}{1-\tau_t^y} - 1 = \frac{\tau_t^y}{1-\tau_t^y}$ and $A_t = 1+\epsilon-\gamma_c$,

$$\frac{\tau_t^y}{1-\tau_t^y} = (1+\epsilon-\gamma_c)\frac{\psi U_c}{\theta f} - \eta_t^s\left[\frac{\sigma}{c} - \frac{\gamma_c}{(1-\tau_t^y)\,y}\right] = A_t\frac{\psi U_c}{\theta f} - \eta_t^s\left[\frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{(1-\tau_t^y)\,y_t}\right]$$

This is the first formula of Proposition 7.4. The bracket is minus the log-derivative of U_c along the utility-compensated earning direction $(dc, dl) = (1-\tau_t^y,\, 1/\theta)$, expressed per net-of-tax dollar of output, the intratemporal image of the delivery-drag statistics of Appendix D. At $\eta_t^s = 0$ the condition is (9).

E.5 Commodity wedge

Dividing the first-order condition for q by U_q, using $U_{ql}l = \gamma_q U_q$,

$$\phi = \frac{(1+\nu_t)f}{U_q} - \frac{\psi}{\theta}\gamma_q + \frac{\eta_t^s}{U_c}\,\frac{U_{cq}}{U_q}\,f$$

Multiplying by $\frac{U_c}{f}$, with $U_c/U_q = 1/(1+\tau^q)$ and $U_{cq}/U_c = -\sigma_{cq}/q$,

$$\frac{U_c\,\phi}{f} = \frac{1+\nu_t}{1+\tau^q} - \gamma_q\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma_{cq}}{(1+\tau^q)\,q} \tag{41}$$

Equating (41) with (36),

$$1 - \gamma_c\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma}{c} = \frac{1+\nu_t}{1+\tau^q} - \gamma_q\frac{\psi U_c}{\theta f} - \eta_t^s\,\frac{\sigma_{cq}}{(1+\tau^q)\,q}$$

After collecting terms, with $1-\frac{1+\nu_t}{1+\tau^q}=\frac{\tau^q-\nu_t}{1+\tau^q}$,

$$\frac{\tau_t^q-\nu_t}{1+\tau_t^q}=(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}+\eta_t^s\left[\frac{\sigma_t}{c_t}-\frac{\sigma_{cq,t}}{(1+\tau_t^q)\,q_t}\right]=(\gamma_{c,t}-\gamma_{q,t})\,\frac{\psi U_{c,t}}{\theta f_t}+\eta_t^s\,\mathcal{I}_t$$

This is the second formula of Proposition 7.4, with $\mathrm{SCC}_t\equiv\nu_t$ and the same income-effect gap $\mathcal{I}_t$ as in Appendix C.3, whose normality reading applies verbatim.

E.6 The costate

Rearranging (36),

$$\phi=\frac{f}{U_c}-\frac{\psi}{\theta}\gamma_c-\eta_t^s\,\frac{\sigma}{c}\,\frac{f}{U_c}$$

Substituting into the costate equation $\dot{\psi}=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\phi$ and moving the γ_c term to the left,

$$\dot{\psi}-\frac{\gamma_c}{\theta}\psi=\lambda_1\bar{\alpha}_t f-\lambda_2 f_2-\left[1-\eta_t^s\,\frac{\sigma}{c}\right]\frac{f}{U_c}$$

Unlike Appendix D.3, the multiplier η_t^s does not depend on the costate, so the integrating factor is that of Appendix A.1 unchanged, and only the source is reweighted. Multiplying by $\exp\left[-\int^\theta\gamma_c(\tilde{x})\,d\tilde{x}/\tilde{x}\right]$ and integrating from θ to ∞ with $\psi(\infty)=0$,

$$\psi(\theta)=\int_\theta^\infty\exp\left[-\int_\theta^x\gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\left[1-\eta_t^s(x)\,\frac{\sigma(x)}{c(x)}\right]\frac{f(x)}{U_c(x)}-\lambda_1\bar{\alpha}_t(x)f(x)+\lambda_2 f_2(x)\right)dx$$

with $\psi(0)=0$ pinning $\lambda_{1,t}$ as in (11). Carrying $U_c(\theta)/U_c(x)$ into the integrand as in identity (12), the product $A_t\frac{\psi U_c}{\theta f}$ inherits the $B_tC_tD_t$ structure of Appendix A.1 with the redistributive weight $1-\eta_t^s(x)\sigma(x)/c(x)-\lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)$ inside C_t: the resource value of taking from type x is corrected by the constraint's valuation of that type's marginal utility. The threat-keeping multiplier is unchanged in form, $\lambda_{2,t}=\beta R\,\psi_{t-1}/f_{t-1}$, with $\psi_{t-1}U_{c,t-1}/(\theta_{t-1}f_{t-1})$ read from the labor-wedge formula of Appendix E.4.

E.7 The social cost of carbon

The first-order condition for S determines the date-t pollution-stock multiplier,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S + \frac{\eta_t^s}{U_c} U_{cS}\, f \right) d\theta$$

Each piece can be expressed in the marginal damage valuation U_S/U_c. Using $U_{Sl} l = \xi U_S$,

$$-\psi U_{Sl} \frac{l}{\theta} = -\frac{\psi}{\theta}\, \xi\, U_S = -\frac{\psi U_c}{\theta f}\, \xi\, \frac{U_S}{U_c}\, f$$

From (36),

$$-\phi U_S = -\left(1 - \chi_t^s\right) \frac{U_S}{U_c}\, f$$

With $U_{cS}/U_c = \xi_c\, (U_S/U_c)/c$,

$$\frac{\eta_t^s}{U_c} U_{cS}\, f = \eta_t^s\, \frac{\xi_c}{c}\, \frac{U_S}{U_c}\, f$$

Hence,

$$\mu_t = -\int_0^\infty \left[\frac{\psi U_c}{\theta f}\, \xi + 1 - \chi_t^s - \eta_t^s\, \frac{\xi_c}{c} \right] \frac{U_S}{U_c}\, f\, d\theta$$

Expanding $1 - \chi_t^s = 1 - \gamma_c \frac{\psi U_c}{\theta f} - \eta_t^s \sigma / c$ and collecting,

$$\frac{\psi U_c}{\theta f}\, \xi + 1 - \gamma_c \frac{\psi U_c}{\theta f} - \eta_t^s\, \frac{\sigma}{c} - \eta_t^s\, \frac{\xi_c}{c} = 1 + (\xi - \gamma_c)\, \frac{\psi U_c}{\theta f} - \eta_t^s\, \frac{\sigma + \xi_c}{c}$$

so that

$$\mu_t = -\int_0^\infty \left[1 + (\xi - \gamma_c)\, \frac{\psi U_c}{\theta f} - \eta_t^s\, \frac{\sigma + \xi_c}{c} \right] \frac{U_S}{U_c}\, f\, d\theta$$

The envelope iteration $\nu_t = \sum_{j=0}^{s} R^{-j} \mu_{t+j} \Phi_j$ is unchanged, giving the damage weight of Proposition 7.4. The income-effect piece is the same $\partial \ln(U_S/U_c)/\partial c = (\sigma + \xi_c)/c$ as in Appendix D.4, now priced at $-\eta_t^s$; at $\eta^s \equiv 0$ the weight collapses to the fiscal factor of Appendix B.2.

E.8 The optimal flat rate

Let the schedule be flat, $\bar{\tau}^s_t(\theta) = \bar{\tau}^s_t$, with the level chosen by the planner. The rate enters the date-0 Lagrangian only through the states $M(\theta)$ passed forward at date t (Appendix E.2). By the chain rule and the envelope condition $\partial V_{t+1}/\partial \hat{M} = -\vartheta_{t+1}$, each date-$t$ node contributes

$$\frac{1}{R}\,\frac{\partial V_{t+1}}{\partial \hat{M}}\,\frac{\partial M}{\partial \bar{\tau}^s_t}\,f = \frac{1}{R}\,(-\vartheta_{t+1})\,\frac{U_{c,t}}{(1-\bar{\tau}^s_t)^2\,\beta R}\,f = -\,\frac{\vartheta_{t+1}\,U_{c,t}}{(1-\bar{\tau}^s_t)^2\,\beta R^2}\,f$$

Aggregating over date-t nodes as in Appendix D.5, with the common factor $(1-\bar{\tau}^s_t)^{-2}\,\beta^{-1}R^{-2}$ pulled out, the envelope theorem gives

$$\frac{dW_0}{d\bar{\tau}^s_t} = -\,\frac{R^{-t}}{(1-\bar{\tau}^s_t)^2\,\beta R^2}\;\mathbb{E}_0[\vartheta_{t+1}\,U_{c,t}]$$

so the optimal level sets $\mathbb{E}_0[\vartheta_{t+1}U_{c,t}] = 0$. For the equivalence in Proposition 7.5, rearranging the definition of the composite multiplier, $\eta^s_t = \vartheta_t U_{c,t} - \vartheta_{t+1}U_{c,t}/((1-\bar{\tau}^s_t)\,\beta R^2)$,

$$\frac{\vartheta_{t+1}\,U_{c,t}}{(1-\bar{\tau}^s_t)\,\beta R^2} = \vartheta_t\,U_{c,t} - \eta^s_t$$

Taking $\mathbb{E}_0$ of both sides, the optimality condition reads $\mathbb{E}_0[\eta^s_t] = \mathbb{E}_0[\vartheta_t U_{c,t}]$; at $t=0$, $\vartheta_0 = 0$ gives $\mathbb{E}_0[\eta^s_0] = 0$. Only the forward component of the multiplier is priced, because the inherited requirements at date t were set by the date-$(t-1)$ rate. Optimization therefore disciplines the multiplier's weighted averages, not its profile: η^s generally remains nonzero node by node, the sign-reversal argument of Appendix E.3 is unaffected, and the carbon-price tilt of Corollary 7.2 survives at the optimal flat path.